GPW WATS 3.01
Market Data Protocol

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1. DISCLAIMER

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In case of sections of documentation at a High level work progress according to the current version of *GPW WATS Advancement of Documentation*, Warsaw Stock Exchange will endeavor to limit changes to these sections of documents to those related to:

1. correcting errors in the documentation or in the software;
2. clarification of the documentation content or removing ambiguity;
3. implementation of approved change requests or;
4. regulatory changes.

2. PREFACE

This document has been prepared by Warsaw Stock Exchange in order to help in the implementation process of GPW WATS trading platform

2.1. TARGET AUDIENCE

This document has been prepared to development staff, Independent Software Vendors who produce software integrated with GPW WATS, analysts, Market participants and all clients who want to deepen their knowledge about GPW WATS.

2.2. DOCUMENT'S PURPOSE

The purpose of this document is to provide a full description of Market Data Gateway, which is part of GPW WATS.

This document contains a description of trading port behavior, message types and their fields.

2.3. ASSOCIATED DOCUMENTS

GPW WATS 3.01 Market Data Protocol is a part of GPW WATS documentation set.

Please check the following documents to learn about the construction of Trading System.

- GPW WATS 1.01 Trading System.

Please check the documentation of the trading protocols supported by GPW WATS.

- GPW WATS 2.01 Native Order Gateway Specification (this document),
- GPW WATS 2.02 FIX Order Gateway Specification.

Please check the description of the communication with Data Distribution Service.

- **GPW WATS 3.01 Market Data Protocol** (this document).

Please check the description of the communication with Internet Data Distribution System.

- GPW WATS 3.02 Internet Data Distribution System,
- GPW WATS 3.03 Streaming Messages for IDDS,
- GPW WATS 3.04 Rest API Messages for IDDS.

Please check the additional documentation, which explains other services provided within GPW WATS.

- GPW WATS 4.01 Drop Copy Gateway,
- GPW WATS 4.02 Post Trade Gateway,
- GPW WATS 5.01 Risk Management Gateway.

Please check the additional documentation describing the following:

- GPW WATS 2.03 Rejection Codes,
- GPW WATS 2.04 BenDec Message Definition Format,
- GPW WATS 4.03 Contract Notes,

- GPW WATS 6.01 Connectivity,
- GPW WATS 6.02 (ENG) Short Code Record Keeping,
- GPW WATS 6.02 (PL) Mapowanie Short Code,
- GPW WATS 6.03 Short-Long Mapper User Guide.

It is recommended to read ***GPW WATS 1.01 Trading System*** document first.

3. Document History

Version	Date	Description
0.51	29.06.2023	The initial publication of the documentation.
0.52	26.07.2023	Messages updated in protocol v0.52: • EndOfSnapshot ○ New field lastBBOSeqNum • Instrument ○ New field referencePrice • ReplayRequest
0.53	16.08.2023	Messages updated in protocol v0.53: • EndOfSnapshot ○ Field lastBBOSeqNum removed
0.54	19.09.2023	Messages updated in protocol v0.54: • TradingPhaseScheduleEntry • Deleted attributes: ○ referencePriceLogicId
0.55	17.10.2023	Section 5.4.2: Updated sequence of messages in the Snapshot service. Messages updated in protocol v0.55 Instrument • status – new dictionary values: ○ Active ○ Inactive ○ MarketOperationsSuspension ○ OutsideCollarsStatic ○ OutsideCollarsDynamic ○ RegulatorySuspension ○ TechnicalHalt • issuerRegCountry added values: ○ PRT ○ SVK InstrumentStatusChange • status – added values: ○ OutsideCollarsStatic ○ OutsideCollarsDynamic MarketStructure • Deleted attributes ○ code ○ structureType • New attributes: • MarketModelType, with values ○ CLOB ○ BLOCK ○ HYBRID ○ CROSS ○ NotApplicable SessionSummary • status – added values: ○ OutsideCollarsStatic ○ OutsideCollarsDynamic
0.56	17.11.2023	Document Changes

Version	Date	Description
		• OrderBookEvent supersedes the functionality of the ClearBook message. This change specifically affects the order of messages after auction uncrossing. Message examples were updated throughout the document to reflect the latest version of the Market Data protocol. The affected messages are: • Trading: o OrderAdd o OrderExecute o OrderModify o OrderDelete • Status: o InstrumentStatusChange o TradeCollars o OrderCollars • Reference data: o MarketStructure o Instrument o TickTableEntry o TradingPhaseScheduleEntry o CalendarException o WeekPlan o CollarTableEntry o CollarGroup • Session management: o StartOfTechnicalSession o ReferenceDataStart o ReferenceDataEnd Deleted messages: • ClearBook • TestMax Added messages: • OrderBookEvent - Order book event for given financial instrument • TestEvent – a test event TestEvent • scenarioName – name of test scenario • eventType – type of test event: o Flush - Flush the test scenario o ScenarioStart - Start of test scenario o ScenarioEnd – End of test scenario OrderBookEvent • instrumentID • eventType: - type of book event o Clear – clear order book o ResendStart - Order book resend start (resend will begin shortly) o ResendEnd - Order book resend end (resend is done) Messages updated in protocol v0.56 Instrument • New attribute: • initialPhaseId - Id of starting phase for the instrument OrderDelete • New attribute:

Version	Date	Description
		• instrumentID – ID of financial instrument OrderModify • New attribute: • instrumentID – ID of financial instrument
0.57	30.11.2023	Document Changes • Section 6.2 was updated to clarify the usage of the orderId and publicOrderId identifiers in Market Data. New messages: OpenPositions: The number of open positions in a financial instrument. • Description of the message and its purpose added to Section 6.4 Messages updated in protocol v0.57 Trade • New attribute: o settlementValue o settlementDate TradingPhaseScheduleEntry • Updated attribute: • tradingPhaseType – added values: o Hybrid – Hybrid normal trading o HybridPause – Hybrid pause • New attribute: • hybridPauseId - ID of Hybrid pause phase.
0.58	18.12.2023	Document Changes • Section 5.3 updated to reflect changes to structure of BBO data, now includes OpenPositions message. • Section 6.3.1 PriceLevelSnapshot, provides a more precise description of the behavior of the message in the BBO during auctions. • Section 6.3.2 describes the sequence of messages during an auction in the BBO stream. Messages updated in protocol v0.58 AuctionSummary • Updated attribute: • auctionType – added values: o UnsuspensionAuction Instrument • Updated attribute: • status – added values: o HybridNoQuotes - Hybrid no valid quotes. o HybridPause - Hybrid pause. InstrumentStatusChange • Updated attribute: • status – added values: o HybridNoQuotes - Hybrid no valid quotes. o HybridPause - Hybrid pause. InstrumentSummary • Updated attribute: • adjustedClosingPriceReason – removed values: o TickTableChange o MultipleCA SessionSummary

Version	Date	Description
		• Updated attribute: • status – added values: ○ HybridNoQuotes - Hybrid no valid quotes. ○ HybridPause - Hybrid pause. TradingPhaseScheduleEntry • Updated attribute: • tradingPhaseType –updated values: ○ HybridPause updated to HybridPreTrade ○ added UnsuspensionAuction - Auction after suspension. • auctionType – added values: ○ UnsuspensionAuction - Auction after suspension. • Deleted attribute: ○ hybridPauseId
0.59	20.01.2024	Messages updated in protocol v0.59 Instrument • nominalValue, type changed to Price from Number • multiplier, type changed to Multiplier from Number
0.60	20.02.2024	Document Changes • Test, TestEvent and Text messages are deprecated • News takes the place of Text message • New description for TopPriceLevelUpdate to emphasize difference from PriceLevelSnapshot • OrderBookEvent supersedes the functionality of the ClearBook message. This change specifically affects the order of messages after auction uncrossing. Message examples were updated throughout the document to reflect the latest version of the Market Data protocol. The affected messages are: • Trading: ○ OrderAdd ○ OrderExecute ○ OrderModify ○ OrderDelete • Auction: ○ OrderBookEvent ○ AuctionUpdate ○ AuctionSummary • Status: ○ InstrumentStatusChange ○ TradeCollars ○ OrderCollars ○ Trade • Reference data: ○ MarketStructure ○ Instrument ○ TickTableEntry ○ TradingPhaseScheduleEntry ○ CalendarException ○ WeekPlan ○ CollarTableEntry ○ CollarGroup ○ AccruedInterestTableEntry

Version	Date	Description
		• Session management: ○ StartOfTechnicalSession ○ ReferenceDataStart ○ ReferenceDataEnd Messages updated in protocol v0.60 Deleted Messages: • Test • TestEvent • Text CalendarException Removed attribute: • calendarExceptionRecurrent Updated attribute: • calendarExceptionType, now contains values: ○ Closed The day is closed for trading. ○ Open The day is open for trading. IndexParams New attribute: • dateValidity Date of validity of index. Instrument Updated attribute: • MarketModelType: reordered dictionary of values, new value IPO MarketStructure Updated attribute: • MarketModelType: Market model type, reordered dictionary with added value: ○ IPO: Initial public offering TradingPhaseScheduleEntry Removed attribute: • tradingPhaseStopTime • auctionUncrossing Updated attribute: • tradingPhaseDetailedType: reordered dictionary with new values, ○ HybridBuyOnly: Hybrid buy only trading ○ Ipo: Initial public offering phase New attribute: • uncrossing: True if phase includes an uncrossing. • lastAuctionPhaseId: ID of last auction phase.
1.0	30.04.2024	Document Changes • Section 6.4 Hybrid Market Model • Section 6.5 IPO/SPO Market Model • Section 6.6 Tender Offer Market Model • Section 7.1.4 changes to Instrument message examples • Chapter 9 Data Redistribution • Chapter 10 Sections 10.1, 10.2TradingSessionStatus and MD Header • PositionReport replaces the OpenPositions message • ProductSummary replaces SessionSummary • Test, TestEvent and Text messages are deprecated Deleted messages: • StartOfTechnicalSession • EndOfTechnicalSession • ReferenceDataStart

Version	Date	Description
		• ReferenceDataEnd • SessionSummary • OpenPositions New messages: • PositionReport replaces the OpenPositions message • ProductSummary replaces SessionSummary • TradingSessionStatus - Information on the status of a Market and on the trading day's events Datas Atomic Datas • Price is now aliased to i64 • Value is now aliased to i64 • Bound is now aliased to i64 • CollarValue is now aliased to i64 • IndexValue is now aliased to i64 • PercentageChange is now aliased to i64 • Multiplier is now aliased to i64 • TickSize is now aliased to i64 Enums Country - Identifier of the country of the registration. Affected messages: Instrument Modification: now features complete country list. ProductType - Type of the product. Affected messages: Instrument Modification: Dictionary completely renamed • FinancialProductShare is now Equity • FinancialProductBond is now FixedIncome • FinancialProductDerivativeFutures is now DerivativeFutures • FinancialProductDerivativeOptions is now DerivativeOptions • FinancialProductIndex is now Index • FinancialProductCurrency is now Currency ProductSubtype - Subtype of the product. Completely new dictionary Affected messages: Instrument • Share Share • AllotmentRight Allotment Right • SubcriptionRight Subcription Right • TenderOffer Tender Offer • IssueRight Issue Right • SubscriptionWarrant Subscription Warrant • BankSecurities Bank Securities • Bond Bond • TreasuryBond Treasury Bond • IssuedBond Bond issued by a government institution • MunicipalBond Municipal Bond • CorporateBankBond Corporate Bond - bank • CorporateFirmBond Corporate Bond - firm • ConvertibleBond Convertible Bond • MortgageBond Mortgage Bond • MortgageBackedBond Mortgage Backed Bond

Version	Date	Description
		• PublicMortgageBond Public Mortgage Bond • Bill Bill • TreasuryBill Treasury Bill • CommercialBill Commercial Bill • ExchangeTradedFund ETF - Exchange Traded Fund • ExchangeTradedNote ETN - Exchange Traded Note • ExchangeTradedCommodity ETC - Exchange Traded Commodity • InvestmentCertificate Investment Certificate • IndexFuturesIndex Futures • StockFuturesStock Futures • CurrencyFutures Currency Futures • BondFutures Bond Futures • InterestRateFutures Interest Rate Futures • IndexOptionsIndex Options • StockOptions Stock Options • OptionWarrants Option warrants • FactorCertificates Factor certificates • TurboCertificates Turbo certificates • CapitalProtectionCertificatesCapital protection certificates • BonusCertificates Bonus certificates • ReverseConvertibleCertificates Reverse convertible certificates • ExpressCertificates Express certificates • DiscountCertificates Discount certificates • IndexTrackerCertificates Index/tracker certificates • OtherInvestmentCertificates Other investment certificates (non leveraged) • StructuredNotes Structured notes • ReferenceRate Reference rate • InterestRate Interest rate • ExchangeRate Exchange rate • PriceIndex Price index • TotalReturnIndex Total return index • SectorPriceIndexSector price index • SectorPriceTotalReturnIndex Sector price total return index • DividendIndex Dividend index • StrategyPriceIndex Strategy price index • StrategyTotalReturnIndex Strategy total return index • CommodityIndex Commodity index • BondTotalReturnIndex Bond total return index MarketModelType - Market Model Affected messages: Instrument, MarketStructure, ProductSummary Modification: new enum value • TenderOffer: Tender offer • Redistribution: Redistribution TradingPhaseType - Type of matching algorithm Affected messages: TradingPhaseScheduleEntry Modification: new enum value • TenderOffer: Tender offer phase ClosingPriceType

Version	Date	Description
		Affected messages: InstrumentSummary Modification: new enum • InitialPrice: Initial price of the instrument • LTP: Last traded price • LastACP: Last adjusted closing price • FairValue: Fair value • DailySettlementPrice: Daily Settlement Price • FinalSettlementPrice: Final Settlement Price AdjustedClosingPriceReason Affected messages: Instrument, MarketStructure Modification: new enum • NotApplicable: Not applicable. • Regular: Regular • Dividend: Dividend • IssueRight: Issue Right • Split: Split • ReverseSplit: Reverse Split • Bonus: Bonus • SpinOff: Spin-Off New Messages PositionReport. • publicProductIdentification.productIdentification Product identification, for example its ISIN code. • publicProductIdentification.productIdentificationType Type of product identification. • posType Type of quantity returned. • openPositions Number of open positions. • tradeID Trade ID. Message changes Header Added 2 variables to the MD header. Header size is now 42 bytes instead of 39 bytes previously: • sessionId: ID of the session. • streamId: ID of the md stream. Instrument New attributes • icebergMinValue: Minimum iceberg order value. Modified attributes • Code is now description: Financial instrument code. Trade New attributes: • pctChange Percentage change. • VWAP Volume-weighted average price. • noTrades Total number of transations. • totalVolume Total transaction volume. • totalValue Total transaction value. • openingPriceThe price of the first trade on the current trading day. • maxPriceHighest price of the instrument on the current trading day. • minPrice Lowest price of the instrument on the current trading day.
1.1	28.06.2024	Document Changes:

Version	Date	Description
		• Section. 7.1.4.2 Instrument definitions: in the table for ProductSubtype, the value SectorPriceTotalReturnIndex corrected to SectorTotalReturnIndex • Section 8.6.1 IndexParams: the description makes it clear that an index can be parametrized with up to 3 ISINs. Enums: • ProductSubtype - Subtype of the product. SectorPriceTotalReturnIndex corrected to SectorTotalReturnIndex - Sector total return index • Affected messages: Instrument Messages: IndexParams: • dateValidity: the description of the attribute changed to “Date of validity of index portfolio” from “Date of validity of index” The document now features uniform examples for equity instrument PKN Orlen with ISIN PLPKN0000018 traded on the CLOB segment. Section 4.1 Information on the distrittribution of InstrumentSummary and ProductSummary messages on the Full Depth and BBO streams added to relevant tables. Section 5.3 Table updated to reflect the publication of InstrumentSummary and ProductSummary messages on the Full Depth and BBO streams. Section 10.4. Snapshot Behaviour Specifics describes details of sequence number behaviour during snapshots. Protocol changes: Messages: • TradingPhaseScheduleEntry • The auctionType attribute is removed Enums: • PriceExpressionType: Price expression type for the financial instrument. • New value in enum: Not applicable. Affected messages: Instrument, Trade
1.1.2	9.08.2024	**Message changes:** • IndexParams length - change from 182 to 186 • Instrument length - change from 588 to 729 **IndexParams** message: indexUnderlyings.count field added. indexUnderlying: new Type (IndexUnderlyingArray), kind (Array (IndexUnderlying)) and values: • productIdentification • productIdentificationType • indexUnderlyingType **Instrument** message: Value added in status field: HybridKnockout. Fields added: • accruedInterestValue • kid • kidIssueDate • valueAtRisk **InstrumentStatusChange** message: Value added in status field: HybridKnockout. **PriceUpdate** message: Values change in priceUpdateType field: • ReferenceIndexPrice • ReferenceFixingIndexPrice • MidPrice

Version	Date	Description
		• FixingPrice • FixingYtm Values change in priceYtmPresence: • None • value • valueBid • valueAsk • YtmNoCalc • YtmBidNoCalc • YtmAskNoCalc **ProductSummary** message: Value added in status field: HybridKnockout.
1.3	16.10.2024	Section 5.4.1 Snapshot with ReferenceData only no longer available Section 9.5 ReplayRequest – how the end of the request is marked Section 10.3 Encryption – new definition of the nonce: • Nonce - The value for the nonce is all zeros (a vector of 12 bytes/96 bits filled with zeros). Section 10.3 Numbers and Decimal Places – new section Section 10.4 Endianness – GPW WATS messages are little-endianChapter 11 Market Data protocol tables – the message tables now feature information on which message attribute is encrypted in Market Data. The messages featuring encrypted attributes are: • OrderAdd • OrderExecute • OrderModify • PriceLevelSnapshot • ProductSummary • TopPriceLevelUpdate **Message changes** Type changes **InstrumentStatus** – new dictionary **PriceUpdateType** New Messages ExternalUnderlying - The list of underlyings of derivatives and structured products not traded in GPW WATS. Modified Messages **IndexParams** New attributes • indexUnderlyings.count How many elements does IndexUnderlyings contain. • indexUnderlying renamed to indexUnderlyings.items with new dictionary **Instrument** New attributes • referenceInstrumentId ElementId Identifier of reference instrument.externalUnderlyingId ElementId External underlying ID.
1.4	30.11.2024	Unpublished version. All changes in this version have been documented in v1.5.
1.5	31.01.2025	Section 10.3 Numbers and Decimal Places: Clarification of the decimal point formatting of certain GPW WATS values. Section 10.6 Snapshot Behavior Specific: All messages on Snapshot service come out with seqNum=0 in the message header. Total size = number of Lots. Price = price per unit (smallest portion) of instrument.

Version	Date	Description
		IndexPortfolioEntry message • suspended field added **ProductSummary** message • quotationSystem field: • NA - 1 - new value added • market field: o Type changed to u8 o Kind changed to Primitive 6.4.1. **AccruedInterestTableEntry** – clarification that 31 messages are published each day during parametrization Section 7.1.7.1. **AuctionUpdate** – added details of behaviour when a single Market Order present on one side of the book during auctions Chapter 8 **Data Redistribution** – general and specific updates Section 8.2 **Status** messages – corrected the status field of the InstrumentStatusChange message Section 8.4 **InstrumentSummary** – corrected the adjustedClosingPriceReason Chapter 9 **Next Session Reference Data** details the mechanism for publishing reference data for the next trading session Section 10.4 **EndofSnapshot** - Description of EndofSnapshot functionality to rebuild the state of the market Section 10.5 **ReplayRequest** - Detailed description of the ReplayRequest message and header **Dictionary changes** • tradingPhaseType – introduces HybridPreTradeBuyOnly ;affected messages InstrumentStatusChange and TradingPhaseScheduleEntry **Message changes** IndexPortfolioEntry New attributes • suspended Is instrument suspended. ProductSummary • market is now u8 instead of enum Defines the market the instrument belongs to. • quotationSystem – new enum TradingSessionStatus – the message now enables the publication of reference data for the next trading session • tradingSessionEvent - new dictionary
1.5.4	30.04.2025	Section 7.3 BBO messages: TopPriceLevelUpdate is deprecated and not used in the System. Section 11.4 Currency: 'XXX' code when currency not defined as per ISO 4217 Section 11.5 Heartbeat: details of the Heartbeat message
1.6	26.05.2025	Section 7.1.4 and 7.1.5 Flow of messages in Auctions: order resend is carried out only when the subsequence phase is Continuous. In case the subsequent is not continuous, resend does not take place. Section 10.2.1 TradingSessionStatus – update on the new MIC feature Section 10.5 Replay Request • Description updated to clarify the marking of the end of the ReplayRequest request: the end of the response to ReplayRequest is marked by 2 null bytes. • Correction: Header length is 4 bytes. Section 11.4 StreamIDs The role of StreamIDs in GPW WATS. **Types** • ProductType – new enum, now includes the Tracker value o Affects: Instrument message

Version	Date	Description
		• ProductSubtype– new enum, now includes the AgencyBond value, which replaces the IssuedBond value; new values include SupranationalBond, WarrantWithKnockOut ○ Affects: Instrument message • InstrumentStatus – new enum, includes the Inactive status - financial instrument is inactive and is not trading. ○ Affects: Instrument message, InstrumentStatusChange, ProductSummary • AdjustedClosingPriceReason - new enum, includes new values TickSizeChange, OrderPurge, OtherReason ○ Affects: InstrumentSummary, ProductSummary • TradingSessionEvent – new enum, added new enum values – StartOfTradingSessionMIC, EndOfTradingSessionMIC ○ Affects: TradingSessionStatus **Messages** • News – includes new NewsId attribute - the news id serves as a unique reference for tracking purposes within GPW WATS. ○ Affects: News message • TradingSessionStatus – tradingSessionState attribute deleted. ○ Affects: TradingSessionStatus
1.6.5	18.06.2025	Section 7.1.3 OrderBookEvent – OrderBookEvent is not disseminated over Snapshor Section 10.2 note that tags PreviousDayRestateStart/PreviousDayRestateEnd are not used in the MD TradingSessionStatus message Section 11.4 StreamID - Improved description of streams and their uses. Section 11.6 OrderBook data - A new section that explains what happens to the order book after the trading in an instrument stops at the end of the trading session.
1.6.6	10.07.2025	New section: 11.12 General Issues – discusses various issues related to the Market Data protocol.
1.6.7	07.08.2025	Section 4.4 A sample GPW WATS trading day – new table presenting the proposed trading schedule on GPW WATS. Sections 9.2 and 10.2 the length of the TradingSessionStatus message corrected in message examples.
1.6.8	14.08.2025	Section 10.5.3 Replay timeout – the replay message must be sent within 100ms otherwise the Replay service drops the connection Chapter 12 – Sending Rules for the WATS Market Data protocol
1.6.12	12.09.2025	Section 5.3 PriceLevelSnapshot is distributed on FullDepth but only for redistributed messages Section 7.2.6.3.1 – A new section that explains TVTIC Section 8.6 – New descriptions for messages related to data redistribution from BondSpot
1.6.15	29.09.2025	New Chapter 11 Aggressive Order Kinematics – the behavior of Market Data with regards to aggressive orders including Icebergs and sequence diagrams of MD messages Section 12.3 TVTIC - Updated description of TVTIC. The section moved from Section 7.2.6.3.1 to a new location in Chapter 12
1.6.16	24.10.2025	Section 7.1.4 Sequence Of Messages In Auctions - The sequence of messages post-auction on Full Depth is changed: InstrumentStatusChange now comes out at the end, after OrderBookEvent messages. Section 7.3.2 Sequence Of Messages On BBO During An Auction - The sequence of messages post-auction on BBO is changed: now PriceLevelSnapshot comes before InstrumentStatusChange (previously, the other way around)

Version	Date	Description
		Important! **The OrderDelete message has been renamed to OrderCancel** Section 4.3.1 General Information Flow – Typo corrected in the Snapshot sequence diagram Section 5.4.2 Sequence Snapshot • Correction: snapshot on BBO includes the Trade message • The section clarifies the order of messages in Snapshot Section 7.1.4 Sequence Of Messages In Auctions – The uncrossing phase is added to the sequence of events during auctions Section 7.1.7.1 AuctionUpdate – example for AuctionUpdate now includes the hexadecimal values for the case when only a market order resides in the book during an auction Section 10.4 Snapshot Operation – The operation of the Snapshot service. Section 12.4 Numbers and Decimal Places – Update to the variable precision table – now contains related types to main attributes Section 7.3.1 PriceLevelSnapshot And TopPriceLevel - Update to BBO levels – WATS now displays up to 5 BBO levels The following tables have been updated: • Section 8.1 Reference Data for Markets and Instruments • Section 8.2 Status messages: Instrument Status Change • Section 8.6.1 Index Params • Section 8.6.2 Index Portfolio Entry • Section 8.6.3 Real Time Index • Section 8.6.4 Index Summary
1.7.1	18.11.2025	**Type changes** **Array** type: now supports 20 characters **Code** type: change in the field's length from 16 to 20. Affected messages: Instrument, ExternalUnderlying in the attribute productCode **auctionType** and **tradingPhaseType** : Unification of auctionType and tradingPhaseType names to be compliant with tradingPhaseType names. • OpeningAuction -> AuctionOpening • ClosingAuction -> AuctionClosing • IntradayAuction -> AuctionIntraday • VolatilityAuctionStatic -> AuctionVolatilityStatic • VolatilityAuctionDynamic -> AuctionVolatilityDynamic • ExtendedVolatilityAuctionStatic -> AuctionExtendedVolatilityStatic • ExtendedVolatilityAuctionDynamic -> AuctionExtendedVolatilityDynamic **ProductType** - addition of Tracker **ProductSubtype** • TenderOffer – deprecated **TradingSessionState** • Deprecated. **Message changes** Length of the messages updated. **Instrument** • Attribute productSubtype = TenderOffer (4) not used in WATS **AuctionSummary, InstrumentStatusChange and TradingPhaseScheduleEntry**

Version	Date	Description
		• Description changes of enums in messages **Login** • The message contains the filter attribute enabling the selection of data to be included in the Snapshot. **OrderCancel** • New attribute – order side **OrderExecute** • executionId renamed to traded • New attribute – order side **OrderModify** • New attribute – order side **TradingSessionStatus** • tradingSessionState attribute deleted • nextSessionDate Date - Date of next trading session. **TradingPhaseScheduleEntry** • lastAuctionPhaseId renamed to clatId **News** • New attribute newsId - The news id serves as a unique reference for tracking purposes within the WATS system. **InstrumentStatusChange** • 2 new values in the status enum: DistributionStart and DistributionEnd **Redistribution messages** **IndexParams** • presenceFlags bitmask for specific variables to distinguish whether they carry a meaningful value or remain empty **RealtimeIndex** • presenceFlags bitmask for specific variables to distinguish whether they carry a meaningful value or remain empty **IndexSummary** • presenceFlags bitmask for specific variables to distinguish whether they carry a meaningful value or remain empty **IndexPortfolioEntry** • indexId InstrumentID of the instrument (index Id). • instrumentId InstrumentID of the financial instrument (index component).
1.7.1.1	12.12.2025	**Section Changes:** Section 5.4.2. Snapshot Sequence – updated sequence of messages in Snapshot Section 6.2.1. TradingPhaseScheduleEntry – the tradingPhaseStartTime is set to 0 in case of unscheduled trading phases Section 6.4.2 IndexationEableEntry – corrected the wording for the information contained in the message - the message contains a bond instrument's indexation coefficient. The original text incorrectly said "indexed nominal value". Section 8.2. Status Messages: InstrumentStatusChange – DistributionStart and DistributionEnd removed due to changes in the InstrumentStatus enum Section 9.2 The Mechanism and Section 13 Sending Rules – the ReferenceSpreadTableEntry message removed from the documentation. Section 10.2.1. StartofTechnicalSession/EndofTechnicalSession - TradingSessionStatus::date=0 in case when TradingSessionStatus::tradingSessionEvent = StartOfTechnicalSession or EndOfTechnicalSession. **Type changes** • InstrumentStatus – DistributionStart and DistributionEnd choices are removed. Affected messages: Instrument, InstrumentStatusChange and ProductSummary

Version	Date	Description
		• tradingPhaseType – new value added to the enum: Distribution - Instrument distribution phase. Affected messages: InstrumentStatusChange , TradingPhaseScheduleEntry
1.7.2-1.7.4		No changes in the document
1.7.5	06.03.2026	Some attributes corrected to camelCase: • AccumulatedInterest renamed to accumulatedInterest (ProductSummary) • tradeID renamed to tradeId (PositionReport). 6.1.4 Example – productType – description has been updated 6.1.4.1. Instrument status = Inactive – additional explanation of the Inactive status for an instrument. 7.1.7. Exceptions In the Auction Process –. the new chapter has been added. A description of the process followed when a volatility auction is followed by an extended volatility auction. 7.7.4.1. Trade Cancellations –the new chapter has been added The details of the Trade message when a trade cancellation occurs. 8.2 Status messages: InstumentStatusChange section - descriptions for phases files have been updated. The new phase type (Distribution) has been added. 8.5. The Distribution Phase – the new section describes the details and use of the Distribution phase for redistributed instruments 10.2.2. InitialReferenceDataStart / InitialReferenceDataEnd – the new section explains the details of the date field in the TradingSessionStatus message. It whould be read together with section 10.2.1. StartOfTechnicalSession / EndOfTechnicalSession 10.5. Replay Operation and ReplayRequest – the start seqnum of the replay sequence should be set to 1 and not to 0 as previously stated - if seqNum==1, the replay starts from the first message, 12.5 Encription chapter has been created as a compilation of previous sections 12.9.(Endianness) and 12.5 (StreamId). The chapter has been enriched with additional information 12.6. Stream management – new chapter combining content related to streams and their management 12.14. (MMT) Market Model Typology Tags Not Used in WATS – the list of MMT attributes that not used in the Trade message on Market Data. 14.37 Trade message – the description of values that are not used has been clarified
1.7.6-1.7.9		No changes in the document
1.7.10	03.06.2026	7.8.2. ProductSummary – better description of the encryption features of the message and the use of the market attribute. 7.1.4. Sequence Of Messages In Auctions –table updated to make clear that OrderExecute messages are not generated for transactions which happen during auctions. 7.2.6.3. OrderExecute and Trade – Clarified that OrderExecutes are published for passive orders only and not for aggressive orders. 11.2.2. The Sequence of Messages in an Aggressor Scenario – no OrderExecutes are published for the hidden part of an Iceberg aggressed by an incoming order. 11.6.3. Aggressive Order Removes Peak and Part of Hidden Quantity – Table corrected. OrderExecute does not come out for the hidden part of the Iceberg. 11.6.4. Aggressive Order Removes Peak and Hidden Quantity - Table corrected. OrderExecute does not come out for the hidden part of the Iceberg.
1.7.11-1.7.12		No changes in the document
1.7.13	15.07.2026	**Updated:** 5.4.2 Shapshot Sequence - for message Type: Continuous Trading/Auctions/BBO the orderAdd message has been clarified 7.1.5. After an Auction - additional information on the sequence of orders during the resend process 7.7.3.2 PriceUpdate section has been updated

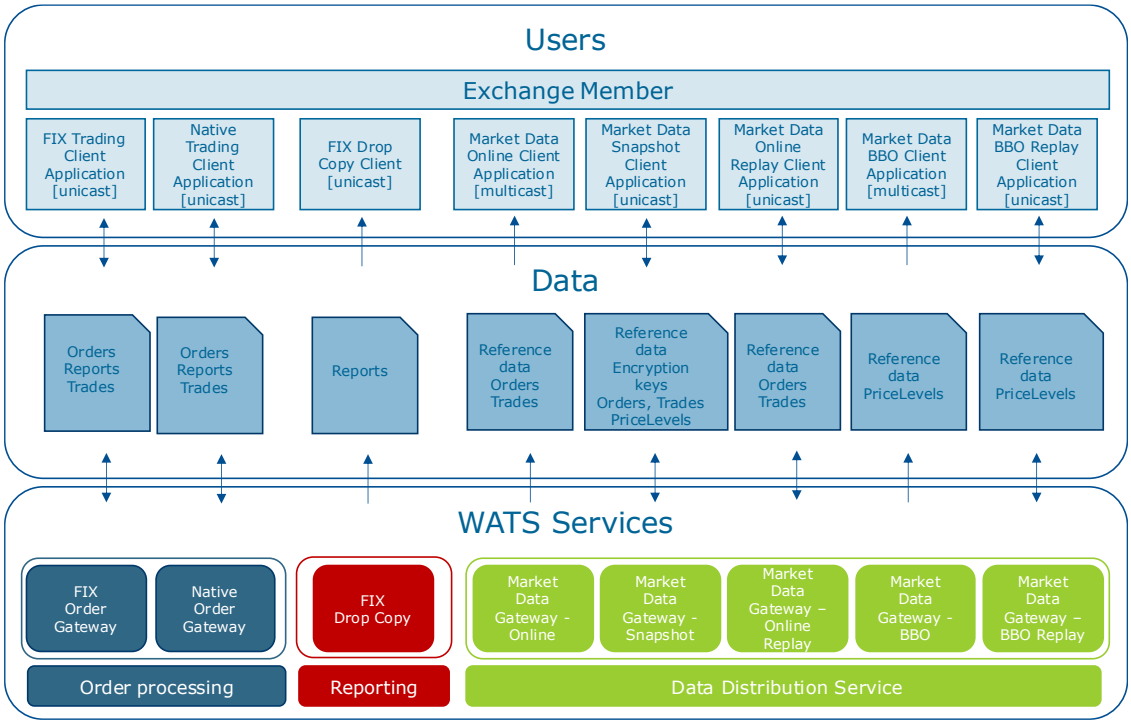
Version	Date	Description
		8. Data Redistribution - in PriceUpdate message for Treasury BondSpot Poland the description has been clarified 8.3 Trading Data section has been changed 8.6 Treasury BondSpot Poland (TBSP) Data Redistribution has been clarified 11.3.1 Inbound aggressive order fully filled against single order - ExecutionId has been change to TradeID 11.3.2 Inbound aggressive order fully filled against multiple orders - ExecutionId has been change to TradeID 11.4.1 Inbound aggressive order partially filled against one order - ExecutionId has been change to TradeID 11.4.2 Inbound aggressive order partially filled against multiple orders - ExecutionId has been change to TradeID 11.4.3 Aggressive order triggers volatility auctions - ExecutionId has been change to TradeID 11.5.1 Aggressive modification of resting order resulting in a full fill - ExecutionId has been change to TradeID 11.5.2 Aggressive modification of resting order resulting in a partial fill - ExecutionId has been change to TradeID 11.6.1 Aggresive order hits peak - ExecutionId has been change to TradeID 11.6.2 Aggresive order removes peak - ExecutionId has been change to TradeID 11.6.3. Aggressive Order Removes Peak and Part of Hidden Quantity – updated kinematics of iceberg orders 11.6.4. Aggressive Order Removes Peak and Hidden Quantity – updated kinematics of iceberg orders 13. Sending Rules - for PriceUpdate message the sending rule has been updated **New:** 10.4.1. Logging into Snapshot – the section describes how to configure which kinds of messages will be published by the Snapshot service. 12.4. PriceUpdate Use Cases – details of the use cases for the PriceUpdate message 12.15 Presence Maps in WATS – Market Data messages are of fixed length, and a presence map mechanism is not implemented.
1.7.17	27.08.2026	**All updates are editorial in nature and do not alter existing system logic.** The following chapters have been corrected: • 5.4.2. Snapshot sequence - description of order priority has been added • 7.1.4. Sequence of messages in auctions – the sequence of messages has been updated • 7.1.6. Auction time suppression - the sequence of messages has been updated • 7.2.5. Orderexecute - clarification of orderExecute for the aggressive side of the transaction • 10.4.4. Snapshot header - descrption of Snapshot Header has been updated • 10.5.1.The replayRequest message header - section has been updated • 13. Sending rules - sending rules for login message type has been updated Minor editorial changes: • 6.1.4.2. Instrument Definitions • 6.3.3.2. Collar Tables • 7.1.7. Exceptions in the Auction Process • 10.2.3. Startoftradingsessionmic / Endoftradingsessionmic • 12.7.2. Streamid and individual streams • 12.13.1. Maximum message size • Update examples (6.1.4.6; 6.2.4.1; 6.3.3.1; 7.1.8.1; 7.1.8.2; 7.2.6.2; 7.2.6.3; 7.2.6.4; 7.3.4.1; 7.7.3.1; 7.7.3.2; 7.7.3.3; 9.2; 10.2)

Version	Date	Description
1.7.6.20	17.09.2026	**All updates are editorial in nature and do not alter existing system logic.** The following chapters have been added/corrected: • Section 7.1.5 After an Auction - features a detailed description of rules for the rebuilding of order books after auctions. • Section 7.1.5.1 ContinuousPriceTime has been added • Section 7.1.5.2. ContinuousRefPriceTime has been added • Section 7.1.5.3. ContinuousLastAuctionTime has been added

4. GPW WATS Market Data

GPW WATS Market Data is a Native protocol application serviced by the Data Distribution Service (DDS) and provides a range of Market data services to users. The DDS is a complex framework designed to provide maximum flexibility in data management to the Market and its users. The scope of the Data Distribution System is highlighted in green in the following diagram.

Figure 1 - GPW WATS Services



GPW WATS Market Data provides users with full-depth and best bid-offer datasets. The data is provided in real time broadcasts through three separate channels, Snapshot, Replay and Stream:

- Snapshot provides reference data and encryption keys for decrypting information sent through Stream and Replay. It can be accessed at any time during the trading day to obtain the full-depth order book and a complete record of execution reports and enriched trades.
- Stream is the principal Market data channel for disseminating real time order and transaction data over multicast. It publishes among others the full depth order book, BBO levels, as well as auction related statistics.
- Replay is a mechanism for recovering information lost by Stream during multicast transmission.

DDS Market Data can be configured in various ways depending on the needs of users and Markets. Different DDS connections can be configured to carry separate streams of information. For example, the Full Depth broadcast can define several connections each conveying full depth data related to specific Markets. Similarly, for BBO data feeds.

4.1. DATASETS

GPW WATS offers two views of the Market: Full Depth and Best Bid-Offer (BBO). The Full Depth dataset provides full information on the current state of the order book including Market depth, while the Best Bid-Offer provides up to 5 levels of the Order Book.

The table below lists the scope of the data distributed in each dataset.

	Real-time data	
Category	**Full Depth**	**Best Bid Offer**
Reference Data messages	Reference data: • Market structure • instruments • Market Phases • tick table entry • calendar exceptions • collar table entry • collar logic • reference prices • trading week plan	Reference data: • Market structure • instruments • Market Phases • tick table entry • calendar exceptions • collar table entry • collar logic • reference prices • trading week plan
Trading messages	Full order book state: • order additions/ modifications/ cancellations • trades • order and trade collar values	BBO information: • 5 BBO price levels • Trades • order and trade collar values
Auction messages	Auctions: • signal to clear book for an upcoming auction • auction type • auction price/quantity • indicative matching price / indicative matching volume • buy/sell quantity • BBO price/quantity if uncrossed book	Auctions: • auction type • auction price/quantity • indicative matching price / indicative matching volume • buy/sell quantity • BBO price/quantity if uncrossed book
Summaries	Instrument summary and Product summary messages	Instrument summary and Product summary messages
Status messages	Changes in Market Phases	Changes in Market Phases

4.1.1. FULL DEPTH DATA

During a continuous Market Phase (Continuous Price and Time) and the trade at last (Continuous Trading at Closing Price), Full Depth Market Data broadcasts all order messages received by the Market and can be used by a participant to reconstruct the state of the Order Book at any point in time.

During Auctions, Full Depth only disseminates auction related statistics such as the Indicative Matching Price and Indicative Matching Volume, or the best buy and sell levels. At the end of each auction (the uncrossing), Full Depth sends auction summary information. During auctions, unlike in continuous time trading, no full depth order book data is published.

In continuous time trading and continuous trading at fixed price (trade-at-last) Full Depth disseminates:

- OrderAdd, OrderModify, OrderCancel messages disseminated by the exchange,
- OrderExecute messages signaling that an order was filled,
- Trade messages which contain enriched trade information.

During Auctions Full Depth only conveys Auction related information:

- During auctions, the AuctionUpdate message continuously updates Market participants on the state of the auction by publishing IMP/IMV or the best buy and sell levels,
- At Uncrossing the AuctionSummary message provides a full auction summary including price and volume.

4.1.2. BEST BID OFFER

In addition to full depth data, GPW WATS provides BBO data to users. By default, up to 5 levels are provided simultaneously. The key features of the BBO dataset are:

- 5 BBO levels for each instrument traded on the platform. Users can subscribe for up to 5 BBO levels of data if they do not need the Full Depth dataset.

4.2. Service Levels

Market participants have the option to subscribe to either the Full Depth, the BBO dataset or both. Access rights will be managed by means of encryption. Through the Snapshot service, users will receive encryption keys enabling them to decrypt the relevant Market Data content which corresponds to Full Depth and BBO data. In addition, the content of the Snapshot will reflect each user's choices and will be customized to relay BBO and Full Depth content.

An example of service-level configuration is the option for a user to receive Trade and OrderCollars/TradeCollars messages in their BBO Stream. By default, these messages are not included in the BBO Stream, but can be activated with the appropriate subscription.

4.3. Channels: Snapshot, Replay and Stream.

4.3.1. GENERAL INFORMATION FLOW

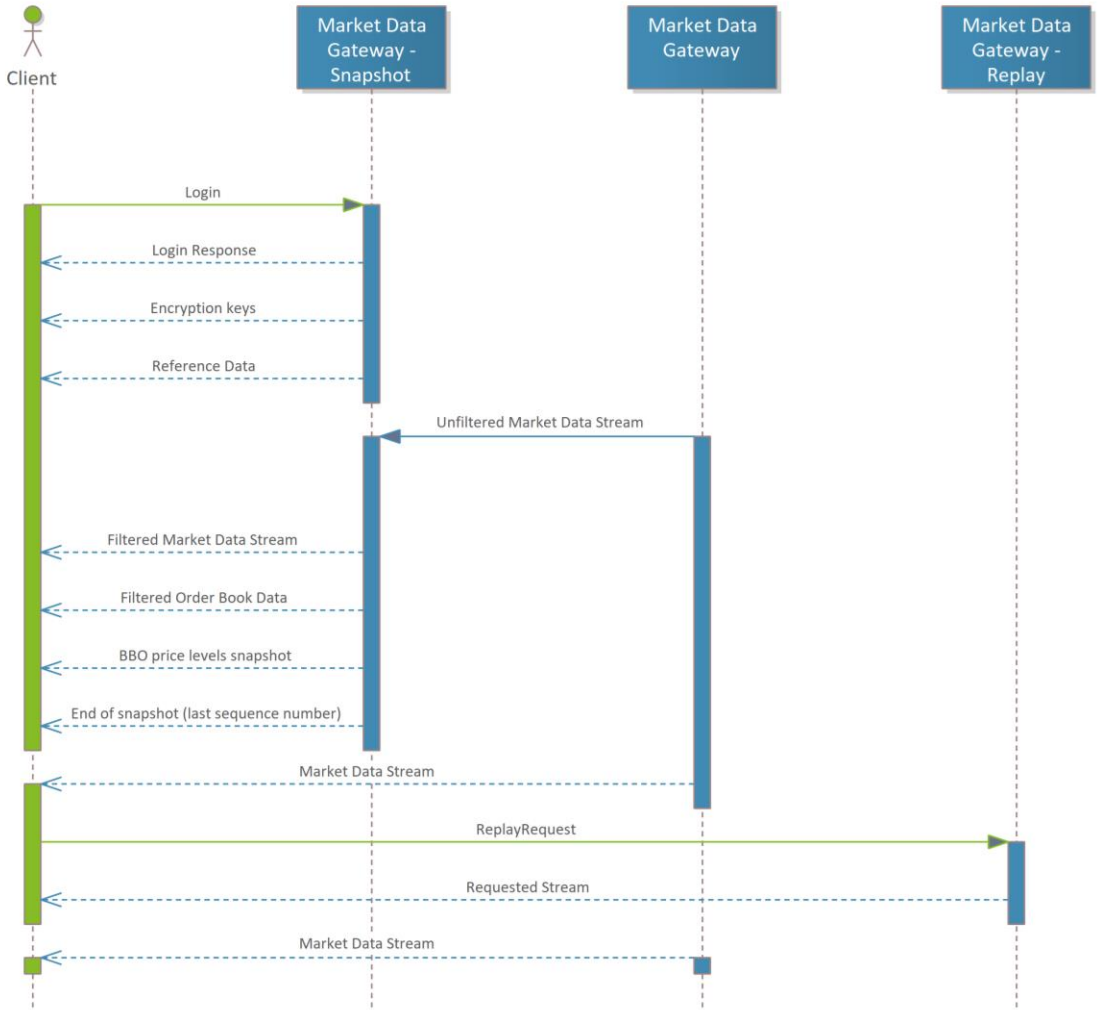
Upon connecting to the Snapshot, the user receives reference data and the full Market state up to that point in time. The user then switches to the Stream service for real time data.

Replay is to be used when gaps in Stream are detected. If there are communication gaps between Snapshot and the real time Stream, a request for the missing data can also be sent to the Replay service.

The dedicated message is ReplayRequest, in which the requested data needs to be specified by a start and end sequence number. In return, the Replay service returns the missing data.

Channel	Delivery method
Stream	Realtime feed, UDP (multicast transmission)
Snapshot	On demand, TCP (unicast transmission)
Replay	On demand, TCP (unicast transmission)

Figure 2 - General Information Flow



4.3.2. SNAPSHOT

The Snapshot service is available via TCP (unicast transmission) for a given instance of Market Data and is implemented as a standalone service.

Snapshot is the first channel users connect to. It conveys reference data such as information about the structure of the Market and the instruments which can be traded as well as schedules. In addition, it disseminates encryption keys for decrypting the information sent through Stream and Replay. Snapshot can be accessed at any time during the trading day to obtain aggregate information about the state of the order book and the full record of trades and transactions up to that time.

Snapshot:

- provides reference data and encryption keys for decrypting the information sent though Stream and Replay. It can be accessed at any time during the trading day to obtain the full depth order book and the complete record of trades and transactions,
- is cumulative. It does not communicate all the OrderAdd, OrderModify and OrderCancel messages sent to System but only their cumulative effect up to the time the snapshot is requested.

The Snapshot is personalized for each user and only sends the data for which the user holds an active subscription. This includes a selection of Instruments, the choice of BBO or Full Depth data and other parameters.

4.3.3. STREAM

Stream is the real time channel for Market data and disseminates order and transaction data over multicast UDP feeds. It publishes among others the full-depth order book, BBO levels as well as auction related statistics.

Stream is designed to operate with minimum technical delay. Publication is executed in a push model, and messages are published on the corresponding multicast group. The user application is responsible for receiving messages from the corresponding multicast group.

4.3.4. REPLAY

Gap detection is an extremely important mechanism for UDP transmissions to deliver complete information reliably and confidently to the user. The purpose of the Replay service is to retransmit message gaps to user applications which missed a part of the Stream.

Replay's mechanism recovers information lost by Stream during a multicast transmission and supports retransmitting of all Stream message types. It is implemented by the ReplayRequest message.

5. Real Time Flow In Market Data

5.1. Message groups

GPW WATS Market Data consists of sets of messages, which taken together enable a Market participant to obtain a Marketplace structure picture, as well as the flow of transactions in instruments. The table below shows the list of messages involved in Market Data broken down by message category. Details of the messages will be elaborated in the following sections and chapters.

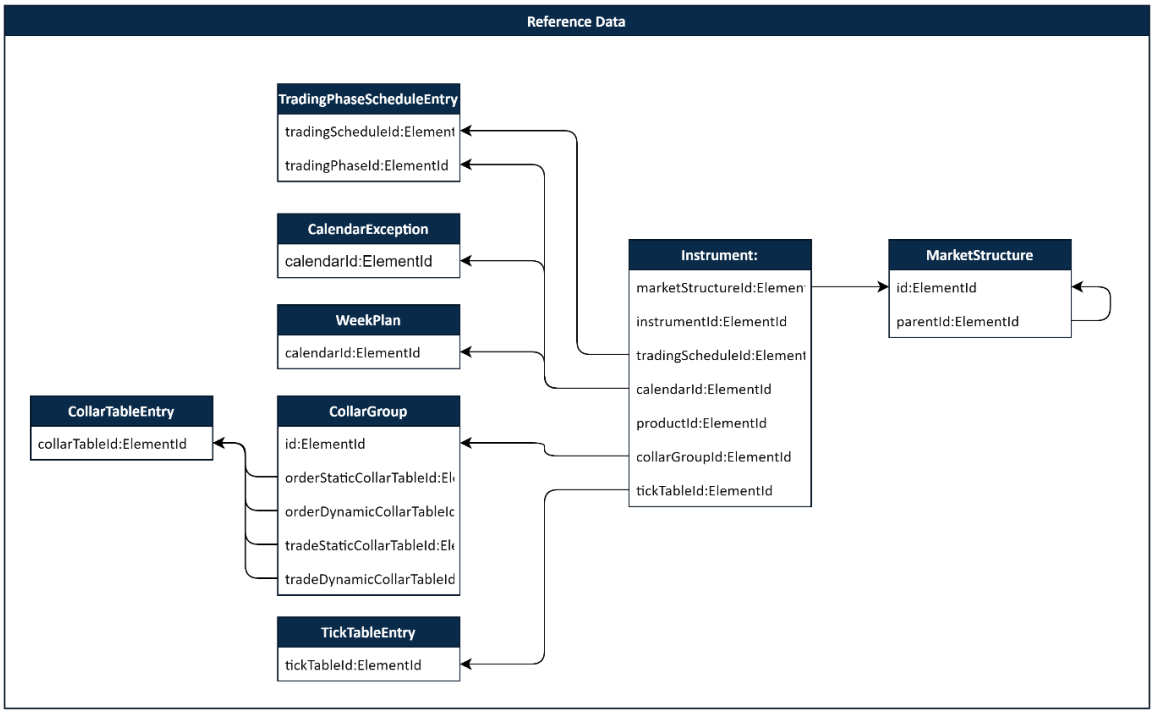
Category	Message type	Message	Description
Trading and execution	Auction	AuctionUpdate AuctionSummary	Messages which communicate trading activity. This includes messages in support of Auctions. This includes notification of executions and trades. BBO Levels are conveyed by PriceLevelSnapshot, the Full Depth order book by Order* and Auction time data by Auction* messages.
	Continuous Trading (Full Depth)	OrderAdd OrderCancel OrderModify OrderBookEvent OrderExecute	
	BBO	PriceLevelSnapshot	
Status messages	Status	InstrumentStatusChange PriceUpdate OrderCollars/TradeCollars Trade News PositionReport	Messages that communicate the status of the Market.
Summary messages	Statistical	InstrumentSummary ProductSummary	Summary statistics about instruments and trading sessions
Reference Data	Market Structure	MarketStructure Instrument TickTableEntry	Group of messages which define Market, instruments, and other Market parameters.
	Trading Calendar	TradingPhaseScheduleEntry CalendarException WeekPlan	
	Order and Trade Collars	CollarTableEntry CollarGroup	
	Pricing Messages	AccruedInterestTableEntry IndexationTableEntry	Information about indexation and accrued interest for fixed income instruments
Index	Index messages	IndexParams IndexPortfolioEntry RealTimeIndex IndexSummary	Group of messages which power the Index feature.
Session Management	Encryption	EncryptionKey	Distribution of encryption keys.
	Technical	TradingSessionStatus EndOfSnapshot EncryptionKey ReplayRequest	Support messages used to frame the contents of Market Data.

5.2. The Big Picture

The diagrams in this section show the extent of inter-relations between different Market Data messages and the way they combine into a coherent whole. The graphs only feature the ID fields (keys) of the individual Market Data messages. The messages carry many more attributes than shown in this illustration. Session management messages together with the EncryptionKey message are omitted.

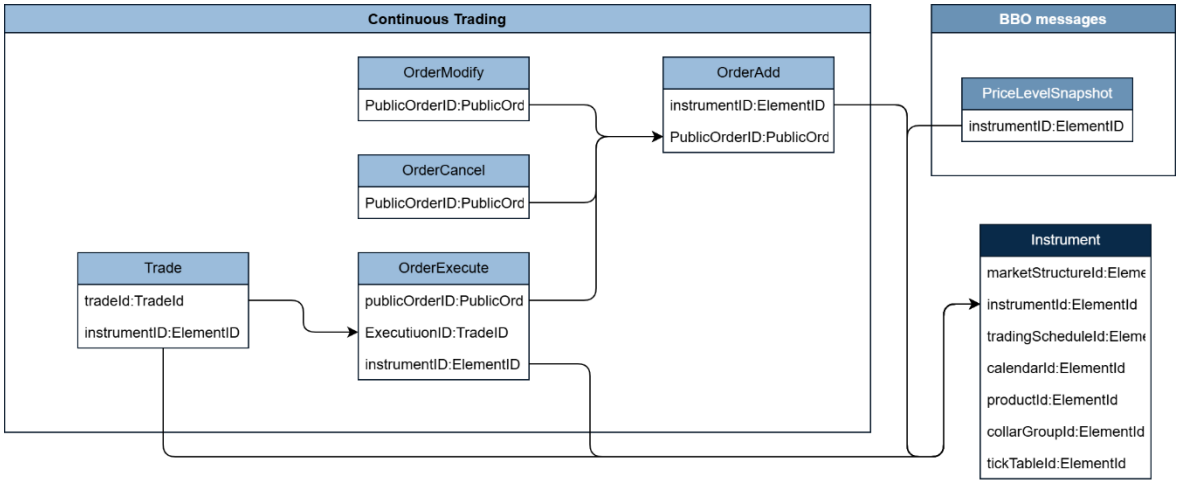
5.2.1. REFERENCE DATA

Figure 3 - Reference data



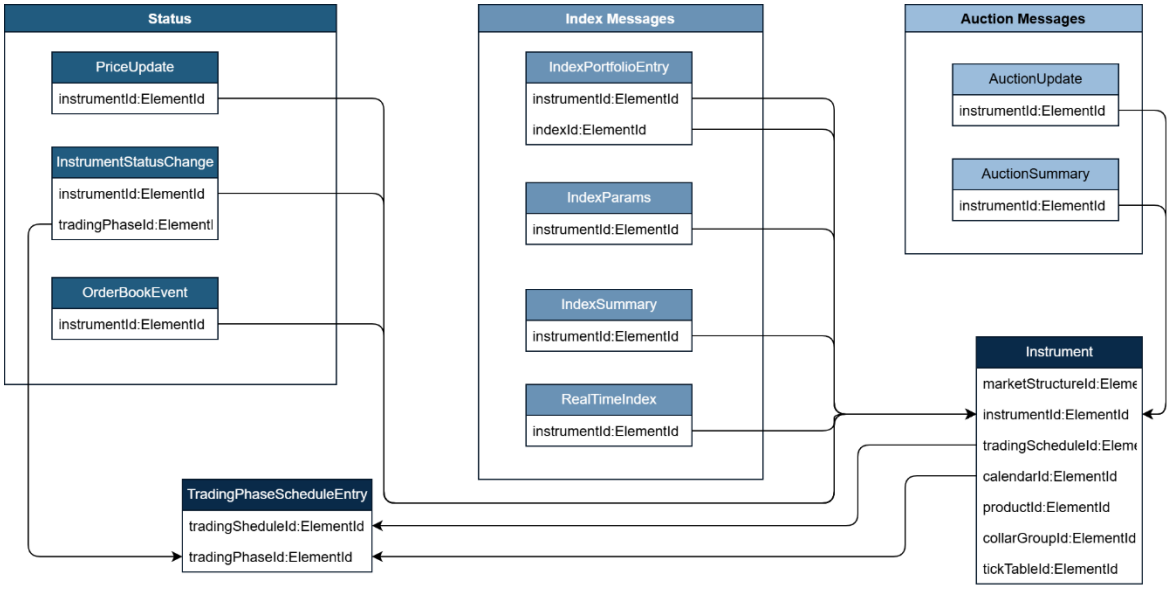
5.2.2. ORDER MESSAGES AND REFERENCE DATA

Figure 4 - Order messages and reference data



5.2.3. STATUS, AUCTION, INDEX MESSAGES AND REFERENCE DATA

Figure 5 - Status, Auction, Index Messages and Reference Data



5.3. STREAM+REPLAY CONTENTS: FULL DEPTH AND BBO

The real time Full Depth and BBO data share many of the same messages and differ in specific content. Full Depth conveys complete Market information, including Auctions. BBO features only bid offer levels and omits OrderAdd, OrderModify, OrderCancel and OrderExecute messages.

The messages and categories of messages involved in the Full Depth Stream and the Full Depth Replay services.

Category	Full Depth Stream + Replay Messages	BBO Stream + Replay Messages
Auction	OrderBookEvent AuctionSummary AuctionUpdate	AuctionSummary AuctionUpdate
Continuous trading	OrderAdd OrderCancel OrderModify	
BBO	PriceLevelSnapshot *	PriceLevelSnapshot **
Execution and Trade	OrderExecute Trade PositionReport	Trade PositionReport
Market Phase	InstrumentStatusChange	InstrumentStatusChange
Trading Collar	OrderCollars/TradeCollars	OrderCollars/TradeCollars *
Summaries	InstrumentSummary ProductSummary	InstrumentSummary ProductSummary
Session Management	TradingSessionStatus	TradingSessionStatus

* published for redistributed instruments only

** optional, based on BBO service level configuration

5.3.1. GENERAL SEQUENCE OF MARKET DATA MESSAGES DURING A TRADING DAY

Stream follows a predetermined structure during a Market Data trading day. The Stream operates within a technical session during which all Market data is disseminated. At the beginning of the technical session, reference data is disseminated to all connected users. After this, once the trading day begins and orders start coming in, Market messages are disseminated according to the rules for different Market Phases. Once the technical session ends, Stream ceases publication of Market data.

It is important to distinguish the technical session which is the framework for Market Data and a trading session. A trading session occurs within a technical session.

Message Type	Sequence of messages in Stream	Note
Session Management	TradingSessionStatus	The technical session starts with TradingSessionStatus.tradingSessionEvent-StartOfTechnicalSession. The GPW WATS system Stream is open and Market Data messages start flowing.
	TradingSessionStatus	Signal for start of reference data. TradingSessionStatus.tradingSessionEvent-InitialReferenceDataStart
Reference Data		Reference data. For information on the structure of the reference data block, please see Section 5.4. Note that reference data messages may arrive in random order and not in related contiguous chunks.
	TradingSessionStatus	Signal for end of reference data. TradingSessionStatus.tradingSessionEvent-InitialReferenceDataEnd
Order Reference Data		OrderAdd messages for persistent orders
Start of Trading Session		
Trading	AuctionUpdate AuctionSummary OrderAdd OrderCancel OrderModify PriceLevelSnapshot OrderExecute Trade InstrumentStatusChange OrderCollars/TradeCollars	Once the trading session starts, trading messages are published as they arrive. The publication of trading messages adheres to the rules of the Market Phases, such as auctions and continuous trading. Market operations related messages such as InstrumentStatusChange and OrderCollars/TradeCollars set the flow of information and convey data related to order control. The list represents a series of possible messages, and not their order of arrival.
End of Trading Session (including per MIC, see Section 10.2)		
Summaries	InstrumentSummary ProductSummary IndexSummary PositionReport	Summaries and statistics about the trading session.
Next Session Reference Data		
Session Management	TradingSessionStatus	The technical session ends with TradingSessionStatus.tradingSessionEvent-EndOfTechnicalSession. The technical session ends. The Flow of Market Data messages is stopped.

5.3.2. REPLAY

The Replay service can re-transmit any of the messages from the Stream service which were lost during transmission. For each Stream the corresponding replay service, Replay Full Depth or Replay BBO, will retransmit the lost messages upon request. The list of messages which can be replayed is the same as in the table above.

5.4. SNAPSHOT: FULL DEPTH AND BBO

5.4.1. SNAPSHOT LEVELS

The Snapshot service offers 2 levels of Snapshots, each customized to the needs of potential users. The Snapshots contain a core of reference data information but differ in the extent of Market data provided. Here are the principal differences:

- Full Depth snapshot, aimed at users who rely on the full picture of the Market for their trading and investment activities. It is the most complete option available and contains Reference data, the full depth order book and auction related information,
- BBO Snapshot, aimed at users who only require best bid-offer levels. It is otherwise identical to the Full Depth snapshot.

5.4.2. SNAPSHOT SEQUENCE

The Snapshot disseminates critical Market structure and whole-day trading information at the start of the trading day as well as during the day upon request. The sequence of the messages in the Snapshot is organized in groups of messages. Together these messages can help the user construct an image of the Market structure and build an image of the order book. The structure of the Snapshot for Full Depth and BBO differs due to the scope of the data and is presented in the table below.

Message Type	Message sequence in Full Depth Snapshot	Message sequence in BBO Snapshot	Message Group
Encryption	EncryptionKey	EncryptionKey	1
Session Management	TradingSessionStatus.tradingSessionEvent =StartOfTechnicalSession	TradingSessionStatus.tradingSessionEvent =StartOfTechnicalSession	
Session Management	TradingSessionStatus.tradingSessionEvent =InitialReferenceDataStart	TradingSessionStatus.tradingSessionEvent =InitialReferenceDataStart	
Trading Calendar	WeekPlan	WeekPlan	
Trading Calendar	CalendarException	CalendarException	
Market Structure	TickTableEntry	TickTableEntry	
Trading Collar	CollarTableEntry	CollarTableEntry	
Trading Calendar	TradingPhaseScheduleEntry	TradingPhaseScheduleEntry	
Market Structure	MarketStructure	MarketStructure	

Message Type	Message sequence in Full Depth Snapshot	Message sequence in BBO Snapshot	Message Group
Trading Collar	CollarGroup	CollarGroup	
External	ExternalUnderlying	ExternalUnderlying	
Market Structure	Instrument	Instrument	
Pricing	AccruedInterestTableEntry	AccruedInterestTableEntry	
Indexes	IndexationTableEntry	IndexationTableEntry	
Session Management	TradingSessionStatus.tradingSessionEvent =InitialReferenceDataEnd	TradingSessionStatus.tradingSessionEvent =InitialReferenceDataEnd	
Status	InstrumentStatusChange	InstrumentStatusChange	3
	News	News	2
	OrderCollars TradeCollars	OrderCollars TradeCollars	3
	PositionReport	PositionReport	2
	InstrumentSummary ProductSummary	InstrumentSummary ProductSummary	2
Session Management	TradingSessionStatus.tradingSessionEvent =StartOfTradingSessionMIC	TradingSessionStatus.tradingSessionEvent =StartOfTradingSessionMIC	1
Continuous Trading / Auctions / BBO	OrderAdd (snapshot OrderAdds feature seqnum=0 instead of real seqnum value)		4
	AuctionSummary AuctionUpdate	AuctionSummary AuctionUpdate	3
	PriceLevelSnapshot	PriceLevelSnapshot	4
	Trade	Trade	2
	PriceUpdate	PriceUpdate	2
Session Management	EndOfSnapshot	EndOfSnapshot	1

The messages in Snapshot are grouped in 4 categories relative to the way they are sent out:

Message Group	Description
1	The order of these messages is fixed and they will always come out in the order mentioned in the table.
2	WATS does not guarantee the order of these messages during the trading day. They are directly copied in the order in which they originally appeared on the Market Data stream. Note: Next Session Reference Data comes out whenever the system loads it at the end of the trading day.
3	Only the last of the series of messages that appear on Market Data is sent out. For example, only the most recent OrderCollars and TradeCollars are sent out and not the entire history of collars from the start of the trading day.
4	Order Book messages. Only those messages that are required to reconstruct the state of the order book at any given time are sent out. For example, this excludes messages related to orders that have been executed during the trading day. Order priority is indicated by the sequence of the incoming messages. Order messages are resent from higher to lower priority, price level by price level.

6. Reference Data Messages

This chapter outlines the set of messages relayed in Reference Data and used to set up the Market trading environment.

6.1. Market Structure Messages

6.1.1. INSTRUMENT

The Instrument message defines an instrument that can be traded on the GPW WATS platform. It is identified by a unique InstrumentId and contains a complete configuration required for trading.

The following messages refer to instruments through the instrumentId field:

Message Category	Message name
Order Messages	OrderAdd, OrderExecute, Trade Note: OrderModify and OrderCancel refer to OrderAdd only.
Auction Messages	Auctionupdate, AuctionSummary
BBO Messages	PriceLevelSnapshot
Status Messages	InstrumentStatusChange, OrderCollars and TradeCollars, OrderBookEvent

6.1.2. MARKETSTRUCTURE

MarketStructure messages combine to create a Market structure for a Market. The Market structure is defined as a tree and starts with specifying a top-level element and continues by defining Market segments. Multiple Markets may be defined simultaneously. Market segments are identified by a Market Identifier Code (MIC) as specified in ISO 10383, named and assigned a Market model type.

The market structure is populated with instruments from Instrument messages and instruments (InstrumentId) are the leaves on the branches of a market structure tree.

6.1.3. TICKTABLEENTRY

TickTableEntry messages combine to produce a tick table which may be assigned to an instrument. A Tick table starts at 0 and specifies the tick sizes for various price levels. lowerBound specifies the price level starting from which a tickSize applies.

6.1.4. EXAMPLES

Market Data broadcasts the Market structure by sending a series of messages to that effect. A Market structure is a set of trees, each of which defines a Market. A Market is a tree with branches which are Market segments, sub-segments, or groups. The leaves on the branches of a Market tree are Instruments.

Instruments are collections of essential information which enable a product to trade. An Instrument incorporates many of the components defined by Reference Data messages (such as trading schedules, collars, tick tables etc.) into a single object. Instruments are allocated to Market Structures.

6.1.4.1. Instrument status = Inactive

Instruments with status=Inactive never appear Market Data reference data.

6.1.4.2. Instrument Definitions

Instruments are the nexus of Reference Data. They are defined using Instrument messages which bind all the Market Data reference messages together and define their behavior by reference to them.


```
{"header":{“length”:741,"msgType":"Instrument","version":4096,"seqNum":342
,"timestamp":1717902001730705137,"sourceTimestamp":1717902001730599335,"is
Encrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,
"streamId":0},"instrumentId":4669,"productType":"Equity","productSubtype":
"Share","firstTradingDate":20231020,"lastTradingDate":0,"productId":669,"c
urrency":"PLN","icebergMinValue":5000000,"lotSize":1,"priceExpressionType"
:"Price","calendarId":3,"marketStructureId":72,"tickTableId":12,"collarGro
upId":17,"tradingScheduleId":2,"description":"PKN XSEE C.SHRS
PLN","mic":"XSEE","nominalValue":125000000,"nominalValueType":"Constant","
multiplier":0,"strikePrice":0,"strikePriceCurrency":"XXX","productIdentifi
cation":"PLPKN0000018","productIdentificationType":"ISIN","settlementCalen
darId":4,"bondCouponType":"NotApplicable","preTradeCheckMinPrice":1000000,
"preTradeCheckMaxPrice":999999999900000000,"preTradeCheckMinQuantity":1,"p
reTradeCheckMaxQuantity":1000000000,"preTradeCheckMaxValue":100000000000,"
preTradeCheckMinValue":0,"liquidity":true,"marketModelType":"CLOB","issuer
":"ORLEN Spolka
Akcyjna","issuerRegCountry":"POL","underlyingInstrumentId":0,"productCode"
:"PKN","cfi":"ESVUFB","fisn":"ORLEN/BRSH SER-A-F
PLN1.25","issueSize":1160942049,"nominalCurrency":"PLN","usIndicator":"Not
Applicable","expiryDate":0,"versionNumber":0,"settlementType":"NotApplicab
le","optionType":"NotApplicable","exerciseType":"NA","productName":"PKNORL
EN","referencePrice":10000000000,"status":"Active","initialPhaseId":12,"th
resholdMax":0,"thresholdMin":0,"isLeverage":"NotApplicable"}
```


Instrument message for the shares of ORLEN Spolka Akcyjna, ticker PKN, productType=Equity (ISIN PLPKN0000018) with product subtype share, traded under MIC XSEE in the central limit order book model – marketModelType=CLOB. This instrument has instrumentId=4669. The FISN for this instrument is ORLEN/BRSH SER-A-F PLN1.25.

MarketStructureId=72 indicates that the Instrument belongs to the element of the Market Structure with Id=72. The instrument follows:

- calendar calendarId=3
- market structure element marketStructureId=72
- tick table tickTableId=12
- collar group collarGroupId=17
- trading schedule tradingScheduleId=2

The key fields of interest in an Instrument message are as follows.

Field name	Refers to Messages [field]	Description
instrumentId		The ID of the Instrument.
productType		Type of financial product: 1. Equity 2. FixedIncome 3. DerivativeFutures 4. DerivativeOptions 5. Index 6. Currency 7. StructuredProduct 9. Tracker
productSubtype		Subtype of the product is a dictionary of different types of products. The productType and productSubtype must be defined together in instrument definition.
calendarId	WeekPlan [calendarId] CalendarException [calendarId]	Trading calendar for the security.
MarketStructureId	MarketStructure [id]	The branch of the Market Structure to which the Instrument belongs.
tickTableId	TickTableEntry [tickTableId]	The tick table for the instrument.
collarGroupId	CollarGroup [id]	The collar group.
tradingScheduleId	TradingPhaseScheduleEntry [tradingScheduleId]	The scheduled Market Phases defined for the instrument.

6.1.4.3. Instrument product sub types

GPW WATS provides a dictionary of product subtypes in the productSubtype attribute of the Instrument message, each related to a specific productType. The table below shows how types of products (productType) are related to subtypes (productSubtype).

productSubtype	productType
Share	Equity
Allotment Right	Equity
Subcription Right	Equity
Tender Offer (not used)	Equity
Issue Right	Equity
Subscription Warrant	Equity
Bank Securities	Fixed Income
Bond	Fixed Income
Treasury Bond	Fixed Income
Bond issued by a government institution	Fixed Income
Municipal Bond	Fixed Income
Corporate Bond - bank	Fixed Income
Corporate Bond - firm	Fixed Income
Convertible Bond	Fixed Income
Mortgage Bond	Fixed Income

productSubtype	productType
Mortgage Backed Bond	Fixed Income
Public Mortgage Bond	Fixed Income
Bill	Fixed Income
Treasury Bill	Fixed Income
Commercial Bill	Fixed Income
ETF - Exchange Traded Fund	Tracker
ETN - Exchange Traded Note	Tracker
ETC - Exchange Traded Commodity	Tracker
Investment Certificate	Equity
Index Futures	Derivative Futures
Stock Futures	Derivative Futures
Currency Futures	Derivative Futures
Bond Futures	Derivative Futures
Interest Rate Futures	Derivative Futures
Index Options	Derivative Options
Stock Options	Derivative Options
Option warrants	Structured Product
Factor certificates	Structured Product
Turbo certificates	Structured Product
Capital protection certificates	Structured Product
Bonus certificates	Structured Product
Reverse convertible certificates	Structured Product
Express certificates	Structured Product
Discount certificates	Structured Product
Index/tracker certificates	Structured Product
Other investment certificates (non leveraged)	Structured Product
Structured notes	Structured Product
Reference rate	Reserved for future use in data redistribution: bank reference rates.
Interest rate	Reserved for future use in data redistribution: WIBOR, SOFR.
Exchange rate	Currency
Price index	Index
Total return index	Index
Sector price index	Index
Sector total return index	Index
Dividend index	Index
Strategy price index	Index
Strategy total return index	Index
Commodity index	Index

productSubtype	productType
Bond total return index	Index
SupranationalBond	Supranational Bond
WarrantWithKnockOut	Warrant with Knock Out

6.1.4.4. Instrument Parametrization for Structured Products

The Instrument message supports structured products by including specific attributes related to this type of product. The table below outlines the elements in the Instrument Market data message characteristic of structured products which are filled in addition to those common to normal financial instruments. The productType enum=StructuredProduct for structured products.

Field name	Description
multiplier	Product value multiplier.
strikePrice	Product strike price.
expiryDate	Expiration date.
settlementType	Identifier of settlement type: cash settlement or physical delivery.
excerciseType	Identifier of exercise style: American or European.
strikePriceCurrency	Currency of the strike price.
thresholdMax	Leveraged products require the definition of a lower and upper barrier. The range is defined by the Market operator upon provision of the barriers data by the issuer or Market maker.
thresholdMin	
isLeverage	True if instrument is leveraged. Not Applicable value refers to other securities types such as equity, debt.

6.1.4.5. Market Structure

A Market Structure is a series of MarketStructure messages which define a tree of trading groups, segments, and sub-segments. An Instrument makes reference to a Market structure branch of which is a leaf.


```
{"header":{"length":105,"msgType":"MarketStructure","version":4096,"seqNum":315,"timestamp":1717902001477070275,"sourceTimestamp":1717902001477014574,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"id":72,"parentId":69,"mic":"","name":"X RM CLOB Equties","marketModelType":"CLOB"}
```


The market structure element under which the PKN Orlen equity instrument is listed – id=72. It is a CLOB segment with parent parentId=69. The name of the segment is X RM CLOB Equties.


```
{"header":{"length":105,"msgType":"MarketStructure","version":4096,"seqNum":250,"timestamp":1717902001467304438,"sourceTimestamp":1717902001467259337,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"id":69,"parentId":0,"mic":"XSEE","name":"X Equities","marketModelType":"NotApplicable"}
```


The parent segment of the X RM CLOB Equties market segment from above is the X Equities market segment. It is a top segment of the market structure because parentId=0. The full market structure in the PKN Orlen example is therefore: X Equities - X RM CLOB Equties - ORLEN Spolka Akcyjna, ticker PKN.

The following table describes details of the MarketStructure message in more detail.

Field name	Description	Message 278 (seqNum=278)
id	The ID of the Market Structure element.	1
parentId	A reference to the parent Market Structure element, i.e. the id field of the parent structure. The root of the structure has parentId=0.	0
mic	Market Identifier Code (MIC) of that particular element of the structure (ISO 10383). Can feature one or more MIC codes.	XWAR
name	The name of the element assigned by the Market Operator.	WSE RM Equities
MarketModelType	The type of Market structure element. The top element of any Market Structure tree is NotApplicable. Possible values for MarketModelType include: 1. NotApplicable 2. CLOB 3. BLOCK 4. HYBRID 5. CROSS 6. IPO 7. TenderOffer 8. Redistribution	NotApplicable.

6.1.4.6. Tick Tables

TickTableEntry messages combine to create Tick Tables. A Tick Table is a set of price intervals with related steps in which prices can increase.

A tick table:

- must include the price 0 in its first interval (lowerBound),
- does not specify an upper bound in the highest interval,
- specifies that prices are ≥ lowerBound within each interval, but < lowerBound in following interval
- is currency agnostic.


```
{"header":{“length”:62,”msgType”:”TickTableEntry”,"version":4096,"seqNum":145,"timestamp":1717902001451145746,"sourceTimestamp":1717902001450608033,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"tickSize":1000000,"lowerBound":0,"tickTableId":12}
```


Tick Table 12 for the PKN Orlen instrument. Prices increase in the interval from 0 to ∞ in steps of 0.01.

A series of TickTableEntry messages with an identical tickTableId define a tick table.


```
{"header":{“length”:62,”msgType”:”TickTableEntry”,"version":4096,"seqNum":160,"timestamp":1705460401072863623,"sourceTimestamp":1705460401072837522,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tickSize":1000000,"lowerBound":0,"tickTableId":11}
{"header":{“length”:62,”msgType”:”TickTableEntry”,"version":4096,"seqNum":161,"timestamp":1705460401072993726,"sourceTimestamp":1705460401072961225,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tickSize":2000000,"lowerBound":1000000000,"tickTableId":11}
{"header":{“length”:62,”msgType”:”TickTableEntry”,"version":4096,"seqNum":162,"timestamp":1705460401073111129,"sourceTimestamp":1705460401073084328,
```

```
"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tickSize":5
000000,"lowerBound":2000000000,"tickTableId":11}
{"header":{“length”:62,”msgType”:”TickTableEntry”,"version":4096,"seqNum":
163,"timestamp":1705460401073259333,"sourceTimestamp":1705460401073231032,
"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tickSize":1
0000000,"lowerBound":5000000000,"tickTableId":11}
{"header":{“length”:62,”msgType”:”TickTableEntry”,"version":4096,"seqNum":
164,"timestamp":1705460401073380436,"sourceTimestamp":1705460401073351235,
"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"tickSize":2
0000000,"lowerBound":10000000000,"tickTableId":11}
```


A series of TickTableEntry messages (unrelated to PKN Orlen) with tickTableId=11 combine to produce a complete tick table shown below.

Tick table identified by tickTableId=11 starts at the price 0.

Sequence number	Price interval		Tick size
	From	To	
160	0	1000000000	1000000
161	1000000000	2000000000	2000000
162	2000000000	5000000000	5000000
163	5000000000	10000000000	10000000
164	10000000000	∞	20000000

6.2. Trading Calendar Messages

Trading calendar messages define a trading calendar and trading schedule for Instruments (InstrumentID). A Trading Schedule (tradingScheduleId) is a collection of Market Phases, each identified by tradingPhaseId. The TradingPhaseScheduleEntry message informs which TradingPhases (identified by tradingPhaseId) are attached to a particular Schedule (tradingScheduleId).

An Instrument (InstrumentID) follows a Trading Schedule (tradingScheduleId) and switches between TradingPhases. Instrument tracks a predefined WeekPlan and applies the tradingScheduleId on valid trading days, except for CalendarException days.

The job of an InstrumentStatusChange message is to report that an Instrument changed phases.

6.2.1. TRADINGPHASESCHEDULEENTRY

Trading schedules are instrument agnostic. They are a collection of phases that an instrument will follow. An Instrument will refer to a particular tradingScheduleId.

TradingPhaseScheduleEntry defines Market Phases (tradingPhaseId) for a trading schedule. TradingPhaseScheduleEntry messages combine to define a complete trading schedule (tradingScheduleId).

TradingPhaseScheduleEntry functions in conjunction with InstrumentStatusChange which signals the beginning of a new Market Phase for an instrument (TradingPhaseId).

Note: for unscheduled trading phases (intraday volatility auctions such as AuctionVolatilityStatic, AuctionVolatilityDynamic, AuctionExtendedVolatilityStatic, AuctionExtendedVolatilityDynamic) the tradingPhaseStartTime is set to 0.

6.2.2. WEEKPLAN

WeekPlan defines the days of the week (Monday-Sunday) during which trading is permitted. The WeekPlan is assigned a particular CalendarID. A customized week plan can be devised for each Instrument.

6.2.3. CALENDAREXCEPTION

Specifies the calendar date for which a trading exception is scheduled. It refers to the CalendarID of the corresponding WeekPlan message.

6.2.4. EXAMPLES

Calendar messages are:

- defined in abstraction – independently of Markets or instruments. They can be assigned to different Markets or instruments,
- linked by the calendarId identifier.

6.2.4.1. Trading Schedules and Phases

A Trading Schedule is a collection of Market Phases. A series of TradingPhaseScheduleEntry messages with the same tradingScheduleId define a Trading Schedule composed of Market Phases (tradingPhaseId). An Instrument refers to a Trading Schedule (tradingScheduleId) and follows it during the trading day. Phase changes in a schedule are signaled on the Market data feed by the InstrumentStatusChange message. Trading Schedules are Instrument agnostic and can be attached to one or more instruments.

The following messages with TradingScheduleID=2 define a trading schedule composed of several market phases, here tradingPhaseId=[12, 13, 14, 15, 16].


```
{"header":{“length”:82,”msgType”:”TradingPhaseScheduleEntry”,"version":4096,"seqNum":214,"timestamp":1717902001461589299,"sourceTimestamp":1717902001461553899,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"tradingScheduleId":2,"tradingPhaseId":12,"tradingPhaseSettlementCycle":0,"tradingPhaseMaxBlockSettlementCycle":0,"staticCollarVolatilityAuctionId":0,"dynamicCollarVolatilityAuctionId":0,"tradingPhaseStartTime":1717884000000000000,"tradingPhaseType":"NoTradingClosed","uncrossing":false,"extStaticCollarVolatilityAuctionId":0,"extDynamicCollarVolatilityAuctionId":0,"lastAuctionPhaseId":0}
{"header":{“length”:82,”msgType”:”TradingPhaseScheduleEntry”,"version":4096,"seqNum":215,"timestamp":1717902001461740103,"sourceTimestamp":1717902001461716102,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"tradingScheduleId":2,"tradingPhaseId":13,"tradingPhaseSettlementCycle":0,"tradingPhaseMaxBlockSettlementCycle":0,"staticCollarVolatilityAuctionId":0,"dynamicCollarVolatilityAuctionId":0,"tradingPhaseStartTime":1717884300000000000,"tradingPhaseType":"NoTradingEarlyMonitoring","uncrossing":false,"extStaticCollarVolatilityAuctionId":0,"extDynamicCollarVolatilityAuctionId":0,"lastAuctionPhaseId":0}
```

```
{"header":{“length”:82,”msgType”:”TradingPhaseScheduleEntry”,"version":4096,"seqNum":216,"timestamp":1717902001461993809,"sourceTimestamp":1717902001461957308,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"tradingScheduleId":2,"tradingPhaseId":14,"tradingPhaseSettlementCycle":2,"tradingPhaseMaxBlockSettlementCycle":0,"staticCollarVolatilityAuctionId":0,"dynamicCollarVolatilityAuctionId":0,"tradingPhaseStartTime":1717884600000000000,"tradingPhaseType":"ContinuousPriceTime","uncrossing":false,"extStaticCollarVolatilityAuctionId":0,"extDynamicCollarVolatilityAuctionId":0,"lastAuctionPhaseId":0}
{"header":{“length”:82,”msgType”:”TradingPhaseScheduleEntry”,"version":4096,"seqNum":217,"timestamp":1717902001462135313,"sourceTimestamp":1717902001462059311,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"tradingScheduleId":2,"tradingPhaseId":15,"tradingPhaseSettlementCycle":0,"tradingPhaseMaxBlockSettlementCycle":0,"staticCollarVolatilityAuctionId":0,"dynamicCollarVolatilityAuctionId":0,"tradingPhaseStartTime":1717969800000000000,"tradingPhaseType":"NoTradingLateMonitoring","uncrossing":false,"extStaticCollarVolatilityAuctionId":0,"extDynamicCollarVolatilityAuctionId":0,"lastAuctionPhaseId":0}
{"header":{“length”:82,”msgType”:”TradingPhaseScheduleEntry”,"version":4096,"seqNum":218,"timestamp":1717902001462143913,"sourceTimestamp":1717902001462100712,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"tradingScheduleId":2,"tradingPhaseId":16,"tradingPhaseSettlementCycle":0,"tradingPhaseMaxBlockSettlementCycle":0,"staticCollarVolatilityAuctionId":0,"dynamicCollarVolatilityAuctionId":0,"tradingPhaseStartTime":1717970100000000000,"tradingPhaseType":"NoTradingClosed","uncrossing":false,"extStaticCollarVolatilityAuctionId":0,"extDynamicCollarVolatilityAuctionId":0,"lastAuctionPhaseId":0}
```


In this series of messages with tradingScheduleId=2 define a several market phases:

- tradingPhaseId=12 – NoTradingClosed
- tradingPhaseId=13 - NoTradingEarlyMonitoring
- tradingPhaseId=14 - ContinuousPriceTime
- tradingPhaseId=15 - NoTradingLateMonitoring
- tradingPhaseId=16 - NoTradingClosed

The Instrument reference data message for PKN Orlen references this trading schedule.

6.2.4.2. Week Plans

WeekPlan specifies the days of the week during which trading can occur. A calendarID may only have a single week-plan.


```
{"header":{"length":53,"msgType":"WeekPlan","version":4096,"seqNum":5,"timestamp":1717902001421973937,"sourceTimestamp":1717902001419986089,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"calendarId":3,"weekdays":[true,true,true,true,true,true,true]}
```

The week plan with calendarId=3, referenced in the related Instrument message stipulates that PKN Orlen shares will be traded 7 days a week. A trading exception is defined by the related CalendarException message and trading will not take place on November 1st 2024.

6.2.4.3. Calendar Exceptions

CalendarException is a means of defining exceptions to the regular trading calendar. CalendarException assigns exceptions to the week plan identified by the same calendarId. It provides a method to define calendar days during which trading will not occur on standard terms. A calendar exception is a calendar date on which the Market may be closed or opened, trading hours may be reduced (half-day), or where a technical exception may take place.

Market Data will usually communicate a series of exception days for a particular CalendarID. In the example below, a series of messages with calendarId=3 combine with a WeekPlan message similarly identified with calendarId=3.


```
{"header":{"length":51,"msgType":"CalendarException","version":4096,"seqNum":8,"timestamp":1717902001422000138,"sourceTimestamp":1717902001420428199,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"calendarId":3,"calendarExceptionDate":20241101,"calendarExceptionType":"Closed"}
```


The CalendarException with calendarID=3 defines the first November 2024 as an exception date during which the market will be closed. This calendarID is referenced by the PKN Orlen Instrument message.

6.3. Order And Trade Collars

6.3.1. COLLARGROUP

A group of collars which includes Static and Dynamic Order collars and Static and Dynamic Trade collars.

6.3.2. COLLARTABLEENTRY

CollarTableEntry defines Order and Trade Collar (collarMode) by providing the range of deviation from reference prices in percentage form or absolute values (collarExpression).

This message only contains the collar ranges, but not the values of the collars. The values of the collars are communicated by the OrderCollars and TradeCollars messages at the beginning of each Market Phase.

6.3.3. EXAMPLES

Market Data communicates the Order and Trade price collars configured by the Market Operator. The key to understanding the difference between the collar types is as follows:

- Order Collars prevent orders from entering the order book whenever their limits exceed predefined limit thresholds (upper and lower limits),
- Trade Collars prevent large deviations in execution prices.

While different in nature, Order and Trade collars, together with their parameters, are communicated by CollarTableEntry Market Data messages. To accurately frame an Instrument's price behavior, collars are

bundled together into groups with the CollarGroup message. The table below provides a short description of various collars and their properties.

Collar Type	Description	Possible Mode	Mode Description
Order Price Collars	Prevent orders with excessive bid or ask prices from entering the Order Book.	Static	Wide collar. Centered around reference price at beginning of day.
		Dynamic	Narrow collar. Follows the Market price in real time.
Trade Price Collars	Ensure that trades in the Order Book occur within reasonable price ranges.	Static	Wide collar. Centered around reference price at beginning of day.
		Dynamic	Narrow collar. Follows the Market price in real time.

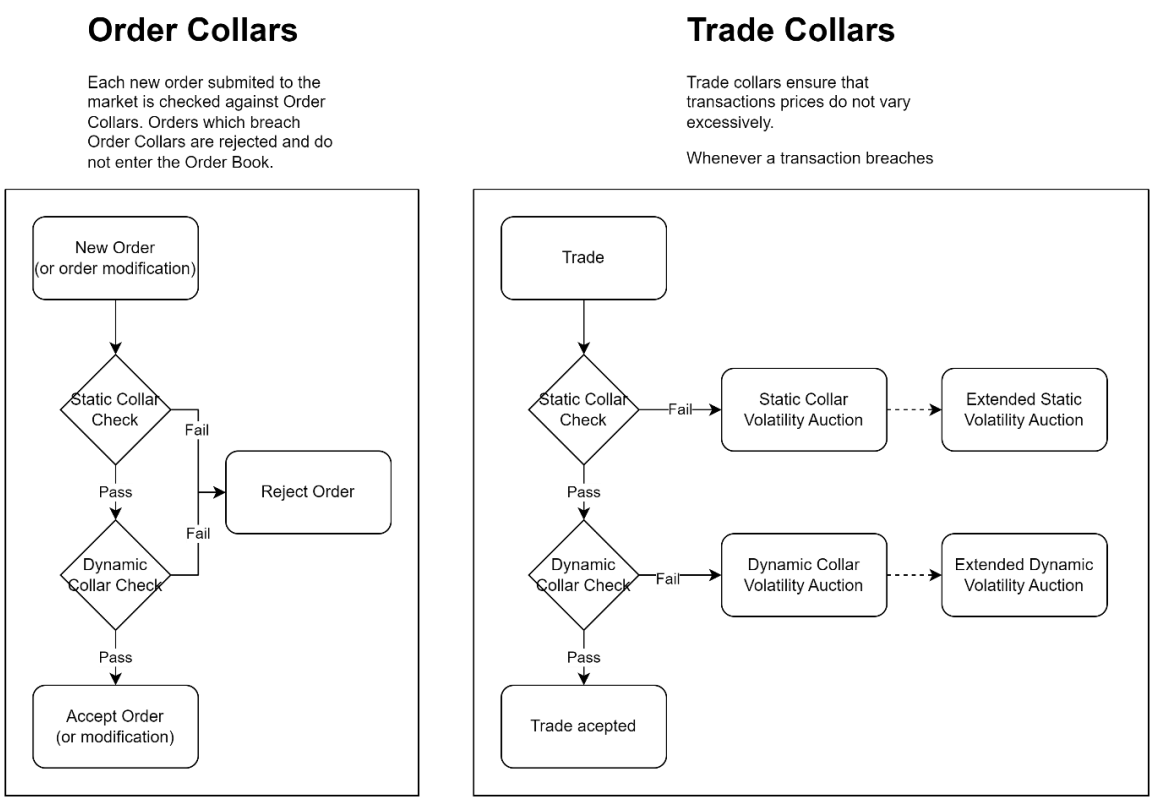
Collars are always expressed as deviations (in percentage form or absolute value) from a reference price or the real time Market price. The collar definition message CollarTableEntry only contains relative deviations and not collar values which may be directly applicable for business purposes. The Market Data messages OrderCollars and TradeCollars convey an Instrument's calculated collar values applicable at any time during the trading session. These values are computed from collar definitions contained in CollarTableEntry and relevant reference prices in effect at the time.

NOTE: because collars are defined through deviations, the same collars assigned to two Instruments will obtain different values if the reference prices for those Instruments differ.

To summarize, Trade and Order price collars are set and configured by two Market Data messages, CollarGroup and CollarTableEntry:

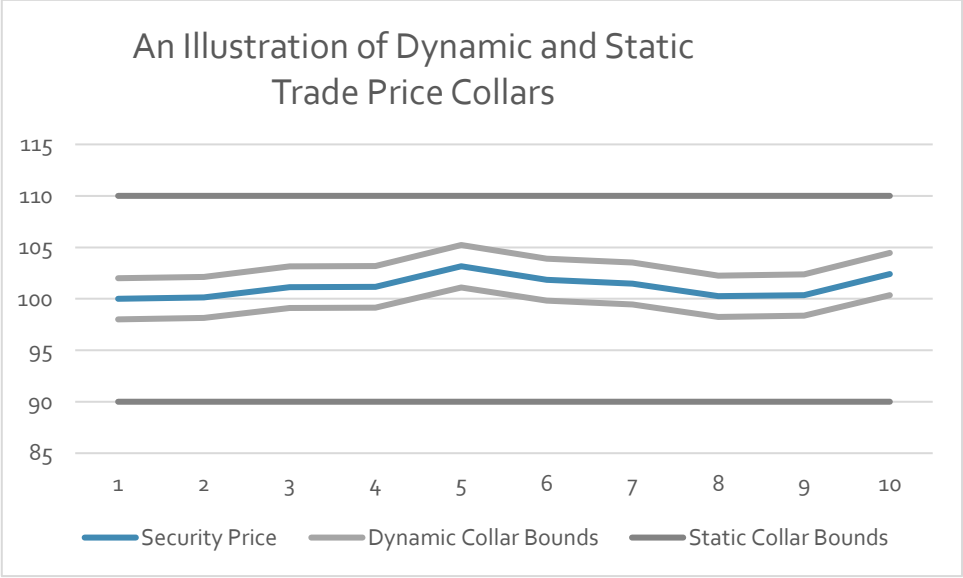
- CollarTableEntry defines collars,
- CollarGroup groups the collars in a single structure used to control trading behavior,
- OrderCollars and TradeCollars convey the calculated collar values for the relevant Instrument.

Figure 6 - Collar Mechanisms



In the figure chart, a Dynamic Trade Price Collar of 2% and a Static Trade Price Collar of 10% around the Reference Price are shown.

Figure 7 - Static and Dynamic Collars



6.3.3.1. Collar Groups

A Collar Group (CollarGroup) binds a set of 4 Collar Tables to produce a complete definition of collars. A Collar Group consists of 4 collar tables specified and defined by CollarTableEntry messages:

- Static collars for Orders,
- Dynamic collars for Orders,
- Static collars for Trades,
- Dynamic collars for Trades.

To fully configure a collar group, a set of 4 Collar Tables must be specified. However, depending on the application, a collar group may contain fewer than 4 collar tables. For example, one Collar Group may only feature Trade Collars, another only Order Collars while still another may only contain Order Static Collars. The choice is driven by regulations, Market rules and the definition of Instruments.

Multiple CollarTableEntry messages may be needed to specify a single Collar Table. In such a case Market Data will broadcast several CollarTableEntry messages with the same collarTableId.

Static and Dynamic collars, for both Orders and Trades, are defined in the same manner but must be encoded in different tables. For example, one CollarTableEntry table needs to be defined for a Static Trade collar and another for a Dynamic Trade collar.

Instruments refer to specified Collar Groups. A single Collar Group can be attached to one or more Instruments and assign the same set of collars to these instruments.

The CollarGroup message only requires the identifiers to relevant CollarTableEntry definitions. These fields are shown below:

Field name	Description	Relevant values from the example below
id	Collar Group identifier.	17
orderStaticCollarTableId	ID of Collar Table.	30
orderDynamicCollarTableId	ID of Collar Table.	0 (undefined)
tradeStaticCollarTableId	ID of Collar Table.	0 (undefined)
tradeDynamicCollarTableId	ID of Collar Table.	0 (undefined)


```
{"header":{“length”:62,”msgType”:”CollarGroup”,"version":4096,"seqNum":325,"timestamp":1717902001728925994,"sourceTimestamp":1717902001728894893,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"id":17,"orderStaticCollarTableId":30,"orderDynamicCollarTableId":0,"tradeStaticCollarTableId":0,"tradeDynamicCollarTableId":0}
```


The Collar Group with id=17 binds only one collar table, Collar Table 30, into a complete collar definition.

6.3.3.2. Collar Tables

CollarTableEntry messages define individual Collar Tables. Those consist of a definition of deviations from a reference price. Collar tables are independent definitions not directly dependent on Instruments which they are meant to service. Each collar table is identified by its own collarTableId.

The table below explains the relevant fields in the CollarTableEntry message and examples below illustrate simple examples of collar definitions.

Field name	Description
collarTableId	The identifier of the Collar Table. Referred to in CollarGroup.
collarMode	OrderPriceCollar for Order Collars or TradePriceCollar for Trade Collars.
collarExpression	Either AbsoluteValue or Percentage. • If AbsoluteValue the fields collarValue, collarLowerBid, collarUpperBid, collarUpperAsk, and collarUpperAsk contain absolute currency values (e.g., USD). • If Percentage, the fields collarValue, collarLowerBid, collarUpperBid, collarUpperAsk, and collarUpperAsk contain percentages.
collarLowerBound	The lower bound of the reference price for which the collar defined in the message is applicable. Several CollarTableEntry may be used to define a "ladder" of collars. If collarLowerBound=0, the collar is applicable in the reference price interval from 0 to ∞, or the collarLowerBound of the following table.
collarValue	Field applicable to Trade price collars (collarMode=TradePriceCollar). Contains the allowed deviation of Trade prices from a reference price. Deviations expressed in absolute values or percentages. The format depends on collarExpression={AbsoluteValue, Percentage}. 0 for Order price collar (collarMode=OrderPriceCollar).
collarLowerBid	Fields applicable to Order price collars only (collarMode=OrderPriceCollar) contain limits on Bids and Asks for incoming orders. Deviations expressed in absolute values or percentages. The format depends on collarExpression={AbsoluteValue, Percentage}. 0 for Trade price collar (collarMode=TradePriceCollar).
collarUpperBid	
collarUpperAsk	
collarLowerAsk	

The table below illustrates collar calculations that are valid for each rung of the ladder defined by collarLowerBound and for Static and Dynamic collars. The calculations depend on the Reference Price (RefPrice) in force for a given Instrument. It is important to note that CollarTableEntry only defines collars in terms of deviations. OrderCollars and TradeCollars convey calculated collar ranges.

Collar Type	Computation of collar ranges for a given Reference Price
Order Price Collar	If collar defined in percentages (collarExpression=Percentage) RefPrice*(1 - collarLowerBid) ≤ Buy OrderPrice ≤ RefPrice*(1 + collarUpperBid) RefPrice*(1 - collarLowerAsk) ≤ Sell OrderPrice ≤ RefPrice*(1 + collarUpperAsk) If collar defined in absolute values (collarExpression=AbsoluteValue) RefPrice - collarLowerBid ≤ Buy OrderPrice ≤ RefPrice + collarUpperBid RefPrice - collarLowerAsk ≤ Sell OrderPrice ≤ RefPrice + collarUpperAsk
Trade Price Collar	If collar defined in percentages (collarExpression=Percentage) RefPrice*(1 - collarValue) ≤ Trade Price ≤ RefPrice*(1 + collarValue) If collar defined in absolute values (collarExpression=AbsoluteValue) RefPrice - collarValue ≤ Trade Price ≤ RefPrice + collarValue


```
{"header":{"length":96,"msgType":"CollarTableEntry","version":4096,"seqNum
":201,"timestamp":1717902001459992861,"sourceTimestamp":17179020014599557
0,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId"
:792,"streamId":0},"collarTableId":30,"collarMode":"OrderPriceCollar","col
```

```
larExpression":"AbsoluteValue","collarLowerBound":1000000,"collarValue":0,
"collarLowerBid":500000000,"collarUpperBid":600000000,"collarUpperAsk":400
000000,"collarLowerAsk":300000000}
{"header":{"length":96,"msgType":"CollarTableEntry","version":4096,"seqNum
":202,"timestamp":1717902001460168665,"sourceTimestamp":171790200146013226
4,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId"
:792,"streamId":0},"collarTableId":30,"collarMode":"OrderPriceCollar","col
larExpression":"Percentage","collarLowerBound":10000000000,"collarValue":0
,"collarLowerBid":900000000,"collarUpperBid":1000000000,"collarUpperAsk":8
00000000,"collarLowerAsk":700000000}
```


The CollarGroup with id=17 from the example above references orderStaticCollarTableId=30. This series of CollarTableEntry messages with collarTableId=30 define static collars for 2 price levels: from 1000000 to 10000000000 and from 10000000000 upwards. The first level of collars is defined in absolute value while the second is defined in percentages of reference price.

These messages are bound in the Collar Group message and that message is bound in turn to the Instrument message for PKN Orlen. In effect, PKN Orlen follows these collars during trading.

Let's decode these messages. The table below highlights the differences between the structure of Order and Trade collar definitions.

Field name	Message 221 Order Collar (seqNum=201)	Message 166 Trade Collar (seqNum=202)
collarTableId	30	30
collarMode	OrderPriceCollar	TradePriceCollar
collarExpression	AbsoluteValue	Percentage
collarLowerBound	1000000	10000000000
collarValue	0	0
collarLowerBid	500000000	900000000
collarUpperBid	600000000	1000000000
collarUpperAsk	400000000	800000000
collarLowerAsk	300000000	700000000

collarTableId=30 is an Order price collar expressed in Absolute Value. collarLowerBound=0 means the collar definition is applicable in the price interval from 1000000 to ∞. As this is an Order Price Collar, the field collarValue=0 (this is always the case for Order Price Collars). The collar absolute deviations are conveyed by the fields collarLowerBid, collarUpperBid, collarUpperAsk, and collarUpperAsk. These are absolute deviations from the reference price and represent currency values. The deviations are Ç50, Ç60, Ç40, and Ç30 from the relevant reference price.

collarTableId=30 is a Trade price collar expressed in Percentages. collarLowerBound=10000000000 means the collar definition is applicable in the price interval from 10000000000 to ∞. If this were a Trade Price Collar the fields collarLowerBid, collarUpperBid, collarUpperAsk, and collarUpperAsk would be left undefined and set to 0.

6.3.4. FURTHER EXAMPLES

As the nature of collars is complex, we include some further examples to clarify the concepts involved.

6.3.4.1. Percentage vs Absolute Value Expressions

This table shows collar definitions in Absolute Values and Percentages. The only difference is the collarExpression field and the change in value types in relevant message fields.

Field name	Order Collar		Trade Collar	
	Percentages	Absolute Value	Percentages	Absolute Value
collarTableId	17	18	23	24
collarMode	OrderPriceCollar	OrderPriceCollar	TradePriceCollar	TradePriceCollar
collarExpression	Percentage	AbsoluteValue	Percentage	AbsoluteValue
collarLowerBound	0	0	0	0
collarValue	0	0	10%	₵100
collarLowerBid	5%	₵50	0	0
collarUpperBid	6%	₵60	0	0
collarUpperAsk	4%	₵40	0	0
collarLowerAsk	3%	₵30	0	0

6.3.4.2. Collar Ranges

Collars can be layered in the same manner as tick tables. Consequently, collar ranges can vary in relation to the value of the reference price. For example, when the reference price is low, collars can be defined to be narrow and broad when the reference price increases. This capability is enabled by the collarTableId and collarLowerBound fields in CollarTableEntry. Collar ladders can be applied to order and trade collars (via collarMode: OrderPriceCollar or TradePriceCollar).

Collar ladders must start at 0 – this is expressed by collarLowerBound=0. In the example below, to define a 3-level collar, 3 CollarTableEntry messages with the same collarTableId are broadcast. They differ in the collarLowerBound value and collar parameters.


```
{"header":{"length":93,"version":4096,"msgType":"CollarTableEntry","seqNum
":208,"timestamp":"2023-02-10T02:32:23.549452Z","sourceTimestamp":"2023-
02-
10T02:32:23.547959Z","isEncrypted":false,"encryptionOffset":0,"encryptionK
eyId":0},"collarTableId":18,"collarMode":"OrderPriceCollar","collarExpress
ion":"Percentage","collarLowerBound":0,"collarValue":0,"collarLowerBid":90
0000000,"collarUpperBid":1000000000,"collarUpperAsk":800000000,"collarLowe
rAsk":700000000}
{"header":{"length":93,"version":4096,"msgType":"CollarTableEntry","seqNum
":209,"timestamp":"2023-02-10T02:32:23.549373Z","sourceTimestamp":"2023-
02-
10T02:32:23.547573Z","isEncrypted":false,"encryptionOffset":0,"encryptionK
eyId":0},"collarTableId":18,"collarMode":"OrderPriceCollar","collarExpress
ion":"AbsoluteValue","collarLowerBound":1000000,"collarValue":0,"collarLow
erBid":500000000,"collarUpperBid":600000000,"collarUpperAsk":400000000,"co
llarLowerAsk":300000000}
{"header":{"length":93,"version":4096,"msgType":"CollarTableEntry","seqNum
":210,"timestamp":"2023-02-10T02:32:23.549452Z","sourceTimestamp":"2023-
02-
```

```
10T02:32:23.547959Z","isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"collarTableId":18,"collarMode":"OrderPriceCollar","collarExpression":"Percentage","collarLowerBound":10000000000,"collarValue":0,"collarLowerBid":900000000,"collarUpperBid":1000000000,"collarUpperAsk":800000000,"collarLowerAsk":700000000}
```


The messages (unrelated to PKN Orlen) with collarTableId=18 imply that the Collar table has 3 different levels. The 3 levels are specified by collarLowerBound={0, 10, 100}.

Let's break down this series of messages.

Field name	Message 208 (seqNum=208)	Message 209 (seqNum=209)	Message 210 (seqNum=210)
collarTableId	18	18	18
collarMode	OrderPriceCollar	OrderPriceCollar	OrderPriceCollar
collarExpression	Percentage	AbsoluteValue	Percentage
collarLowerBound	0	10	100
collarValue	0	0	0
collarLowerBid	9%	Ç50	9%
collarUpperBid	10%	Ç60	10%
collarUpperAsk	8%	Ç40	8%
collarLowerAsk	7%	Ç30	7%

The three messages each refer to collarTableId=18. This signals that they form a collar ladder. They also define Order Price Collars. Message 208 states that collarLowerBound=0, message 209 features collarLowerBound=10 and message 210, collarLowerBound=100. This series of messages defines a 3-level collar. The final spread of the collars clearly depends on the value of the reference price - when the price breaks through $10 or $100, the range of the collars changes. The calculated collar values in currency units are communicated by OrderCollars and TradeCollars as the reference price varies relative to the collarLowerBounds defined.

Collar level	Value of Reference Price	Collar Calculation reported by OrderCollars and TradeCollars
1	0 < RefPrice < $10	Collar is defined in percentages (collarExpression=Percentage) RefPrice*(1 - 9%) ≤ Buy OrderPrice ≤ RefPrice*(1 + 10%) RefPrice*(1 - 7%) ≤ Sell OrderPrice ≤ RefPrice*(1 + 8%)
2	$10 ≤ RefPrice < $100	Collar is defined in absolute values (collarExpression=AbsoluteValue) RefPrice - Ç50≤ Buy OrderPrice ≤ RefPrice + Ç60 RefPrice - Ç30 ≤ Sell OrderPrice ≤ RefPrice + Ç40
3	$100 ≤ RefPrice < ∞	Collar is defined in percentages (collarExpression=Percentage) RefPrice*(1 - 9%) ≤ Buy OrderPrice ≤ RefPrice*(1 + 10%) RefPrice*(1 - 7%) ≤ Sell OrderPrice ≤ RefPrice*(1 + 8%)

6.4. Pricing Messages

6.4.1. ACCRUEDINTERESTTABLEENTRY

AccruedInterestTableEntry shows the accrued interest for a financial product (ProductID). A financial product is the archetype of an instrument (identified by InstrumentID). An instrument (InstrumentID) refers to a financial product (ProductID) in its Instrument definition message.

Parametrization sends out interest tables for the current day and up to 30 days in the future. In total, 31 AccruedInterestTableEntry messages are sent for each financial product at one time (one for the current day t, and 30 for days t+1 to t+30).


```
{"header":{"length":58,"msgType":"AccruedInterestTableEntry","version":4096,"seqNum":417,"timestamp":1717902001736581880,"sourceTimestamp":1717902001736503778,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"productId":954,"accruedInterestTableEntryDate":20240609,"accruedInterestTableEntryValue":2842000000}
```


A sample AccruedInterestTableEntry message.

6.4.2. INDEXATIONTABLEENTRY

IndexationTableEntry behaves similarly to AccruedInterestTableEntry. It shows a bond instrument's indexation coefficient. An instrument (InstrumentID) refers to a financial product (ProductID) in its Instrument definition message.

7. TRADING MESSAGES

7.1. AUCTIONS

7.1.1. AUCTIONUPDATE

AuctionUpdate is sent during an auction whenever IMP/IMV values change and may include the best buy and sell levels. As for an uncrossed book, only the best buy and sell levels are sent.

7.1.2. AUCTIONSUMMARY

Disseminated at the end of an auction after uncrossing. Contains the final auction price and quantity. The type of auction is specified by auctionType.

As for an uncrossed book, AuctionSummary sends zeros in the price and quantity fields.

7.1.3. ORDERBOOKEVENT

OrderBookEvent is a multipurpose message. Among other things it instructs Participants to clear the order book in anticipation of an upcoming auction. Its particular use follows auction uncrossing. It services the following events:

eventType	
Clear	Clear order book
ResendStart	Order book resend start (resend will begin shortly)
ResendEnd	Order book resend end (resend is done)

OrderBookEvent is not disseminated on Snapshot for recovery because Snapshot contains the current state of the Order Book and not historical data.

7.1.4. SEQUENCE OF MESSAGES IN AUCTIONS

Auctions are announced and preceded by a special sequence of messages and InstrumentStatusChange is only one of the messages involved in the process. The following table outlines the sequence of messages leading up to an auction:

Sequence	Message	Category	Note
1	InstrumentStatusChange	Message sequence to set up the Auction.	Signals the change of Market Phase to Auction.
2	OrderBookEvent		OrderBookEvent with eventType=Clear signals the need to clear the book in preparation for an auction. Auction starts and participants enter orders on the trading port.
3	OrderCollars/TradeCollars		OrderCollars/TradeCollars messages arrive in series to define dynamic or static collars.
4	AuctionUpdate	Auction Call	Informs participants on the state of the Auction Call.
5	InstrumentStatusChange	Change of Market Phase	Auction enters the Uncrossing phase: tradingPhaseType=Uncrossing

Sequence	Message	Category	Note
6	AuctionSummary	Auction Uncrossing	Auction price publication after Uncrossing.
7	OrderBookEvent	Post-auction book rebuilding process. The Resend process takes place only when the subsequent phase is a Continuous Trading phase.	OrderBookEvent with eventType= ResendStart This message functions as a signal to prepare rebuilding the order book and is followed by OrderAdd messages.
8	OrderBookEvent		OrderBookEvent with eventType= ResendEnd Orders related to rebuilding the order book have been sent. Trading can commence.
9	InstrumentStatusChange	Change of Market Phase	Switching into another Market Phase after Uncrossing (Continuous Trading or Auction).
10	Trade	Trade	Enriched Trade reports from the auction follow. Please note that no Execution reports (OrderExecute) are generated for auctions.

The messages surrounding an Auction for instrumentId=7 are reported in the table below. It is assumed that TradingPhaseScheduleEntry with tradingPhaseId=4 exists which identifies a tradingPhaseType="AuctionOpening" with a type auctionType="AuctionOpening" and that it is attached to an Instrument with InstrumentId=7.

Message	Message Contents
InstrumentStatusChange	{"header":{“length”:53,”msgType”:”InstrumentStatusChange”,"version":4096,"seqNum":26050,"timestamp":1705476600008972078,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"tradingPhaseId":4,"status":"Active","stressedMarket":false}
OrderBookEvent	{"header":{“length”:47,”msgType”:”OrderBookEvent”,"version":4096,"seqNum":26056,"timestamp":1705476600009119282,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"eventType":"Clear"}
OrderCollars	{"header":{“length”:87,”msgType”:”OrderCollars”,"version":4096,"seqNum":26054,"timestamp":1705476600009042780,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"collarType":"Static","price":261000000,"lowerBid":130600000,"upperBid":391500000,"lowerAsk":130600000,"upperAsk":391500000}
TradeCollars	{"header":{“length”:71,”msgType”:”TradeCollars”,"version":4096,"seqNum":26052,"timestamp":1705476600008994278,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"collarType":"Dynamic","price":261000000,"lower":245500000,"upper":276500000} {"header":{“length”:71,”msgType”:”TradeCollars”,"version":4096,"seqNum":26053,"timestamp":1705476600009042780,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"collarType":"Static","price":261000000,"lower":209000000,"upper":313000000}

Message	Message Contents
AuctionUpdate	{"header":{“length”:110,”msgType”:”AuctionUpdate”,"version":4096,"seqNum":26057,"timestamp":1705476600009132382,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"indicativeMatchingPrice":0,"indicativeMatchingVolume":0,"totalSellQuantityAtImp":0,"totalBuyQuantityAtImp":0,"bestSellLevel":0,"bestSellLevelQuantity":0,"bestBuyLevel":0,"bestBuyLevelQuantity":0}
AuctionSummary	{"header":{“length”:63,”msgType”:”AuctionSummary”,"version":4096,"seqNum":41080,"timestamp":1705478400003537367,"sourceTimestamp":1705478400003460765,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"price":0,"quantity":0,"auctionType":"AuctionOpening"}
OrderBookEvent	{"header":{“length”:47,”msgType”:”OrderBookEvent”,"version":4096,"seqNum":41081,"timestamp":1705478400003655271,"sourceTimestamp":1705478400003460765,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"eventType":"ResendStart"}
Book rebuilding process	OrderAdd messages involved in book rebuilding come out between the OrderBookEvent message with "eventType":"ResendStart" and OrderBookEvent message with "eventType":"ResendEnd".
OrderBookEvent	{"header":{“length”:47,”msgType”:”OrderBookEvent”,"version":4096,"seqNum":41082,"timestamp":1705478400003728273,"sourceTimestamp":1705478400003460765,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"eventType":"ResendEnd"}
InstrumentStatusChange	{"header":{“length”:53,”msgType”:”InstrumentStatusChange”,"version":4096,"seqNum":41083,"timestamp":1705478400003545268,"sourceTimestamp":1705478400003460765,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"tradingPhaseId":5,"status":"Active","stressedMarket":false}

7.1.5. AFTER AN AUCTION

To help participants rebuild the order book after an auction, Market Data publishes the state of the order book at the beginning of each market phase of continuous type. We distinguish several types of continuous phases that may follow an auction: ContinuousPriceTime, ContinuousRefPriceTime and ContinuousLastAuctionTime. The resending rules for each of these situations are explained next.

7.1.5.1. ContinuousPriceTime

Post-auction, buy orders are published first, followed by sell orders:

1. Buy orders are published from the highest to the lowest price level. Each price level is ordered in time-priority starting with orders with the highest to those with lowest time-priority.
2. Sell orders are published from the lowest to the highest price level. Each price level is ordered in time-priority starting with orders with the highest to those with lowest time-priority.

7.1.5.2. ContinuousRefPriceTime

Post-auction, buy orders are published first, followed by sell orders.

1. Buy orders are published first.
    a) Buy orders with prices ≥ RefPrice are arranged in time-priority starting with orders with the highest to those with lowest time-priority. In this case, price levels are ignored and only time-priority is taken into account.
    b) Buy orders with prices < RefPrice are published from the highest to the lowest price level. Each price level is ordered in time-priority starting with orders with the highest to those with lowest time-priority.
2. Sell orders are published second.
    a) Sell orders with prices ≤ RefPrice are arranged in time-priority starting with orders with the highest to those with lowest time-priority. In this case, price levels are ignored and only time-priority is taken into account.
    b) Sell orders with prices > RefPrice are published from the lowest to the highest price level. Each price level is ordered in time-priority starting with orders with the highest to those with lowest time-priority.

7.1.5.3. ContinuousLastAuctionTime

Following a closing auction (AuctionClosing), the order book is published in time-priority.

1. Buy orders are published first.
    a) Buy orders with prices = RefPrice are arranged in time-priority starting with orders with the highest time-priority to those with lowest time-priority. In this case, price levels are ignored and only time-priority is taken into account.
    b) Buy orders with prices < RefPrice are published from the highest to the lowest price level. Each price level is ordered in time-priority starting with orders with the highest to those with lowest time-priority.
2. Sell orders are published second.
    a) Sell orders with prices = RefPrice are arranged in time-priority starting with orders with the highest to those with lowest time-priority. In this case, price levels are ignored and only time-priority is taken into account.
    b) Sell orders with prices > RefPrice are published from the lowest to the highest price level. Each price level is ordered in time-priority starting with orders with the highest to those with lowest time-priority.

7.1.6. AUCTION TIME SUPPRESSION

During Auctions, orders inbound on the trading port are suppressed on Market Data. Specifically, orders will be suppressed starting from the InstrumentStatusChange message which signals the start of the auction. Orders will restart following the end of an Auction phase which is communicated through a suitable InstrumentStatusChange message. The list of messages not published on Market Data during Auctions is:

Messages Suppressed	Note
OrderAdd	These messages are not published during auctions in line with the auction regulations.

Messages Suppressed	Note
OrderCancel **OrderModify**	
OrderExecute **Trade**	These messages are not published during auctions. OrderExecute is not broadcast during auction calls or uncrossing. Trade messages are published as normal after auction uncrossing to communicate the outcomes of the auction. The message sequence is: AuctionSummary, InstrumentStatusChange out of the Auction and Trade messages.

7.1.7. EXCEPTIONS IN THE AUCTION PROCESS

The sequence of Market Data messages differs in situations when Extended Volatility Auctions are initiated.

In the trading context, there may arise a phase change scenario such as,

Continuous trading → Volatility Auction → Extended volatility Auction → Continuous trading.

The table below illustrates the sequence of messages in this particular scenario.

tradingPhaseType	Message	Note
ContinuousPriceTime	An InstrumentStatusChange was sent at the start of continuous trading. It is not repeated here.	Continuous trading is taking place. In a moment, upon breach of dynamic trade collars, the system enters the Dynamic volatility Auction. The auction is *unscheduled*.
AuctionVolatilityDynamic	InstrumentStatusChange	Upon trade collars breach, the system enters the volatility auction.
AuctionVolatilityDynamic	AuctionUpdate	AuctionUpdates are published as the auction progresses.
Uncrossing	InstrumentStatusChange	The volatility auction has predetermined duration. After this time elapses, an attempt to uncross takes place and an instrument status change is sent out.
Uncrossing	AuctionUpdate	Late auction updates may propagate past the announcement of the uncrossing. These auction updates relate to the dynamic volatility auction.
	(AuctionSummary is not published)	Since the volatility auction failed, and transitined to an extended auction no auction summary is available. The AuctionSummary is not published.
AuctionExtendedVolatility Dynamic	InstrumentStatusChange	The system determines the state of the book and calls for an extended auction. Past this point, only auction updates related to the extended volatility auction come out.
AuctionExtendedVolatility Dynamic	AuctionUpdate	Auction updates relevant to the extended auction are published.
Uncrossing	InstrumentStatusChange	A successful attempt is made to uncross..
Uncrossing	AuctionSummary	The auction result is published.
Uncrossing	OrderBookEvent	Post-auction book rebuilding process. OrderBookEvent with eventType= ResendStart

tradingPhaseType	Message	Note
Uncrossing	OrderBookEvent	OrderBookEvent with eventType= ResendEnd
ContinuousPriceTime	InstrumentStatusChange	Switching into Continuous trading. Enriched Trade (Trade messages) reports from the auction follow.

The sequence of steps may change depending on circumstances.

In general, the AuctionSummary message will not be published in cases when the auction uncrossing fails. This may happen in situations when

- Normal or scheduled auction → volatility auction,
- Volatility auction → extended volatility auction.

In such cases, there will be no AuctionSummary at the end of the first leg of the auction sequence. An AuctionSummary will be published following the successful uncrossing in the last auction.

For example, in the case of extended volatility auction cascades, that is when an extended volatility auction fails and transitions into another extended volatility auction, there will be no intervening AuctionSummaries. An AuctionSummary will only appear at the end of the last extended auction in the case of a successful uncrossing.

7.1.8. EXAMPLES

7.1.8.1. AuctionUpdate

AuctionUpdate sends updates in the IMP/IMV or BBO during an Auction.


```
{"header":{“length”:110,”msgType”:”AuctionUpdate”,"version":4096,"seqNum":26057,"timestamp":1705476600009132382,"sourceTimestamp":1705476600008820474,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":7,"indicativeMatchingPrice":0,"indicativeMatchingVolume":0,"totalSellQuantityAtImp":0,"totalBuyQuantityAtImp":0,"bestSellLevel":0,"bestSellLevelQuantity":0,"bestBuyLevel":0,"bestBuyLevelQuantity":0}
```


A sample update message during the auction call for instrumentId=7.

The behavior of the AuctionUpdate message is illustrated below:

Field name	Description	Behavior
instrumentId	A reference to the instrument.	
indicativeMatchingPrice	Indicative Matching Price and Indicative Matching Volume. Disseminated whenever a change in IMP/IMV occurs.	These fields are only filled in the case of a Crossed Book.
indicativeMatchingVolume		
totalSellQuantityAtImp	The buy and sell quantity.	
totalBuyQuantityAtImp		
bestSellLevel		

Field name	Description	Behavior
bestSellLevelQuantity	BBO Level 1 components.	In case there is only one order on the buy or sell side and that order is a Market Order, the following values are provided in the bestBuyLevel / bestSellLevel fields: • bestBuyLevel:i64::MAX =9223372036854775807 (hexadecimal 0x7FFF FFFF FFFF FFFF) • bestSellLevel:i64::MIN = -9223372036854775808 (hexadecimal 0x8000 0000 0000 0000) In case of an Uncrossed Book, only the first BBO level is broadcast.
bestBuyLevel		
bestBuyLevelQuantity		

7.1.8.2. AuctionSummary

AuctionSummary is sent at the conclusion of an Auction and provides the results of the Auction Uncrossing.


```
{"header":{“length”:63,”msgType”:”AuctionSummary”,"version":4096,"seqNum":41070,"timestamp":1705478400003252460,"sourceTimestamp":1705478400003121256,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":4,"price":0,"quantity":0,"auctionType":"AuctionOpening"}
```


AuctionSummary is sent at the end of an auctionType=AuctionOpening, communicates the price and quantity reached during the auction for instrumentId=4.

7.2. Full Depth Order Messages

7.2.1. ORDERID AND PUBLICORDERID

GPW WATS uses the publicOrderId attribute to identify orders on Market Data. A publicOrderId is assigned to every order entered into System.

GPW WATS assigns each order with two distinct attributes: orderId and publicOrderId. The publicOrderId identifies the order on Market Data while the orderId remains private for each trading party.

A trading party obtains these identifiers upon the successful submission of an order on the trading port by means of response messages. On the Native trading port an OrderAddResponse message will contain the orderId and publicOrderId corresponding to an order successfully submitted by the OrderAdd message. The FIX trading port follows a similar logic.

The Market Data publicOrderIds are reset daily and start each day with the value 1, each new order being assigned a sequentially increasing value. publicOrderIds are unchangeable during the trading day and are only valid within each trading session.

The Market Data publicOrderId cannot be used to modify or cancel an order. To modify or cancel an order, the orderId from the OrderAddResponse message needs to be employed.

7.2.2. ORDERADD

The message specifies the side, instrument (InstrumentId), (limit) price and quantity. In the Hybrid Market model, a Market maker quote will be identified by mmQuote=TRUE.

The publicOrderId referenced within the message corresponds to the publicOrderId received in the OrderAddResponse message in response to an OrderAdd on the Native or in the Execution Report message in response to a New Order Single on the FIX trading ports.

7.2.3. ORDERCANCEL

The publicOrderId identifies the order on the Market Data stream.

7.2.4. ORDERMODIFY

The Price and Quantity fields represent the new values for the order. The PriorityFlag indicates whether the order retained or lost priority and is set following time-priority rules.

For further information about the impact of modifications on order priority in the CLOB, please refer to the *GPW WATS 1.01 Trading System* document.

7.2.5. ORDEREXECUTE

OrderExecutes specify the execution price and quantity (executionQuantity). OrderExecute generates a report for the aggressive side of the transaction. The publicOrderId refers to an OrderAdd message.

The Quantity field represents the quantity of the order left over in the Order Book. If Quantity=0, the order has been fully filled and can be removed from the Order Book.

The tradeId in both OrderExecute messages is the same and refers to a unique TradeID in the Market Data Trade message.

OrderExecute messages are not published on the BBO Stream.

7.2.6. EXAMPLES

7.2.6.1. OrderAdd

OrderAdd informs that a limit order was added to the order book. It is sent immediately upon the entry of the order into the order book. The publicOrderId uniquely identifies the order for that given trading day.


```
{"header":{"length":72,"msgType":"OrderAdd","version":4096,"seqNum":50572,
"timestamp":"2024-06-10 07:51:01.930993741","sourceTimestamp":"2024-06-10
07:51:01.930826537","isEncrypted":false,"encryptionOffset":0,"encryptionKe
yId":0,"sessionId":793,"streamId":0},"publicOrderId":1,"instrumentId":4669
,"side":"Buy","price":10000000000,"quantity":100,"mmQuote":false}
```


A buy order for Instrument 4669 PKN Orlen at price 100 and quantity 100. The order is identified by publicOrderId=1.

7.2.6.2. OrderModify and OrderCancel

OrderModify and OrderCancel are related to OrderAdd. They refer to previous orders by means of publicOrderId and InstrumentIDs. OrderModify informs that an order with publicOrderId was modified and provides the new price and quantity. In addition, it confirms if the order has lost or retained priority. OrderCancel reports that an order was deleted from the order book. The message refers to the deleted order through publicOrderId.


```
{"header":{“length”:72,"msgType":"OrderModify","version":4096,"seqNum":298
84,"timestamp":1699289311630633951,"sourceTimestamp":1699289311630280464,"
isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":79
```

```
3,"streamId":0},"instrumentId":4672,"publicOrderId":2,"price":10000000000,
"quantity":2000,"priorityFlag":"Lost"}
```


The price for order number 2 was modified. The order lost priority. The new price, quantity and priority flag are published.


```
{"header":{“length”:55,"msgType":"OrderCancel","version":4096,"seqNum":505
74,"timestamp":"2024-06-10 07:51:44.107256946","sourceTimestamp":"2024-06-
1007:51:44.107058141","isEncrypted":false,"encryptionOffset":0,"encryption
KeyId":0,"sessionId":793,"streamId":0},"instrumentId":4669,"publicOrderId"
:1}
```


The order for PKN Orlen with publicOrderId=1 which was entered in the example above was deleted from the order book.

7.2.6.3. OrderExecute and Trade

OrderExecute is sent when a passive order executes. OrderExecutes are not produced for aggressive orders. The message refers to an order by the publicOrderId field. The execution may be full, in which case quantity=0, or partial. The ID of the related Trade is provided by tradeId.

OrderExecute is published immediately upon a match. Trade differs from OrderExecute in that it is the official record of the transaction and appears with a slight delay due to its size and contents.


```
{"header":{"length":83,"msgType":"OrderExecute","version":2089,"seqNum":81
603,"timestamp":"2026-08-24 09:25:50.345035480","sourceTimestamp":"2026-
08-24
09:25:50.337826637","isEncrypted":false,"encryptionOffset":0,"encryptionKe
yId":0,"sessionId":1785,"streamId":0},"quantity":0,"instrumentId":4669,"pu
blicOrderId":11,"tradeId":1,"executionPrice":13500000000,"executionQuantit
y":11,"side":"Buy"}
```


The OrderExecute messages inform of an execution in PKN Orlen shares, instrumentID=4669. Order with publicOrderId=11 executed fully (“remaining” quantity=0) at a price of 135. The trade will be reported by the Trade message with tradeId=tradeId=1.


```
{"header":{"length":220,"msgType":"Trade","version":2089,"seqNum":81607,"t
imestamp":"2026-08-24 09:27:23.531524086","sourceTimestamp":"2026-08-24
09:27:23.528743748","isEncrypted":false,"encryptionOffset":0,"encryptionKe
yId":0,"sessionId":1785,"streamId":0},"instrumentId":4669,"settlementValue
":148500,"settlementDate":20260826,"tradingTimestamp":"2026-08-24
09:25:50.337826637","publicProductIdentification":{"productIdentification"
:"PLPKN0000018","productIdentificationType":"ISIN"},"price":13500000000,"p
riceCurrency":"PLN","priceNotation":"Price","quantity":11,"nominalValue":1
48500,"nominalCurrency":"PLN","mic":"XWAR","publicationTimestamp":"2026-
08-24
09:25:50.337826637","tradeId":5,"tradeToBeCleared":true,"pctChange":100000
00000,"vwap":13500000000,"noTrades":1,"totalVolume":11,"totalValue":148500
,"openingPrice":13500000000,"maxPrice":13500000000,"minPrice":13500000000,
"mmtMarketMechanism":"CentralLimitOrderBook","mmtTradingMode":"ContinuousT
rading","mmtTransactionCategory":"None","mmtNegotiationIndicator":"NoNegot
```

```
iatedTrade","mmtAgencyCrossTradeIndicator":"NoAgencyCrossTrade","mmtModificationIndicator":"TradeCancellation","mmtBenchmarkReferencePriceIndicator":"NoBenchmarkorReferencePriceTrade","mmtSpecialDividendIndicator":"NoSpecialDividendTrade","mmtOffBookAutomatedIndicator":"UnspecifiedOrNotApply","mmtOrdinaryTradeIndicator":"PlainVanillaTrade","mmtAlgorithmicIndicator":"NoAlgorithmicTrade","mmtPostTradeDeferralReason":"ImmediatePublication","mmtPostTradeDeferralType":"NotApplicable"}
```


A trade tradeId=1 in PKN Orlen was made at a price of 135. This trade is related to the previous OrderExecute message which references it with tradeId=tradeId=1,

7.2.6.4. OrderBookEvent

OrderBookEvent can signal the need to clear an Instrument's order book for an upcoming Auction.


```
{"header":{"length":47,"msgType":"OrderBookEvent","version":2089,"seqNum":81613,"timestamp":"2026-08-24 09:30:00.008197546","sourceTimestamp":"2026-08-24
09:30:00.000146283","isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":793,"streamId":0},"instrumentId":4669,"eventType":"Clear"}
```


This OrderBookEvent message instructs participants to clear the order book for an upcoming Auction in the Instrument with instrumentId=4669 PKN Orlen.

7.3. BBO Messages

7.3.1. PRICELEVELSNAPSHOT AND TOPPRICELEVEL UPDATE

PriceLevelSnapshot and TopPriceLevelUpdate are specific to the BBO dataset.

They contain the Best Bid and Offer data such as prices, quantities, and order counts for a security (InstrumentID) at any given time.

- PriceLevelSnapshot contains up to 5 BBO levels
- TopPriceLevelUpdate contains only the best bid and offer.
- *Note: TopPriceLevelUpdate is not currently used in GPW WATS*

The depth of the BBO is defined by the Market operator. By default, GPW WATS provides 5 BBO levels (parameter maxDepth). The data encrypted for each BBO level is price, quantity and OrderCount.

7.3.2. SEQUENCE OF MESSAGES ON BBO DURING AN AUCTION

The table below presents the general flow of BBO messages around auctions. At auction start, a PriceLevelSnapshot filled with zeros follows an InstrumentStatusChange but precedes AuctionUpdate messages. Inside auctions, PriceLevelSnapshots are muted and AuctionUpdate messages update BBO recipients on auction progress. AuctionSummary recaps auction results. PriceLevelSnapshot resume upon completion of auctions after rebuilding of the order book.

Sequence	Message	Category	Note
1	InstrumentStatusChange	Sequence of messages setting up the Auction.	Signals the change of Market Phase to Auction.
2	OrderCollars/TradeCollars		OrderCollars/TradeCollars define dynamic or static collars.
3	PriceLevelSnapshot		PriceLevelSnapshot with zeros (0) in all fields signals the start of the auction.
4	AuctionUpdate	Auction Call	Updates on the state of the Auction.
5	AuctionSummary	Auction Uncross	Auction price publication after Uncrossing.
6	PriceLevelSnapshot	Post-auction	PriceLevelSnapshots resume after the auction.
7	InstrumentStatusChange	Change of Market Phase	Switching into another Market Phase after Uncrossing (Continuous Trading or Auction). Enriched Trade (Trade messages) reports from the auction follow this message.

7.3.3. BBO MESSAGE SCHEDULE

Unlike Order Messages (OrderAdd, OrderCancel, and OrderModify), BBO messages are not disseminated in real time. The rules for broadcasting BBO messages are as follows:

- PriceLevelSnapshot is generated at regular time intervals measured in milliseconds. PriceLevelSnapshot contains all the changes to price levels which occurred since its last publication. The publication interval is currently set at 100ms and can be configured at System operator level. The recommended practice is to choose a delay of 50-100ms,
- PriceLevelSnapshot can be configured to send either the best bid and offer or 5 BBO.

7.3.4. EXAMPLES

7.3.4.1. PriceLevelSnapshot

The PriceLevelSnapshot communicates up to 5 BBO levels in the form of price, quantity and orderCount. The message also shows the bit and ask price levels at which the Market maker is present: mmBuyQuoteLevel and mmSellQuoteLevel.

The BBO depth is set by the maxDepth parameter, which in the following message is set to 2. PriceLevelSnapshot is sent out at regular intervals of 100ms.


```
{"header":{“length”:359,”msgType”:”PriceLevelSnapshot”,"seqNum":22538,"tim
estamp":"2023-03-28 11:34:13.439440733","sourceTimestamp":"2023-03-28
11:34:13.410688751","isEncrypted":false,"encryptionOffset":0,"encryptionKe
yId":0},"instrumentId":145,"maxDepth":2," mmBuyQuoteLevel":1,"
mmSellQuoteLevel":1,
"buy":[{"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"pric
e":10000000000,"quantity":7,"orderCount":1},{"isEncrypted":false,"encrypti
onOffset":0,"encryptionKeyId":0,"price":0,"quantity":0,"orderCount":0},{"i
sEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"price":0,"quan
tity":0,"orderCount":0},{"isEncrypted":false,"encryptionOffset":0,"encrypt
ionKeyId":0,"price":0,"quantity":0,"orderCount":0},{"isEncrypted":false,"e
ncryptionOffset":0,"encryptionKeyId":0,"price":0,"quantity":0,"orderCount"
:0},{"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"price":
0,"quantity":0,"orderCount":0},{"isEncrypted":false,"encryptionOffset":0,"
```

```
encryptionKeyId":0,"price":0,"quantity":0,"orderCount":0},{"isEncrypted":f
alse,"encryptionOffset":0,"encryptionKeyId":0,"price":0,"quantity":0,"orde
rCount":0},{"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"
price":0,"quantity":0,"orderCount":0},{"isEncrypted":false,"encryptionOffs
et":0,"encryptionKeyId":0,"price":0,"quantity":0,"orderCount":0}],"sell":[
{"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"price":0,"q
uantity":0,"orderCount":0},{"isEncrypted":false,"encryptionOffset":0,"encr
yptionKeyId":0,"price":0,"quantity":0,"orderCount":0},{"isEncrypted":false
,"encryptionOffset":0,"encryptionKeyId":0,"price":0,"quantity":0,"orderCou
nt":0},{"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"pric
e":0,"quantity":0,"orderCount":0},{"isEncrypted":false,"encryptionOffset":
0,"encryptionKeyId":0,"price":0,"quantity":0,"orderCount":0},{"isEncrypted
":false,"encryptionOffset":0,"encryptionKeyId":0,"price":0,"quantity":0,"o
rderCount":0},{"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":
0,"price":0,"quantity":0,"orderCount":0},{"isEncrypted":false,"encryptionO
ffset":0,"encryptionKeyId":0,"price":0,"quantity":0,"orderCount":0},{"isEn
crypted":false,"encryptionOffset":0,"encryptionKeyId":0,"price":0,"quantit
y":0,"orderCount":0},{"isEncrypted":false,"encryptionOffset":0,"encryption
KeyId":0,"price":0,"quantity":0,"orderCount":0}]}
```


Instrument 145, on 2023-03-28 at 11:34:13.439440733, featured only one set of prices on the buy side price=10000000000, quantity=7, orderCount=1. The Market maker quotes on the first price level.

7.4. Hybrid Market Model

The hybrid Market employs the standard set of Market data messages to convey information on the state of the hybrid Market. One should note that a Hybrid Market is defined on a hybrid instrument - Instrument with MarketModelType=Hybrid.

There are 3 types of Market Phases (apart from Monitoring and Closed Phases) relevant to the Hybrid Market as shown in the table below. A Hybrid trading schedule will be composed of the HybridPreTrade phase in combination with one of the Hybrid or HybridBuyOnly phases.

Message	Fields	Note
TradingPhaseScheduleEntry	tradingPhaseType	Hybrid – the standard hybrid Market model phase in which the Market Maker offers two sided quotes. HybridBuyOnly – the phase where the Market Maker only offers buy-side quotes. The Market Maker can switch between the Hybrid and HybridBuyOnly phases. HybridPreTrade – the pre-trade session of the hybrid model. Enables modification of transient orders.

The status of an instrument on the Hybrid Market is signaled by the InstrumentStatusChange message. An instrument is Active when the Market Maker maintains a full set of quotes. If for any reason, a full set of quotes is not provided, the instrument will change to a HybridNoQuotes state. When an instrument is in the HybridBuyOnly phase, the full set of quotes means that a valid buy quote is maintained by the Market maker.

Message	Fields	Note
InstrumentStatusChange	status	Active – the instrument is active and trading is possible.

		HybridNoQuotes – the status which indicates that no valid quotes are provided by the Market Maker.

7.4.1. MARKET MAKING AND ORDER MESSAGES

Market data broadcasts Market Maker engagement in full depth and BBO messages and the OrderAdd and PriceLevelSnapshot messages feature attributes specific to Market Maker activity.

Message	Market Maker specific tags	Description
OrderAdd	mmQuote	True if the order is a quote entered by a Market Maker. Only used if the instrument is trading in the hybrid Market model.
PriceLevelSnapshot	mmBuyQuoteLevel	The BBO buy and sell levels on which the Market Maker is actively quoting.
	mmSellQuoteLevel	

7.5. IPO/SPO MARKET MODEL

In the IPO/SPO Market model (hereinafter IPO), Market Data can operate in a hidden mode only.

Hidden mode

In the hidden mode Market Data publishes the following information:

- reference data such as instrument's reference data, tick tables, trading schedule entries, collars etc.

Information about orders, trades as well as any related information summarizing the activity on an instrument is not published.

In the IPO Market model, the instrument enters an IPO specific Market Phase scheduled by a TradingPhaseScheduleEntry with tradingPhaseType=IPO.

IPO phase can be divided into two IPO's separate sub-phases. During the first IPO phase Exchange Members having the appropriate rights can place buy orders. An appointed Exchange Member can place sell order.

Once the first IPO phase ends, the second IPO phase starts and during this phase sell order book is accessible for appointed Exchange Member only who can input or amend the details of their sell order if necessary. No operation is allowed on buy side during this phase.

At the end of this phase an order book is uncrossed and buy orders are executed against sell order following the appropriate allocation algorithm.

Unexecuted or partially executed orders are cancelled during the transition to the next Market Phase.

If the collection of orders lasts more than one trading day, only one IPO phase may be defined during this period in the trading schedule where only buy and sell orders may be entered and the phase itself does not end with an uncross mechanism.

7.6. Tender Offer Market Model

In the Tender Market model Market Data can operate in an open mode.

Open mode

In the open mode, Market Data publishes relevant information in a way similar to the CLOB Market model:

- reference data such as instrument reference data, tick tables, trading schedule entries, collars etc.
- orders,
- auction's summary,
- trades,
- instrument's summary.

In the Tender Market model (MarketModelType=TenderOffer), the instrument enters a Tender specific Market Phase scheduled by a TradingPhaseScheduleEntry with tradingPhaseType= TenderOffer.

Tender phase can be divided into two Tender's separate sub-phases. During the first Tender phase Exchange Members having the appropriate rights can place sell orders. An appointed Exchange Member can place buy order.

Once the first Tender phase ends, the second Tender phase starts and during this phase buy order book is accessible for appointed Exchange Member only who can input or amend the details of their buy order if necessary. No operation is allowed on sell side during this phase.

At the end of this phase an order book is uncrossed and sell orders are executed against buy order at fixed price. All the orders must be fully executed, no partial fills are expected.

Irrespective of the duration of this scheduled phase, if the volume of orders on the buy and sell side is equal, the Market Operator may force the shortening of this phase and the execution of orders will take place immediately.

If the collection of orders lasts more than one trading day, only one Tender phase may be defined during this period in the trading schedule where only sell and buy orders may be entered and the phase itself does not end with an uncross mechanism.

7.7. Status Messages

Status change messages signal important events in the state of the Market. They comprise of:

Message	Description
InstrumentStatusChange	Signals the change of Market Phase for a given Instrument. May indicate stressed Market conditions.
PriceUpdate	Communicates new reference price for an Instrument.
OrderCollars	Communicates calculated Order price collars for an Instrument.
TradeCollars	Communicates calculated Trade price collars for an Instrument.

7.7.1. INSTRUMENTSTATUSCHANGE

InstrumentStatusChange signals that an Instrument (InstrumentID) has entered a new Market Phase (tradingPhaseId).

A secondary role of the message is to communicate Stressed Market Conditions for particular instruments. This is accomplished by means of the stressedMarket=true/false flag.

InstrumentStatusChange refers to the Market Phase defined by the TradingPhaseScheduleEntry message via the tradingPhaseId field. TradingPhaseScheduleEntry is part of the Reference Data disseminated by the Snapshot service.

Instruments can be in the states described in the table below. Only one state is in force at any given time. The Market Operations team has the responsibility of setting and releasing instruments from states.

Name	Description
Active	Trading is allowed. Order placement and trades may take place as long as the Market Phase allows for such activity.
MarketOperationsSuspension	Manual suspension by Market Operations in case of an extraordinary event impacting trading.
OutsideCollarsStatic / OutsideCollarsDynamic	Activated automatically whenever an attempt is observed to conclude a trade outside the collars (order or trade collars, when applicable). Once the status is triggered, a Volatility Auction takes place.
RegulatorySuspension	Reserved for Financial Supervisory Authority decisions.
HybridKnockout **HybridKnockoutByIssuer**	Hybrid knockout (by issuer).
HybridNoQuotes	Statuses used in the hybrid Market model.

7.7.2. ORDERCOLLARS AND TRADECOLLARS

OrderCollars and TradeCollars communicate the calculated values of the order and trade price collars for a particular security (InstrumentId) based on the ranges defined in the CollarGroup message.

The "price" field represents the value of the reference price used to compute the collars while "collarType" describes whether the collar in question is static or dynamic.

Collar values are contained in the remaining parts of the message.

Message	Fields	Note
OrderCollars	lowerBid, upperBid lowerAsk, upperAsk	OrderCollars disseminates the lower and upper bid and ask limits for order entry
TradeCollars	lower upper	TradeCollars disseminates the lower and upper bounds within which trades can be made

The functioning of price collars is explained in more detail in the *GPW WATS 1.01 Trading System* document.

7.7.3. EXAMPLES

7.7.3.1. InstrumentStatusChange - Switching between Phases

When an instrument switches between scheduled (or unscheduled) phases, an InstrumentStatusChange message is broadcast on Market Data. In the example below, a ContinuousPriceTime is scheduled to take place. An InstrumentStatusChange will be sent on Market Data to signal the change of phases and at the relevant time instrument PKN Orlen which follows this schedule will switch to ContinuousPriceTime.


```
{"header":{“length”:82,”msgType”:”TradingPhaseScheduleEntry”,"version":4096,"seqNum":216,"timestamp":1717902001461993809,"sourceTimestamp":1717902001461957308,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"tradingScheduleId":2,"tradingPhaseId":14,"tradingPhaseSettlementCycle":2,"tradingPhaseMaxBlockSettlementCycle":0,"staticCollarVolatilityAuctionId":0,"dynamicCollarVolatilityAuctionId":0,"tradingPhaseStartTime":1717884600000000000,"tradingPhaseType":"ContinuousPriceTime","uncrossing":false,"extStaticCollarVolatilityAuctionId":0,"extDynamicCollarVolatilityAuctionId":0,"lastAuctionPhaseId":0}
```


The TradingPhaseScheduleEntry message defines a trading phase with tradingPhaseId=14. It is part of tradingScheduleId=2 referenced by the Instrument message for PKN Orlen InstrumentID=4669.

The relevant InstrumentStatusChange change message which signals the transition to ContinuousPriceTime for PKN Orlen.


```
{"header":{"length":53,"msgType":"InstrumentStatusChange","version":4096,"seqNum":344,"timestamp":1717902001731087046,"sourceTimestamp":1717902001730951043,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"instrumentId":4669,"tradingPhaseId":14,"tradingPhaseType":"ContinuousPriceTime","status":"Active","stressedMarket":false}
```


Instrument PKN Order InstrumentID=4669 enters market phase 14. The stressed market flag is off.

7.7.3.2. PriceUpdate


```
{"header":{“length”:73,”version":4096,"msgType":"PriceUpdate","seqNum":23299,"timestamp":"2023-02-10T12:09:59.191830Z","sourceTimestamp":"2023-02-10T12:09:59.188603Z","isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":155,"priceType":"ReferencePrice","price":10000000000}
```


This PriceUpdate reports that the value of the reference price for Instrument 155 (`instrumentId=155`) is `price=100`. It is a reference price, and not a midpoint price.

7.7.3.3. OrderCollars and TradeCollars

OrderCollars and TradeCollars communicate values of order and trade collars for a given Instrument based on the ranges defined in CollarTableEntry.

When an Instrument features both Dynamic and Static collars, 2 OrderCollars/TradeCollars messages appear on Market Data, one for each type. When only one collar type is configured, one OrderCollars/TradeCollars message (either static or dynamic) appears. Whether 1 or 2 OrderCollars/TradeCollars messages are sent depends on the configuration of the Instrument.

```
{"header":{“length”:87,”msgType”:”OrderCollars”,"version":4096,"seqNum":123894,"timestamp":1705507500206244876,"sourceTimestamp":1705507500206120973,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":1489,"collarType":"Static","price":595000000,"lowerBid":298000000,"upperBid":890000000,"lowerAsk":298000000,"upperAsk":890000000}
```


This OrderCollars reports that the value of the reference price for Instrument 1489 (instrumentId=1489) is price=595000000.
The type of collars in the message is Static (collarType=Static).
The range of Static Buy Order Collars in the currency of Instrument 1407 is [298000000, 890000000]
The range of Static Sell Order Collars in the currency of Instrument 1407 is [298000000, 890000000]
The collar values and are expressed in the instrument's trading currency.
This is only one of two possible OrderCollars messages for Instrument 1489. If configured with 2 collar types, collarType=Dynamic and collarType=Static, a second OrderCollars would also appear.


```
{“header”:{“length”:71,”msgType”:”TradeCollars”,"version":4096,"seqNum":5788,"timestamp":1705460401848694429,"sourceTimestamp":1705460401848434122,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0},"instrumentId":1489,"collarType":"Static","price":595000000,"lower":476000000,"upper":710000000}
```


TradeCollars reports that the value of the reference price for Instrument 1489 (instrumentId=1489) is price=595000000. TradeCollars sends the values of the collars, and not their deviations as in the case in CollarTableEntry.
The type of collars in the message is Static (collarType=Static).
The value of Static Trade Collars in the trading currency of Instrument 1489 is [476000000, 710000000]
These ranges are collar values and are expressed in a unit of the instrument's currency, here assumed PLN.
This is only one of two possible TradeCollars messages for Instrument 1489. If configured with 2 collar types, collarType=Dynamic and collarType=Static, a second TradeCollars would also appear.

7.7.4. TRADE

Trade is an enriched message which sends trade-related information after the order is processed. Trade messages include MIFID flags which follow the MMT specification. The TradeID attribute refers to the original OrderExecute message. Trade messages are also published on the BBO Stream.


```
{"header":{"length":220,"msgType":"Trade","version":4096,"seqNum":50589,"timestamp":"2024-06-10 08:33:00.779987344","sourceTimestamp":"2024-06-10 08:33:00.779908042","isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":793,"streamId":0},"instrumentId":4669,"settlementValue":148500,"settlementDate":20240612,"tradingTimestamp":"2024-06-10 08:33:00.779484731","publicProductIdentification":{"productIdentification":"PLPKN0000018","productIdentificationType":"ISIN"},"price":13500000000,"priceCurrency":"PLN","priceNotation":"Price","quantity":11,"nominalValue":148500,"nominalCurrency":"PLN","mic":"XWAR","publicationTimestamp":"2024-06-10 08:33:00.779484731","tradeId":1,"tradeToBeCleared":true,"pctChange":10000000000,"vwap":13500000000,"noTrades":1,"totalVolume":11,"totalValue":148500,"openingPrice":13500000000,"maxPrice":13500000000,"minPrice":13500000000,"mmtMarketMechanism":"CentralLimitOrderBook","mmtTradingMode":"ContinuousTrading","mmtTransactionCategory":"None","mmtNegotiationIndicator":"NoNegotiatedTrade","mmtAgencyCrossTradeIndicator":"NoAgencyCrossTrade","mmtModifi
```

cationIndicator":"NewTrade","mmtBenchmarkReferencePriceIndicator":"NoBenchmarkorReferencePriceTrade","mmtSpecialDividendIndicator":"NoSpecialDividendTrade","mmtOffBookAutomatedIndicator":"UnspecifiedOrNotApply","mmtOrdinaryTradeIndicator":"PlainVanillaTrade","mmtAlgorithmicIndicator":"NoAlgorithmicTrade","mmtPostTradeDeferralReason":"ImmediatePublication","mmtPostTradeDeferralType":"NotApplicable"}

A trade tradeId=1 in PKN Orlen was made at a price of 135. This trade is related to the previous OrderExecute message which references it with tradeId=1,

The Trade message also includes a range of trade statistics related to the financial instrument. The statistics are updated with every trade of in financial instrument.

Statistic	Description
pctChange	Percentage change in price from last day's adjusted closing price.
VWAP	Volume-weighted average price.
noTrades	Total number of transactions in the financial instrument.
totalVolume	Total transaction volume.
totalValue	Total transaction value.
openingPrice	The price of the first trade on the current trading day.
maxPrice	Highest price of the instrument on the current trading day.
minPrice	Lowest price of the instrument on the current trading day.

7.7.4.1. Trade Cancellations

Market Data announces trade cancellations by means of the Trade message. In case of a Trade cancellation, the relevant flags are set in the following manner:

MD::Trade Message Attribute	Value
MD::Header::timestamp	The time the Trade message was published
MD::Header::sourceTimestamp	The time the cancellation occurred
mmtModificationIndicator	TradeCancellation
tradingTimestamp	Copied from the original Trade message
publicationTimestamp	Copied from the original Trade message

7.7.5. NEWS

The News message enables the Market operator to send messages addressed to a specific Market. The Market is identified by a MarketStructureID from the MarketStructure message. The body of the message consists of a title and a text string and can be up to 800 characters long.

7.7.6. POSITIONREPORT

The PositionReport message communicates the number of open positions in a financial product identified by an ISIN. The position may be communicated at the start of day, end of day or intraday. A reference to the last trade in the related derivative instrument is provided.

The PositionReport message may be sent out at predetermined time intervals as well as following each transaction on a derivative instrument. The publication frequency varies by source of the data.

The first publication of PositionReport takes place at session start and continues during trading.

7.8. Summary Messages

7.8.1. INSTRUMENTSUMMARY

InstrumentSummary presents various statistical information about an instrument's trading session (instrumentId). The message is sent at the end of each trading session after Market close. If a corporate action is planned for an instrument, the message is distributed once all necessary information is collected.

Name	Description
lastTradedPrice	Last Traded Price (LTP)
closingPrice	Closing Price (CP)
closingPriceType	Closing Price Type: • InitialPrice, • LTP, • LastACP, • FairValue, • DailySettlementPrice, • FinalSettlementPrice.
adjustedClosingPrice	Adjusted Closing Price (ACP).
adjustedClosingPriceReason	Reason for adjusting the Closing Price. Potential reasons: • NotApplicable • Dividend • IssueRight • Split • ReverseSplit • Bonus • SpinOff • TickSizeChange • OrderPurge • OtherReason
pctChange	Percentage change in price from last day's adjusted closing price.
VWAP	Volume-weighted average price
noTrades	Total number of transactions in the current trading day
totalVolume	Total transaction volume
totalValue	Total transaction value
openingPrice	The price of the first trade in the current trading day
maxPrice	Highest price of the instrument in the current trading day
minPrice	Lowest price of the instrument in the current trading day
isCorrection	True if current message is correction of previous message.

```
{"header":{"length":137,"msgType":"InstrumentSummary","version":4096,"seqNum":50710,"timestamp":"2024-06-10 13:00:00.003225276","sourceTimestamp":"2024-06-10 13:00:00.003200275","isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":793,"streamId":0},"instrumentId":4669,"lastTradedPrice":0,"closingPrice":2780000000,"closingPriceType":"InitialPrice","adjustedClosingPrice":2780000000,"adjustedClosingPriceReason":"NotApplicable","pctChange":0,"vwap":0,"noTrades":0,"totalVolume":0,"totalValue":0,"openingPrice":0,"maxPrice":0,"minPrice":0,"isCorrection":false}
```


An example InstrumentSummary message for instrumentID=4669 PKN Orlen.

7.8.2. PRODUCTSUMMARY

ProductSummary presents end-of-day summary statistics for a financial product.. It contains enriched summary information calculated from trading day statistics for the individual instruments stemming from that financial product which trade under that Market model. By default, it provides statistics for the CLOB, BLOCK, CLOSS and Hybrid instruments of a single financial product on a specific MIC code. It also provides several option-related statistics.

ProductSummary is encrypted only on the encrypted channel.

ProductSummary:market is an indicator of the market to which the instrument belongs. The following list presents the markets that are currently defined. This list is designed to be flexible and extensible and may change. For this reason, it is not devised as an enum in the Market Data protocol.

Value	Name	Description
1	Primary	Primary market.
2	Parallel	Parallel market.
3	NewConnectPriceDriven	New Connect Market (price driven market).
4	NewConnectOrderDriven	New Connect Market (order driven market).
5	CatalystRegulated	Catalyst Regulated Market.
6	CatalystAso	Catalyst ASO.
7	Other	Other market.

8. Data Redistribution

GPW WATS implements the redistribution of Market data from outside Markets and other sources such as indices.

GPW WATS' data redistribution framework enables the redistribution of trading data from any Market or trading platform. The flexible design enables the easy redistribution of key Market data from different trading venues without the need for costly customizations. The framework reuses GPW WATS parametrization and trading messages to set up the Market context and redistribute Market data and comprises messages specific to certain Markets, such as Treasury BondSpot Poland.

Please note that:

- trading is not allowed in redistributed instruments (instruments are "read-only"),
- attributes related to GPW WATS traded instruments are left empty in redistributed instrument messages.

The scope of messages used in data redistribution is as follows.

Type	Message	Description
Reference data	MarketStructure	The Market structure of the foreign-Market can be replicated using MarketStructure messages. MarketModelType=Redistribution will identify a redistributed instrument.
	Instrument	Each redistributed instrument is defined by an Instrument message attached to a specific redistributed Market segment. The instrument will contain an instrumentID and ISIN, MarketModelType=Redistribution will identify a redistributed instrument.
Trading Data	Trade	Trade in the redistributed instrument.
	PriceLevelSnapshot	BBO price levels and volumes. instrumentId refers to the redistributed instrument defined by an Instrument message.
Status	InstrumentStatusChange	The messages used to signal the market phase and status of the redistributed instrument.
Treasury BondSpot Poland	PriceUpdate	A message used to redistribute prices of fixed-income instruments traded on Treasury BondSpot Poland. The key fields in the message reflect this purpose: priceUpdateType: pricing method used to establish the security price priceYtmPresence: type of data being provided.
GPW Benchmark Market Indices	IndexParams IndexPortfolioEntry IndexSummary RealTimeIndex	Messages intended to service the distribution of index statistics.
Summary	InstrumentSummary	Summary statistics for every instrument broadcast on GPW WATS.

8.1. Reference Data for Markets and Instruments

GPW WATS parametrizes redistributed instruments and their Markets in the same way as natively traded

GPW WATS replicates the Market structure of the foreign Market in the same way as it defines the Market structure of a native Market using MarketStructure messages. Foreign instruments are replicated within

GPW WATS using Instrument messages. The MarketModelType flag of a redistributed Market or instrument will be set to Redistribution. For further details of parametrization please refer to the reference data section.

For redistributed instruments the Instrument message carries required and optional attributes. The table below presents the list of attributes for the Instrument message in the redistribution context.

Name	Required / Optional / NA	Description
instrumentId	R	ID of financial instrument.
productType	R	Type of the product.
productSubtype	R	Subtype of the product.
firstTradingDate	R	First trading day of the financial instrument.
lastTradingDate	O	Last trading day of the financial instrument.
productId	R	Reference to financial product definition.
currency	R	Trading currency (e.g. USD).
icebergMinValue	O	Minimum iceberg order value.
lotSize	O	Lot size for the instrument.
priceExpressionType	R	Price expression type for the financial instrument.
calendarId	R	ID of trading calendar for the financial instrument.
MarketStructureId	R	ID of the financial instrument's Market segment.
tickTableId	O	ID of the tick table used for the financial instrument.
referenceInstrumentId	O	Identifier of reference instrument.
collarGroupId	O	Collar group ID.
tradingScheduleId	O	Trading schedule ID.
description	R	Financial instrument code.
mic	R	Market structure's Market Identifier Code (MIC) as specified in ISO 10383.
nominalValue	O	Nominal value of the financial instrument.
nominalValueType	R	Type of the product's nominal value (no nominal, constant or unknown).
multiplier	O	Product value multiplier.
strikePrice	O	Product strike price.
strikePriceCurrency	O	Strike Price Currency
productIdentification	R	Product identification, e.g. ISIN number.
productIdentificationType	R	Type of product identification.
settlementCalendarId	O	Settlement calendar ID.
bondCouponType	O	Coupon type for a bond.
preTradeCheckMinPrice	O	Minimum order price for pre-trade check.
preTradeCheckMaxPrice	O	Maximum order price for pre-trade check.
preTradeCheckMinQuantity	O	Minimum volume for pre-trade check.
preTradeCheckMaxQuantity	O	Maximum volume for pre-trade check.

Name	Required / Optional / NA	Description
preTradeCheckMaxValue	O	Maximum value for pre-trade check.
preTradeCheckMinValue	O	Minimum value for pre-trade check.
liquidity	R	Liquidity flag, true - liquid, false - Illiquid.
MarketModelType	R	Market Model.
issuer	R	Name of the issuer.
issuerRegCountry	R	Identifier of the country of the registration.
underlyingInstrumentId	O	Identifier of the underlying instrument.
productCode	R	Unique code of the product.
cfi	O	CFI (Classification of Financial Instrument) code.
fisn	O	Product FISN (Financial Instrument Short Name) code.
issueSize	O	Size of the issue (depends on the issue size type)
nominalCurrency	O	Identifier of the nominal currency (ISO-4217).
usIndicator	O	Identifier of the us indicator.
expiryDate	O	Expiration date.
versionNumber	O	Version of the derivative. Version is incremented when corporate event adjusting underlying closing price occurs.
settlementType	O	Identifier of settlement type.
optionType	O	Identifier of option type.
exerciseType	O	Identifier of exercise style.
productName	O	Name of the product.
referencePrice	O	Reference price
status	R	Instrument status.
initialPhaseId	O	Id of starting phase for the instrument
thresholdMax	O	Upper threshold for leveraged instruments
thresholdMin	O	Lower threshold for leveraged instruments
isLeverage	O	Identifier of leverage instruments
accruedInterestValue	O	Accrued iterest value of the instrument
kid	O	KID.
kidIssueDate	O	KID issue date.
valueAtRisk	O	Value at risk.
externalUnderlyingId	O	External underlying ID.

8.2. STATUS MESSAGES: INSTRUMENTSTATUSCHANGE

For some redistributed data, the instrument's status on the original Market is disseminated using the **InstrumentStatusChange** message as is the case for natively traded instruments. The message will specify the type of Market Phase (tradingPhaseType) together with the instrument's status (status).

<table>
<tr><th>Name</th><th>Description</th></tr>
<tr><td>instrumentId</td><td>ID of financial instrument.</td></tr>
<tr><td>tradingPhaseId</td><td>Market Phase ID.</td></tr>
<tr><td>tradingPhaseType</td><td>Type of matching algorithm.
<table>
<tr><th>Name</th><th>Description</th></tr>
<tr><td>ContinuousPriceTime</td><td>Continuous price and time.</td></tr>
<tr><td>ContinuousRefPriceTime</td><td>Continuous time at reference price.</td></tr>
<tr><td>ContinuousLastAuctionTime</td><td>Trade at last.</td></tr>
<tr><td>AuctionOpening</td><td>Opening auction.</td></tr>
<tr><td>BlockExactMatching</td><td>Block transaction, exact matching.</td></tr>
<tr><td>NoTradingClosed</td><td>Trading disabled, no matching.</td></tr>
<tr><td>Uncrossing</td><td>Uncrossing.</td></tr>
<tr><td>AuctionClosing</td><td>Closing auction.</td></tr>
<tr><td>AuctionIntraday</td><td>Intraday auction.</td></tr>
<tr><td>AuctionVolatilityStatic</td><td>Volatility auction after static collars breach.</td></tr>
<tr><td>AuctionVolatilityDynamic</td><td>Volatility auction after dynamic collars breach.</td></tr>
<tr><td>AuctionExtendedVolatilityStatic</td><td>Additional volatility auction after static collar breach. It can be activated independently or as an extension of the previous AuctionVolatilityStatic phase.</td></tr>
<tr><td>AuctionExtendedVolatilityDynamic</td><td>Additional volatility auction after dynamic collar breach. It can be activated independently or as an extension of the previous AuctionVolatilityDynamic phase.</td></tr>
<tr><td>NoTradingEarlyMonitoring</td><td>Early Monitoring</td></tr>
<tr><td>NoTradingLateMonitoring</td><td>Late Monitoring</td></tr>
<tr><td>Hybrid</td><td>Hybrid normal trading</td></tr>
<tr><td>HybridBuyOnly</td><td>Hybrid buy only phase</td></tr>
<tr><td>HybridPreTrade</td><td>Hybrid pre trade phase</td></tr>
<tr><td>UnsuspensionAuction</td><td>A phase triggered whenever an instrument's changes its status from RegulatorySuspension or MarketOperationSuspension before entering into continuous phase.</td></tr>
<tr><td>Ipo</td><td>Initial public offering phase</td></tr>
<tr><td>TenderOffer</td><td>Tender offer phase</td></tr>
<tr><td>HybridPreTradeBuyOnly</td><td>Hybrid pre trade BuyOnly phase</td></tr>
<tr><td>Distribution</td><td>Instrument distribution phase</td></tr>
</table></td></tr>
<tr><td>status</td><td>Financial instrument status.
<table>
<tr><th>Name</th><th>Description</th></tr>
<tr><td>Active</td><td>Financial instrument is active.</td></tr>
<tr><td>Inactive</td><td>Financial instrument is inactive and is not trading.</td></tr>
<tr><td>MarketOperationsSuspension</td><td>Manual suspension by Market Operations.</td></tr>
</table></td></tr>
</table>

Name	Description	
	OutsideCollarsStatic	Outside static trade price collars.
	OutsideCollarsDynamic	Outside dynamic trade price collars.
	RegulatorySuspension	Regulatory suspension
	HybridNoQuotes	Hybrid no valid quotes.
	HybridKnockout	Hybrid knockout.
	HybridKnockoutByIssuer	Hybrid knockout by issuer.
stressedMarket	Stressed Market conditions is called for when instrument experience high and short term intraday volatility.	

8.3. Trading Data

Once a foreign Market and its instruments are parametrized, trading messages can redistribute prices from the foreign Market on Market data. There are two types of transaction and price messages: general and custom.

General messages are normal trading-time GPW WATS messages reused to carry foreign Market prices. These messages are unchanged from their original versions. The only difference is that selected fields may be left unfilled for redistributed instruments contrasted to native instruments. Example general purpose messages are Trade and PriceLevelSnapshot.

8.4. InstrumentSummary

The InstrumentSummary message provides end-of-day statistics for each instrument on the platform.

Name	Description	
instrumentId	Identifier of the instrument.	
lastTradedPrice	Last Traded Price (LTP).	
closingPrice	Closing Price (CP).	
closingPriceType	Closing Price Type. For OffBook means last transaction price.	
	Name	**Description**
	InitialPrice	Initial price of the instrument
	LTP	Last traded price
	LastACP	Last adjusted closing price
	FairValue	Fair value
	DailySettlementPrice	Daily Settlement Price
	FinalSettlementPrice	Final Settlement Price
adjustedClosingPrice	Adjusted Closing Price (ACP).	
adjustedClosingPriceReason	Reason for adjusting the Closing Price.	
	Name	**Description**
	NotApplicable	1 - Not applicable.
	Dividend	2 – Dividend

Name	Description	
	IssueRight	3 - Issue Right
	Split	4 – Split
	ReverseSplit	5 - Reverse Split
	Bonus	6 – Bonus
	SpinOff	7 - Spin-Off
	TickSizeChange	8 - Tick Size Change
	OrderPurge	9 - Order Purge
	OtherReason	10 - Other reason
pctChange	Percentage change in price from last day's adjusted closing price.	
vwap	Volume-weighted average price for the Market model in which there is an instrument.	
noTrades	Total number of transactions on the current trading day.	
totalVolume	Total transaction volume.	
totalValue	Total transaction value.	
openingPrice	The price of the first trade on the current trading day.	
maxPrice	Highest price of the instrument on the current trading day.	
minPrice	Lowest price of the instrument on the current trading day.	
isCorrection	Previous message correction flag.	

8.5. The Distribution phase

The Distribution phase (tradingPhaseType=Distribution) is a phase specifically dedicated to redistributed instruments, that is those not quoted on WATS, i.e. indices from GPW Benchmark or obligations quoted by BondSpot on the TBSP market. Redistribution applies only to Instruments and Market Structures for which marketModelType= Redistribution.

The phase is meant to inform market participants of the start and end of the redistribution process:

- The start of data redistribution is marked by the transition from tradingPhaseType= NoTradingClosed to tradingPhaseType=Distribution
- The end of redistribution is marked by the transition from the tradingPhaseType=Distribution to tradingPhaseType= NoTradingClosed

During redistribution WATS solely retransmits messages originating from outside systems. WATS does not conduct any operations on the information received and its role is only that of an intermediary. The phase transition from Distribution to NoTradingClosed may occur at any time because it is controlled by the schedule in external systems:

- In case of redistributed instruments unrelated to instruments quoted on WATS, the phase change may occur at any time during 24-hour system operation and 7-day cycle (if the instrument is quoted in such a schedule).
- In the case of GPW Benchmark instruments that refer to instrument quoted on WATS, the transition to the Distribution phase should coincide with the schedule of the instrument that is part of the index.

8.6. Treasury BondSpot Poland (TBSP) Data Redistribution

GPW WATS redistributes information related to fixing prices from Treasury BondSpot Poland using the PriceUpdate message on Market Data.

The PriceUpdate message transcribes market prices data from TBSP and republishes it within GPW WATS.

Note that the PriceUpdate message is also used for other purposes.

8.7. GPW Benchmark Market Indices

The Index GPW WATS facility conveys the value of Market indices published by GPW Benchmark throughout the trading day together with end-of-day statistics. This information is published on GPW WATS by the data redistribution framework.

The order of Index messages during a trading day is as follows:

Index Message	Message Type	Note
IndexParams	Parametrization	The first Index message sent during parametrization. It frames the index.
IndexPortfolioEntry	Parametrization	Sequence of messages following an IndexParams message. Lists the Instruments which compose of the index.
RealTimeIndex	Realtime data	The value of the index broadcast during the trading day.
IndexSummary	End of day data	Index statistics at session close.

8.7.1. INDEXPARAMS

Carries metadata related to an index. This information contains parameters such as the base value of the index, the number of components and currency.

The IndexParams message can carry information about up to 3 instruments in the indexUnderlying attribute array, each identified by its own ISIN.

The IndexParams message is sent before the start of the session, in reference data, but in emergency cases it may also be sent during the session.

Name	Description
header	Message header.
instrumentId	ID of financial instrument (index ID).
indexBaseValue	Index base value, usually the first value of index.
indexBaseCapitalisation	Index base capitalization, market value of index portfolio corresponding to the first index value.
indexBaseDate	Index base date, usually the first date of publication.
correctionCoefficient	Index adjustment coefficient K(t). Indicator that allows for maintaining continuity of index value after periodic and extraordinary adjustments. Component of algorithm of indices.
numOfComponents	Total number of instruments in the index portfolio.

<table>
<tr><th>Name</th><th>Description</th></tr>
<tr><td>indexPublicationSchedule</td><td>Mode of index dissemination.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Continuous</td><td>1</td><td>Index is disseminated in determined time intervals.</td></tr>
<tr><td>Fixing</td><td>2</td><td>Index is disseminated on determined hours.</td></tr>
<tr><td>Closing</td><td>3</td><td>Index is disseminated on closing.</td></tr>
</table></td></tr>
<tr><td>indexType</td><td>Index product classification
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>W</td><td>1</td><td>WSE index from the main or alternative markets.</td></tr>
<tr><td>Z</td><td>2</td><td>External index.</td></tr>
<tr><td>V</td><td>3</td><td>Another WSE information product.</td></tr>
<tr><td>M</td><td>4</td><td>WSE strategy dividend point index.</td></tr>
<tr><td>S</td><td>5</td><td>WSE strategy index of short type.</td></tr>
<tr><td>L</td><td>6</td><td>WSE strategy index of leverage type.</td></tr>
</table></td></tr>
<tr><td>presenceFlags</td><td>Indicates the filling of fields: daysSinceLastPublication, numberOfDividends.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>None</td><td>0b00000000</td><td></td></tr>
<tr><td>HasDaysSinceLastPublication</td><td>0b00000001</td><td></td></tr>
<tr><td>HasNumberOfDividends</td><td>0b00000010</td><td>.</td></tr>
</table></td></tr>
<tr><td>daysSinceLastPublication</td><td>Number of calendar days between current session and the previous session day. The field is filled for strategy indices of short/leverage type (indexType S/L) and is a component of algorithm of above mentioned indices.</td></tr>
<tr><td>numberOfDividends</td><td>Number of dividends included in the index calculation. The field is filled for strategy dividend point indices (indexType M).</td></tr>
<tr><td>indexUnderlyings.count</td><td>How many elements does IndexUnderlyings contain.</td></tr>
<tr><td>indexUnderlyings.items</td><td>IndexUnderlying Array.
IndexUnderlying:
<table>
<tr><th>Name</th><th>Description</th></tr>
<tr><td>productIdentification.productIdentification</td><td>Product identification, for example its ISIN code.</td></tr>
<tr><td>productIdentification.productIdentificationType</td><td>Type of product identification.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>1</td><td>Not Applicable</td></tr>
<tr><td>ISIN</td><td>2</td><td>Financial instrument's ISI</td></tr>
</table></td></tr>
<tr><td>indexUnderlyingType</td><td>Underlying type of index.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>1</td><td>Not Applicable</td></tr>
<tr><td>Index</td><td>2</td><td>Index.</td></tr>
<tr><td>ReferenceRate</td><td>3</td><td>Reference Rate.</td></tr>
</table></td></tr>
</table></td></tr>
<tr><td>publicationOrder</td><td>Index publication presentation order.</td></tr>
</table>

Name	Description
currency	Index currency (e.g. USD).
dateValidity	Date of validity of index portfolio.

8.7.2. INDEXPORTFOLIOENTRY

Defines an index by reference to the instruments traded in the GPW WATS platform. The definition of the index is made by giving a list of InstrumentId relating to specific instruments. The index is defined using a series of IndexPortfolioEntry messages, one for each security composing the index. For example, in order to specify an index composed of 30 securities, 30 IndexPortfolioEntry messages wil l be broadcast on Market Data. The series of IndexPortfolioEntry messages for a given index is sent after IndexParams message for this index. The rules for sending the series of IndexPortfolioEntry messages in emergency cases are the same as for the IndexParams message, which means it may also be sent during the session.

Name	Description
header	Message header.
id	The identifier of the entry within the given index.
indexId	Instrument Id of the redistribution index.
presenceFlags	Indicates the filling of fields: instrumentId. **Name** / **Value** / **Description** **None** / 0b00000000 / **HasInstrumentId** / 0b00000001 /
instrumentId	InstrumentID of the financial instrument (index component). Applies only to instruments listed on GPW WATS markets.
publicProductIdentification.productIdentification	Product identification, for example its ISIN code of the instrument participating in the index portfolio.
publicProductIdentification.productIdentificationType	Type of product identification. **Name** / **Value** / **Description** **NotApplicable** / 1 / Not applicable **ISIN** / 2 / Financial instrument's ISIN.
mic	Market structure's Market Identifier Code (MIC) as specified in ISO 10383 of the instrument participating in the index portfolio..
currency	Quotation currency (e.g. USD) of the instrument participating in the index portfolio.
instrumentPacket	Instrument packet in index portfolio / of information product. Number of units (usually stocks/shares) of a given instrument in the index portfolio or in the information product.
suspended	Is instrument suspended. Suspended instrument is not taken into account when calculating the index, but remain its participants.

8.7.3. REALTIMEINDEX

Communicates the real time value of the index and related statistics.

Index information is sent at intervals appropriate for each respective Market index and the publication intervals are parametrized at system level. A variety of publication intervals can be pre-configured ranging from near-real time publication (15 seconds or less) to several times per day.

The publication frequency is dependent on the specification of the index.

<table>
<tr><th>Name</th><th>Description</th></tr>
<tr><td>header</td><td>Message header.</td></tr>
<tr><td>instrumentId</td><td>ID of the instrument (index ID).</td></tr>
<tr><td>indexLevelCode</td><td>Level of the index.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>PreviousClosingValue</td><td>1</td><td>Closing index value from previous session (reference index).</td></tr>
<tr><td>TheoreticalValueBeforeOpening</td><td>2</td><td>Theoretical opening index before opening the session.</td></tr>
<tr><td>ValueBeforeOpening</td><td>3</td><td>Theoretical value of the index after opening the session but before index opening.</td></tr>
<tr><td>OpeningValue</td><td>4</td><td>Index opening value.</td></tr>
<tr><td>CurrentValue</td><td>5</td><td>Current index value.</td></tr>
<tr><td>PreClosingValue</td><td>6</td><td>Preliminary closing value.</td></tr>
<tr><td>ClosingValue</td><td>7</td><td>Closing index value.</td></tr>
<tr><td>ValueWithoutOpening</td><td>8</td><td>Index value without opening.</td></tr>
</table></td></tr>
<tr><td>numOfActiveInstruments</td><td>Number of instruments – participants of the index, for which the quotation was set during the session.</td></tr>
<tr><td>indIndexOpeningPortfolio</td><td>Indicator of the index opening portfolio W(t). Percentage share of instruments with at least one trade during the session.</td></tr>
<tr><td>indexValue</td><td>The value of the last level for the index according to the definition provided in the indexLevelCode. The field can be set to zero for strategy dividend point indices (indexType M).</td></tr>
<tr><td>pctChangeIndexValPrevSession</td><td>Percentage change of the index value in relation to the value of the closing index from the previous session.</td></tr>
<tr><td>sessionLow</td><td>Lowest value of the day. The field can be set to zero for strategy dividend point indices (indexType M).</td></tr>
<tr><td>sessionHigh</td><td>Highest value of the day. The field can be set to zero for strategy dividend point indices (indexType M).</td></tr>
<tr><td>presenceFlags</td><td>Indicates the filling of fields: tradingValue, indexValueBid, indexValueAsk, midSpreadIndex, differenceCentralSpread.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>None</td><td>0b00000000</td><td></td></tr>
<tr><td>HasTradingValue</td><td>0b00000001</td><td></td></tr>
<tr><td>HasIndexValueBid</td><td>0b00000010</td><td>.</td></tr>
<tr><td>HasIndexValueAsk</td><td>0b00000100</td><td></td></tr>
<tr><td>HasMidSpreadIndex</td><td>0b00001000</td><td></td></tr>
<tr><td>HasDifferenceCentralSpread</td><td>0b00010000</td><td></td></tr>
</table></td></tr>
<tr><td>tradingValue</td><td>Value of trading in the instruments from the given index.</td></tr>
</table>

Name	Description
indexValueBid	Value of the buy index based on the best sell offers of instruments. The field is filled for messages with indexLevelCode 3 (ValueBeforeOpening), 4 (OpeningValue) and 5 (CurrentValue) for underlying indices for derivatives. In other cases, the field is set to zero.
indexValueAsk	Value of the sell index based on the best sell offers of instruments. The field is filled for messages with indexLevelCode 3 (ValueBeforeOpening), 4 (OpeningValue) and 5 (CurrentValue) for underlying indices for derivatives. In other cases, the field is set to zero.
midSpreadIndex	Mean of indexValueBid and indexValueAsk. The field is filled for messages with indexLevelCode 3 (ValueBeforeOpening), 4 (OpeningValue) and 5 (CurrentValue) for underlying indices for derivatives. In other cases, the field is set to zero.
differenceCentralSpread	The point difference between midSpreadIndex and indexValue. The field is filled for messages with indexLevelCode 3 (ValueBeforeOpening), 4 (OpeningValue) and 5 (CurrentValue) for underlying indices for derivatives. In other cases, the field is set to zero.

8.7.4. INDEXSUMMARY

This is an end-of-trading-day message. It provides the end-of-day value of the index and summarizes the variation in the index during the trading day with select statistics including high, low and closing value. The message is sent twice a day with the RealTimeIndex message with indexLevelCode 6 (PreClosingValue) and 7 (ClosingValue).

Name	Description
header	Message header.
instrumentId	ID of the instrument (index ID).
openingIndexValue	Opening index value. The field can be set to zero for strategy dividend point indices (indexType M).
sessionLow	Lowest value of the day. The field can be set to zero for strategy dividend point indices (indexType M).
sessionHigh	Highest value of the day. The field can be set to zero for strategy dividend point indices (indexType M).
closingIndexValue	Closing index value. The field can be set to zero for strategy dividend point indices (indexType M).
presenceFlags	Indicates the filling of fields: sessionAvg. **Name** / **Value** / **Description** **None** / 0b00000000 / **HasSessionAvg** / 0b00000001 /
sessionAvg	Average value of the day. The field can be set to zero for indices for which the value is not calculated.
pctChangeIndexValPrevSession	Percentage change of the index value in relation to the value of the closing index from the previous session.
pctChangeIndexValPrevYear	Percentage change of the index value in relation to its value at the end of the previous year
closingCapitalisationPortfolio	Capitalization of the portfolio for the closing index expressed in the currency of the index, on basis of which the closing index is calculated.
indIndexOpeningPortfolio	Indicator of the index opening portfolio W(t). Percentage share of instruments with at least one trade during the session.

9. Next Session Reference Data

Reference data configures GPW WATS for the current trading session (see Chapter 6). GPW WATS Next Session reference data is a feature that aims to inform the market of GPW WATS configuration for the next trading session. The next trading day is the next business day on which trading takes place on GPW WATS. It does not configure or preconfigure WATS for the next trading day but serves an informational purpose. GPW WATS is parametrized for the next trading session in the morning of that session's calendar day prior to the start of trading. The rules of parametrization with same-day reference data are described in Chapter 6. This chapter describes the Next Session Reference Data.

Note: Next Session Reference Data does not cover redistributed instruments. It focuses exclusively on instruments traded on GPW WATS.

9.1. The next session calendar

Next Session Reference Data is published on Market Data towards the end of the trading session with the aim to help members prepare for the next trading session.

In this context, the next trading session means the next day on which the instruments parametrized will be traded excluding non-trading days and holidays.

For example on a Friday, Next Session Reference Data will generate reference data for the set of instruments that are planned to be traded on the following Monday (if it is a trading day). On Saturday morning, prior to trading, GPW WATS will produce the current day's reference data on Market Data. On Monday morning it will do the same for the instruments that will trade that day. In addition, on Saturday evening (assuming Sunday is not a trading day) GPW WATS Next Session Reference Data will produce reference data for the set of instruments that will trade the next trading day, Monday.

9.2. The mechanism

It enables the parametrization of the market structure and instruments that will be traded on the next trading day. It consists of the following

The Next Session reference data is produced by means of the same Market Data messages as the reference data.

Next day reference data is published using the TradingSessionStatus mechanism. TradingSessionStatus uses tradingSessionEvent=NextSessionReferenceDataStart and NextSessionReferenceDataEnd to wrap reference data messages. An example of data for the next session is presented below.

Event	Message	
Trading	Trading session messages.	The ongoing trading session.
Beginning of next session data	{"header":{“length”:55,”msgType”:”TradingSessionStatus”,"version":4096,"seqNum":2,"timestamp":1717902001420610304,"sourceTimestamp":1717902001416863413,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"marketId":"","marketStructureId":0,"tradingSessionState":"Open","tradingSessionEvent":" NextSessionReferenceDataStart"}	TradingSessionStatus message is sent to mark the start of the reference data for the next trading day. TradingSessionEvent=NextSessionReferenceDataStart marks the start of the series of messages containing reference data for the next trading session. Note that tradingSessionState=Open.
Next session reference data	Reference data messages.	Next trading session reference data messages serialized on Market Data.
End of next session reference data	{"header":{“length”:55,”msgType”:”TradingSessionStatus”,"version":4096,"seqNum":2214,"timestamp":1717902002008936596,"sourceTimestamp":1717902002008172778,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"marketId":"","marketStructureId":0,"tradingSessionState":"Open","tradingSessionEvent":" NextSessionReferenceDataEnd"}	TradingSessionStatus with TradingSessionEvent=NextSessionReferenceDataEnd marks the end of the series of messages containing reference data for the next trading session.
Trading	Trading messages continue.	Current-day trading messages published on Market data.

The trading messages which are included in the next day parametrization are shown in the following table.

Category	Message type	Message	Description
Reference Data	Market Structure	MarketStructure Instrument TickTableEntry	Group of messages which define Market, instruments, and other Market parameters.
	Trading Calendar	TradingPhaseScheduleEntry CalendarException WeekPlan	
	Order and Trade Collars	CollarTableEntry CollarGroup	
	Pricing Messages	AccruedInterestTableEntry IndexationTableEntry	Information about indexation and accrued interest for fixed income instruments
	Other	ExternalUnderlying	

10. Session Management Messages

10.1. Header

The Market data header is incorporated into every Market data message. It contains several fields of interest, among which the key are outlined below.

Name	Description
length	Message length.
msgType	Type of message (e.g. OrderExecute).
version	Version of the Market Data protocol.
seqNum	Sequence number of the message added by the Market Data Sequencer.
timestamp	Timestamp indicating when the message was sequenced.
sourceTimestamp	Timestamp added by the service which sent the message.
isEncrypted	True if message is encrypted.
encryptionOffset	Encryption byte offset.
encryptionKeyId	Reference to an EncryptionKey message.
sessionId	ID of the session.
streamId	ID of the md stream.

10.2. TradingSessionStatus

TradingSessionStatus updates the Market on the status of trading sessions. The message is addressed to entire MIC codes, and not to specific instruments. The message is sent automatically by System and can't be solicited.

The message is sent to mark specific events related to the trading session:

- the beginning and end of the technical session,
- the beginning and end of reference data within a technical session,
- Next day parametrization
- Potential other uses.
- **Note: tags PreviousDayRestateStart/PreviousDayRestateEnd are not used.**

The following table illustrates the usage of the TradingSessionStatus in the context of the trading day for MIC code XWAR.

Event	Message	
Start of the technical session	{"header":{“length”:55,”msgType”:”TradingSessionStatus”,"version":4096,"seqNum":1,"timestamp":1717873216964202862,"sourceTimestamp":1717873214793870267,"isEncrypted":false,"encryptionOffset":0,"encryptionK	StartOfTechnicalSession - The first message sent by the Stream service during the daily session is a TradingSessionStatus with TradingSessionEvent=StartOfT

Event	Message	
	eyId":0,"sessionId":792,"streamId":0},"marketId":"","marketStructureId":0,"tradingSessionState":"Unknown","tradingSessionEvent":"StartOfTechnicalSession","date":0}	echnicalSession marking the start of that session. The event date is set to 0.
Beginning of reference data	{"header":{"length":55,"msgType":"TradingSessionStatus","version":4096,"seqNum":2,"timestamp":1717902001420610304,"sourceTimestamp":1717902001416863413,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"marketId":"","marketStructureId":0,"tradingSessionState":"Open","tradingSessionEvent":"InitialReferenceDataStart","date":2024-06-09}	InitialReferenceDataStart - TradingSessionStatus with TradingSessionEvent=InitialReferenceDataStart marks the start of the series of messages containing reference data. It comes out immediately after the previous message, hence seqNum=2. The event date is set to the date of the trading session.
Reference data	Reference data messages.	Reference data messages serialized on Market Data.
End of reference data	{"header":{"length":55,"msgType":"TradingSessionStatus","version":4096,"seqNum":2214,"timestamp":1717902002008936596,"sourceTimestamp":1717902002008172778,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"marketId":"","marketStructureId":0,"tradingSessionState":"Open","tradingSessionEvent":"InitialReferenceDataEnd","date":2024-06-09}	InitialReferenceDataEnd - TradingSessionStatus with TradingSessionEvent=InitialReferenceDataEnd marks the end of the series of messages containing reference data. The event date is set to the date of the trading session.
Trading	Trading messages.	Trading messages published on Market data.
End of technical session	{"header":{"length":55,"msgType":"TradingSessionStatus","version":4096,"seqNum":2215,"timestamp":1717959601376129958,"sourceTimestamp":1717959601376019556,"isEncrypted":false,"encryptionOffset":0,"encryptionKeyId":0,"sessionId":792,"streamId":0},"marketId":"","marketStructureId":0,"tradingSessionState":"Close","tradingSessionEvent":"EndOfTechnicalSession","date":0}	EndOfTechnicalSession - The last message sent by the Stream service during the technical session marks the end of that session. After this message only Heartbeats appear on Market Data. The event date is set to 0.

10.2.1. STARTOFTECHNICALSESSION / ENDOFTECHNICALSESSION

TradingSessionStatus messages with TradingSessionEvent=StartOfTechnicalSession and EndOfTechnicalSession are published with a date field set to 0, TradingSessionStatus:: date=0.

10.2.2. INITIALREFERENCEDATASTART / INITIALREFERENCEDATAEND

TradingSessionStatus messages with TradingSessionEvent= InitialReferenceDataStart and InitialReferenceDataEnd are published with a date field set to the date of the session to which the reference data apply.

10.2.3. STARTOFTRADINGSESSIONMIC / ENDOFTRADINGSESSIONMIC

StartOfTradingSessionMIC and EndOfTradingSessionMIC are status flags within the structure of the TradingSessionStatus message which monitor the state of instruments belonging to a MIC.

The TradingSessionStatus message with these flags is sent in 2 cases:

- StartOfTradingSessionMIC is sent whenever *any* instrument in a MIC transitioned from a NoTradingClosed phase to NoTradingEarlyMonitoring
- EndOfTradingSessionMIC is sent whenever *all* instruments in a MIC transitioned from the NoTradingLateMonitoring phase to NoTradingClosed

10.3. ENCRYPTIONKEY

The encryption keys used to decrypt the relevant Market Data messages are disseminated daily by the Snapshot service.

Each Key is accompanied by a label (key ID) which identifies the key. Messages will refer to particular KeyIDs to be decrypted.

10.4. SNAPSHOT OPERATION

The Snapshot service enables the user to download the reference and latest trading data at any time during the trading session.

The user interacts with the Snapshot by means of 3 messages:

Message	Note	Details
Login	Login into the Snapshot service	The Filter attribute enables the user to select which types of data to transmit.
LoginResponse	Response to the snapshot service	The Result attribute communicates the type of error upon logging in.
Logout	Logout message ending Snapshot transmission	

10.4.1. LOGGING INTO SNAPSHOT

Logging into Snapshot is performed by means of the Login message.

The Snapshot can be filtered by passing the appropriate flag in the *filter* attribute of the Login message. This attribute specifies the type of data that the snapshot will send to the user. The filter is a bitmask which accepts mixing of the 3 values provided as choices.

Filter setting	Value	Details
None	0b0000	Retransmit all order book related data but disable Trades and Reference Data.
Trades	0b0001	All order book data and Trades
RefData	0b0010	All order book data and Reference Data

To receive both the Trade messages and Reference Data messages in the snapshot, the user needs to specify the correct bitflag combination in the Login message:

- filter=0b0011 - Snapshot sends Reference Data and Trade messages for all Instruments.
- filter=0b0000 – neither Reference Data nor Trade messages are published on Snapshot. Other messages are published.

In case WATS is not parametrized, Snapshot will send a LoginResponse with a result= LoginNotAllowed.

10.4.2. ENDOFSNAPSHOT

This message sent by the Snapshot service indicates the end of the snapshot. Following the message, the Snapshot service disconnects automatically.

EndOfSnapshot contains the last sequence number – lastSeqNum.

This number is used to synchronize with the messages from the real time Stream service. Under optimal conditions, after retrieving the Snapshot, the user should capture the message numbered lastSeqNum+1 on Stream.

If the Snapshot presents gaps relative to the current sequence number (found in the header of each Market Data message), the Replay service can be used to complete the state of the market. The ReplayRequest message will request messages starting from EndOfSnapshot:lastSeqNum+1 to the current sequence number.

10.4.3. SEQUENCE NUMBERS IN SNAPSHOT

All messages on Snapshot service come out with seqNum=0 in the message header.

10.4.4. SNAPSHOT HEADER

The header in the snapshot is the same as that in the MD protocol. For inbound messages, the header should be filled in the manner outlined in the table below.

Name	Description
length	Message length – the length of the inbound message
msgType	Type of inbound message
version	Version of the Market Data protocol
seqNum	Set to 0
timestamp	Message sending time
sourceTimestamp	Set to 0
isEncrypted	False.
encryptionOffset	Set to 0
encryptionKeyId	Set to 0
sessionId	Set to 0
streamId	Set to 0

10.5. REPLAY OPERATION AND REPLAYREQUEST

The ReplayRequest message supports the single use case of recovering missing messages from Stream. In this situation ReplayRequest repeats messages referenced by a range of sequence numbers.

A replay request contains two parameters: seqNum (starting sequence number) and endSeqNum (ending sequence number):

- if seqNum==1, the replay starts from the first message,
- if endSeqNum==0, the replay ends on the last message.

The Replay service sets an upper bound on the number of messages that can be replayed in one request: (endSeqNo -seqNo) < SystemMaximum. The SystemMaximum amount is defined by the System Operator and is currently set to 10,000,000 sequence numbers (per service, full depth and BBO). When there is a large message gap or when a Participant joins the trading platform late during the Market session, a Snapshot should be used.

The end of the response to ReplayRequest is marked by 2 null bytes.

10.5.1. THE REPLAYREQUEST MESSAGE HEADER

The ReplayRequest is an incoming message and its header is 4 bytes in length (note that the regular header for outgoing Market Data messages is 42 bytes in length.). The ReplayRequest header structure is the following:

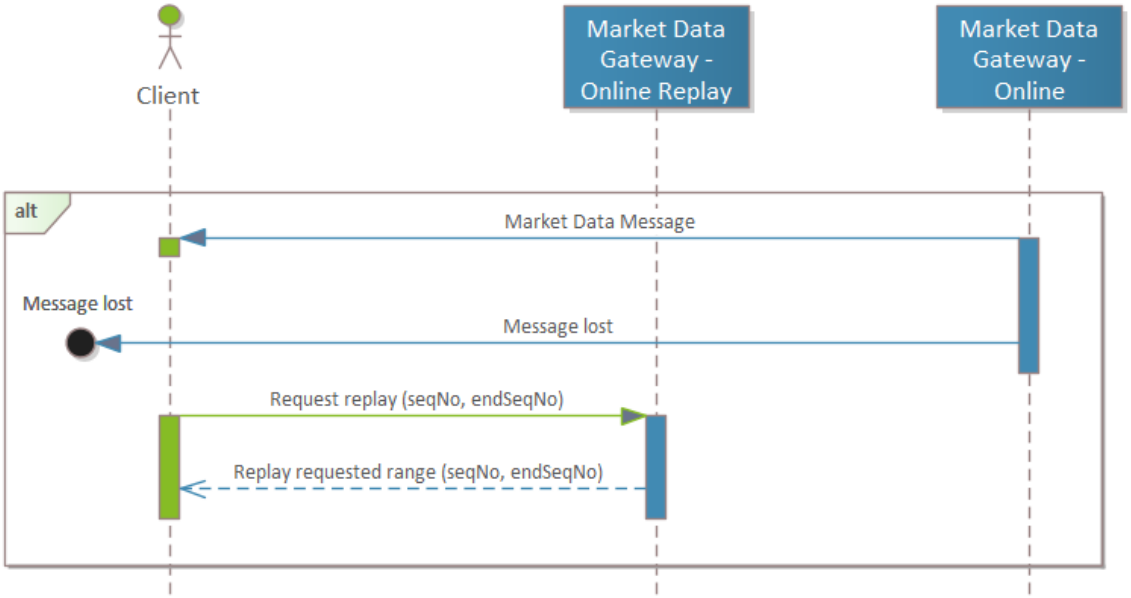
Name	Type	Kind	Length	Description
length	MsgLength	Alias (u16)	2	Message length.
replayMsgType	ReplayMsgType	Enum	2	Type of the message. **Name** / **Value** / **Description** **ReplayRequest** / 1 / Replay service request

10.5.2. THE REPLAYREQUEST MESSAGE

The replay request message carries the start and end sequence numbers for the sequence to be resent. The header is the replay header and not the standard Market Data header.

Name	Type	Kind	Length	Description
header	Header	Struct	4	Message header.
seqNum	SeqNum	Alias (u32)	4	The sequence number of the first missing message.
endSeqNum	SeqNum	Alias (u32)	4	The sequence number of the last missing message in the requested range.
streamId	StreamId	Alias (u8)	1	ID of the MD stream.

Figure 8 - OMD Replay Behavior



10.5.3. REPLAY TIMEOUT

Upon connection to Market Data Replay, the ReplayRequest message must be sent within 100ms.

Otherwise Replay will drop the connection. This applies to BBO and FD.

11. Aggressive Order Kinematics

11.1. Introduction

This chapter details the kinematics of WATS Market Data related to aggressive orders.

The WATS Market Data implementation directly relays the state of the matching engine and does not require clients to implement complex matching logics. Aggressive orders that match immediately upon entry into the system, such as market orders and aggressive limits, are treated in one consistent manner. This approach enables clients to easily maintain a mirror image of the state of the order book.

It also ensures that recipients of Market Data never see a crossed book on the Market Data stream.

11.2. Context

The normal state for the order book is an uncrossed state with passive orders resting on the buy and sell sides.

WATS Market Data publishes a feed which always conveys an uncrossed order book and enables users to reconstruct the complete state of the book without relying on complex matching procedures.

This chapter outlines scenarios when an aggressive order enters the book and immediately matches orders on the opposite side.

We treat 2 main cases:

1. An aggressive market or limit order enters the book
2. A passive (limit) order is modified into an aggressive order

This paper covers the kinematics of the Market Data protocol during the continuous trading phase only. The behaviour of Market Data in other types of market phases such as auctions or hybrid is not discussed and follows generally accepted principles.

11.2.1. WATS SCHEME FOR PROCESSING AGGRESSIVE ORDERS

The WATS framework for the processing of aggressive orders, market or limit, ensures that the order book never appears to be crossed on the Market Data feed. The general principles followed by Market Data kinematics are:

- Aggressive OrderAdds (market or limit) are not published on Market Data. WATS Market Data does not publish aggressive orders (IOC or FOK). These orders are matched immediately upon entering the order book and only the matches and trades are published on the Market Data feed.
- Only OrderExecutes for the resting orders matched with the aggressor are published. OrderExecutes related to the aggressive order are omitted.
- OrderAdds for remaining quantities (if any) of the aggressive orders are published
- Trade messages related to the order executions are published after the matching ends

11.2.2. THE SEQUENCE OF MESSAGES IN AN AGGRESSOR SCENARIO

The general sequence of messages On Market Data in an aggressive match scenario is the following:

1. OrderExecutes are published for the passive orders hit by the aggressive order. No OrderExecute/s are/is published for the aggressor. No OrderExecutes are published for the hidden part of an Iceberg.
2. OrderAdd for the leftover part of the aggressive order is published. If the aggressive order is fully filled no OrderAdd is published.
3. Trade messages conveying final trade information are published.

11.3. COMPLETE FILLS

Here we present the kinematics of the WATS Market Data feed in the situation when an aggressive limit order enters the order book during the continuous trading market phase. Two cases are treated,

1. the aggressor is fully filled against a single order,
2. the aggressor sweeps several orders and is fully filled.

The scenarios are enhanced with examples of order books illustrating the execution of the aggressive order.

11.3.1. INBOUND AGGRESSIVE ORDER FULLY FILLED AGAINST A SINGLE ORDER

This example examines the case when an aggressive limit order enters the order book and executes fully with a single opposite resting order.

The table below shows two views of an order book: the initial (uncrossed) state which naturally occurs (left panel) and the state after the aggressive order is matched with resting orders (right panel). The aggressive order is shown in the lower part of the left panel together with its limit price and quantity.

In this particular example, three orders rest on the sell side and one order rests on the buy side. The numbers on the Bids and Asks columns represent order quantity. The numbers in the parentheses are the individual orders' PublicOrderIDs. Order priority is assessed from left to right. For example, there are 2 orders on the 200 price level – the order with quantity 100 has higher priority than the order with quantity 200. The book is uncrossed.

An aggressive buy limit order with a price of 200 and volume 30 enters the book and is assigned a PublicOrderID=160. The aggressor is matched with resting opposite-side order with PublicOrderID=101 and is fully filled. The final state of the order book is presented in the right-side panel. Order with PublicOrderID=101 is reduced to a quantity 70. The rest of the order book remains unchanged.

Initial Order Book State (Instrument A)			**Final Order Book State**		
Asks (sell)	**Price**	**Bids (buy)**	**Asks (sell)**	**Price**	**Bids (buy)**
50(102)	201		**50(102)**	201	
100(101),200(115)	200		**70(101),200(115)**	200	
	199	**50(98)**		199	**50(98)**
Inbound aggressive order (limit price, side and quantity)					

Initial Order Book State (Instrument A)			Final Order Book State		
Asks (sell)	**Price**	**Bids (buy)**	**Asks (sell)**	**Price**	**Bids (buy)**
	200	**30(160)**			

The example above produces a flow on the WATS Market Data. The Market Data flow follows the WATS rules pertaining to aggressive orders – see sections above. The steps followed by Market Data are presented in the summary table below.

Because incoming order 160 is aggressive, its OrderAdd message is suppressed on Market Data and the order is matched with resting orders without being published. Specifically, it matches resting order 101 and is fully filled. See step 1 in the table below.

Market Data suppresses publication of the OrderExecute message associated with aggressive order (step 2). Only an OrderExecute related to passive order 101 is published (step 3). The anatomy of the OrderExecute message reveals that a match took place at a price of 200 (ExecutionPrice=200) with passive order 101 and that an TradeId=31 was assigned to the match. The TradeId will be referred to by the subsequent Trade message in the TradeId attribute (see step 4). The OrderExecute further reveals that the match reduced the volume of the resting order by 30 (ExecutionQuantity=30). Order 101 was left on the book with quantity 70 (Quantity=70). Had passive order 101 been exhausted, Quantity would equal 0.

The long Trade bookkeeping message follows the OrderExecute and refers to the TradeId of the OrderExecute by means of the TradeId=31 field. A TotalVolume=30 was traded in this scenario.

Message flow on WATS Market Data

Step	**Message**	**Message Details**	**Note**
1	OrderAdd	N/A	OrderAdd for the aggressive buy limit is suppressed (order 160).
2	OrderExecute	N/A	OrderExecute attendant to the aggressive order is suppressed (order 160).
3	OrderExecute	Message=OrderExecute Quantity=70 PublicOrderId=101 TradeId=31 ExecutionPrice=200 ExecutionQuantity=30	OrderExecute for passive order 101, crossed by the inbound aggressive order.
4	Trade	Message=Trade Price=200 Quantity=30 TradeId=31	There is one Trade message in this execution since the inbound order crossed only one resting order and was fully filled.

11.3.2. INBOUND AGGRESSIVE ORDER FULLY FILLED AGAINST MULTIPLE ORDERS

In this example, an incoming aggressive order is fully filled after crossing several passive orders. The initial state of the order book is unchanged. Aggressive limit order 160 enters the book with volume 325. It sweeps price levels 200 and 201 and leaves part of resting order 102 on the book. The aggressor is fully filled and orders 101 and 115 are exhausted. The final state of the book shows order 102 left on the book with volume 25.

Initial Order Book State (Instrument A)			**Final Order Book State**		
Asks (sell)	**Price**	**Bids (buy)**	**Asks (sell)**	**Price**	**Bids (buy)**
50(102)	201		**25(102)**	201	
100(101),200(115)	200			200	
	199	**50(98)**		199	**50(98)**
Inbound aggressive order (limit price, side and quantity)					
	201	**325(160)**			

The message flow on Market Data reflects the fact that the aggressive order touched 3 passive orders in total. Following WATS rules, only OrderExecute messages related to the passive orders are published.

Message flow on WATS Market Data

Step	Message	Message Details	Note
1	OrderAdd	N/A	OrderAdd for the aggressive buy limit is suppressed (order 160).
2	OrderExecute	N/A	3 OrderExecutes attendant to the aggressive order are suppressed (order 160).
3	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=101 TradeId=31 ExecutionPrice=200 ExecutionQuantity=100	OrderExecute for passive order 101, crossed by the inbound aggressive order. The order is exhausted.
4	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=115 TradeId=32 ExecutionPrice=200 ExecutionQuantity=200	OrderExecute for passive order 115, crossed by the inbound aggressive order. The order is exhausted.
5	OrderExecute	Message=OrderExecute Quantity=25 PublicOrderId=102 TradeId=33 ExecutionPrice=201 ExecutionQuantity=25	OrderExecute for passive order 102, crossed by the inbound aggressive order.
6	Trade	Message=Trade Price=200 Quantity=100 TradeId=31	Trade message related to TradeId=31 on order 101.

Message flow on WATS Market Data

Step	Message	Message Details	Note
7	Trade	Message=Trade Price=200 Quantity=200 TradeId=32	The second trade message for the second passive order.
8	Trade	Message=Trade Price=201 Quantity=25 TradeId=33	The third and final trade message.

11.4. PARTIAL FILLS

Here we present the kinematics of the WATS Market Data feed in the situation when an aggressive limit order enters the order book during the continuous trading market phase. Three cases are treated,

1. the aggressor is *partially* filled against a single order,
2. the aggressor sweeps several orders but is *partially* filled,
3. the aggressor triggers a volatility auction.

11.4.1. INBOUND AGGRESSIVE ORDER PARTIALLY FILLED AGAINST ONE ORDER

The partial fill scenario roughly corresponds to the full fill scenario with the allowance that the incoming aggressive order is only partially executed.

Order 160 aggresses order 101 and is partially filled. The leftover enters the book at price 200 with volume 20.

Initial Order Book State (Instrument A)			Final Order Book State		
Asks (sell)	Price	Bids (buy)	Asks (sell)	Price	Bids (buy)
50(101)	200			200	**20(160)**
	199	**50(98)**		199	**50(98)**
Inbound aggressive order (limit price, side and quantity)					
	200	**70(160)**			

WATS Market Data suppresses the aggressive OrderAdd and only produces an OrderExecute for the passive order 101. A Trade message follows.

The only modification to the message flow is the insertion of an OrderAdd to reflect the leftover quantity in the aggressor order. The OrderAdd is published after the OrderExecute message.

Message flow on WATS Market Data

Step	Message	Message Details	Note
1	OrderAdd	N/A	OrderAdd for the aggressive buy limit is suppressed (order 160).
2	OrderExecute	N/A	OrderExecute attendant to the aggressive order is suppressed (order 160).
3	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=101 TradeId=31 ExecutionPrice=200 ExecutionQuantity=50	OrderExecute for passive order 101, crossed by the inbound aggressive order.
4	OrderAdd	Message=OrderAdd PublicOrderId=160 Side=Buy Price=200	OrderAdd for the unfilled quantity in the aggressive order.

Message flow on WATS Market Data

Step	Message	Message Details	Note
		Quantity=20	
5	Trade	Message=Trade Price=200 Quantity=50 TradeId=31	There is one Trade message in this execution since the aggressor crossed only one resting order.

11.4.2. INBOUND AGGRESSIVE ORDER PARTIALLY FILLED AGAINST MULTIPLE ORDERS

Here, the aggressor sweeps several orders but is not fully filled. Aggressive limit 160 touches passive sell orders 101 and 115 and rests on the buy side with volume 20 at price 200.

Initial Order Book State (Instrument A)			Final Order Book State		
Asks (sell)	**Price**	**Bids (buy)**	**Asks (sell)**	**Price**	**Bids (buy)**
50(101),30(115)	200			200	**20(160)**
	199	**50(98)**		199	**50(98)**
Inbound aggressive order (limit price, side and quantity)					
	200	**100(160)**			

The message flow on Market Data reflects the fact that the aggressive order touched 2 passive orders in total. Following WATS rules, only OrderExecute messages related to the passive orders are published.

Message flow on WATS Market Data

Step	Message	Message Details	Note
1	OrderAdd	N/A	OrderAdd for the aggressive buy limit is suppressed (order 160).
2	OrderExecute	N/A	2 OrderExecutes attendant to the aggressive order are suppressed (order 160).
3	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=101 TradeId=31 ExecutionPrice=200 ExecutionQuantity=50	OrderExecute for passive order 101, crossed by the inbound aggressive order.
4	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=115 TradeId=32 ExecutionPrice=200 ExecutionQuantity=30	OrderExecute for passive order 115.
5	OrderAdd	Message=OrderAdd PublicOrderId=160 Side=Buy Price=200 Quantity=20	OrderAdd for the unfilled quantity in the aggressive order 160.
6	Trade	Message=Trade Price=200 Quantity=50 TradeId=31	Trade 1 related to OrderExecute for order 101.
7	Trade	Message=Trade Price=200 Quantity=30 TradeId=32	Trade 2 related to OrderExecute for order 115.

11.4.3. AGGRESSIVE ORDER TRIGGERS VOLATILITY AUCTION

In this example, the aggressive limit is partially filled and triggers a volatility auction. Limit order 160 attempts to transact at a price of 201 which exceeds trade collars. The aggressor sweeps order 101 and upon hitting order 115 at price 201 which exceeds trade collars, triggers a volatility auction.

Initial Order Book State (Instrument A)			Final Order Book State		
Asks (sell)	**Price**	**Bids (buy)**	**Asks (sell)**	**Price**	**Bids (buy)**
30(115)	201			201	
50(101)	200			200	
	199	**50(98)**		199	
Inbound aggressive order (limit price, side and quantity)					
	201	**100(160)**			

The message flow on Market Data shows the transaction effected with order 101 and at that point transitions to a volatility auction phase.

Message flow on WATS Market Data

Step	Message	Message Details	Note
1	OrderAdd	N/A	OrderAdd for the aggressive buy limit is suppressed (order 160).
2	OrderExecute	N/A	The OrderExecute attendant to the aggressive order is suppressed (order 160).
3	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=101 TradeId=31 ExecutionPrice=200 ExecutionQuantity=50	OrderExecute for passive order 101.
4	Trade	Message=Trade Price=200 Quantity=50 TradeId=31	Trade related to order 101.
5	InstrumentStatusChange	tradingPhaseType= AuctionVolatilityDynamic	Signals the change of Market Phase to a volatility auction.
6	OrderBookEvent	eventType=Clear	Signal to clear the book in preparation for an auction.
7	At this point the amount leftover in order 160 enters the auction.		
8	The auction continues with participants entering orders. For more details on the sequence of Market Data messages, please consult the Market Data document (GPW WATS 3.01 Market Data Protocol).		

11.5. Aggressive Order Modifications

11.5.1. AGGRESSIVE MODIFICATION OF RESTING ORDER RESULTING IN A FULL FILL

A passive limit order (98) is resting on book. It is modified to an aggressive order by changing the limit price to 200. The aggressive modification matches 2 passive orders and is fully filled.

Initial Order Book State (Instrument A)			Final Order Book State		
Asks (sell)	Price	Bids (buy)	Asks (sell)	Price	Bids (buy)
100(101),200(115)	200		**150(115)**	200	
	199	**150(98)**		199	
Aggressive modification of order 98 to price 200					
	200	**150(98)**			

The message flow on Market Data reflects the fact that the aggressive modification touched 2 passive orders in total. The OrderModify message for order 98 is not communicated to the market and instead the updated order is matched by the matching engine.

At the end of the process an OrderCancel message is sent signaling that order 98 has been fully filled and can be deleted from the book. This ensures that the book remains uncrossed.

Message flow on WATS Market Data

Step	Message	Message Details	Note
1	OrderModify	N/A	OrderModify for order 98 is suppressed since it is aggressive in nature.
2	OrderExecute	N/A	2 OrderExecutes attendant to the aggressive order are suppressed (order 98).
3	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=101 TradeId=31 ExecutionPrice=200 ExecutionQuantity=100	OrderExecute for passive order 101 crossed by modified order 98.
4	OrderExecute	Message=OrderExecute Quantity=150 PublicOrderId=115 TradeId=32 ExecutionPrice=200 ExecutionQuantity=50	OrderExecute for passive order 115 crossed by modification.
5	Trade	Message=Trade Price=200 Quantity=100 TradeId=31	Trade message related to TradeId=31 on order 101.
6	Trade	Message=Trade Price=200 Quantity=50	The second trade message for the second passive order.

Message flow on WATS Market Data

Step	Message	Message Details	Note
		TradeId=32	
7	OrderCancel	publicOrderId=98	Order 98 is deleted from the book following its modification and fill.

11.5.2. AGGRESSIVE MODIFICATION OF RESTING ORDER RESULTING IN A PARTIAL FILL

This example is similar to the previous one, with the exception that the order modified into an aggressor I is not fully filled and at the end of the matching process its unfilled part rests on the book at the updated price level.

Initial Order Book State (Instrument A)			Final Order Book State		
Asks (sell)	**Price**	**Bids (buy)**	**Asks (sell)**	**Price**	**Bids (buy)**
100(101),20(115)	200			200	**30(98)**
	199	**150(98)**		199	
Aggressive modification of order 98 to price 200					
	200	**150(98)**			

In this example, the modified order collides with 2 orders and rests on book with the leftover quantity at a new price level.

The difference from the prior example rests with the insertion of the leftover quantity of the modified order into the order book by means an OrderModify in step 5.

Message flow on WATS Market Data

Step	Message	Message Details	Note
1	OrderModify	N/A	OrderModify for order 98 is suppressed at this point in the sequence. It will be published in step 5.
2	OrderExecute	N/A	2 OrderExecutes attendant to the aggressive order are suppressed (order 98).
3	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=101 TradeId=31 ExecutionPrice=200 ExecutionQuantity=100	OrderExecute for passive order 101 crossed by modified order 98.
4	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=115 TradeId=32 ExecutionPrice=200 ExecutionQuantity=20	OrderExecute for passive order 115 crossed by modification.
5	OrderModify	PublicOrderId=98 Price=200 Quantity=30	The OrderModify for the modified order is injected. It features the new price and a quantity corresponding to the unfilled part.
6	Trade	Message=Trade Price=200 Quantity=100 TradeId=31	Trade message related to TradeId=31 on order 101.
7	Trade	Message=Trade Price=200	The second trade message for the second passive order.

Message flow on WATS Market Data

Step	Message	Message Details	Note
		Quantity=20 TradeId=32	

11.6. ICEBERGS

11.6.1. AGGRESSIVE ORDER HITS PEAK

In this example, an incoming aggressive order is fully filled after partially matching an iceberg's peak. The first order on the sell side of the book is an iceberg with peak 100. Upon matching with the incoming order, the peak is reduced to 70 and an OrderExecute message together with a Trade are generated. No new PublicOrderID is asigned since the peak is not exhausted. We use the following notation: 100/500(101) is an iceberg with peak 100, hidden 500 (therefore, total volume 600) and PublicOrderID 101.

Initial Order Book State (Instrument A)			**Final Order Book State**		
Asks (sell)	**Price**	**Bids (buy)**	**Asks (sell)**	**Price**	**Bids (buy)**
50(102)	201		**50(102)**	201	
100/500(101),200(115)	200		**70/500(101),200(115)**	200	
	199	**50(98)**		199	**50(98)**
Inbound aggressive order (limit price, side and quantity)					
	200	**30(160)**			

The messages from this order flow are shown in the next table.

Message flow on WATS Market Data

Step	Message	Message Details	Note
1	OrderAdd	N/A	OrderAdd for the aggressive buy limit is suppressed (order 160).
2	OrderExecute	N/A	OrderExecute attendant to the aggressive order is suppressed (order 160).
3	OrderExecute	Message=OrderExecute Quantity=70 PublicOrderId=101 TradeId=31 ExecutionPrice=200 ExecutionQuantity=30	OrderExecute related to the part of the peak (order 101). Only the message related to the passive side is published.
4	Trade	Message=Trade Price=200 Quantity=30 TradeId=31	Trade message for the partial execution of peak.

11.6.2. AGGRESSIVE ORDER REMOVES PEAK

Here, an aggressive order sweeps the peak. The iceberg of the peak is reloaded with the part of the hidden quantity and a new PublicOrderID is assigned.

Initial Order Book State (Instrument A)			Final Order Book State		
Asks (sell)	**Price**	**Bids (buy)**	**Asks (sell)**	**Price**	**Bids (buy)**
50(102)	201		**50(102)**	201	
70/500(101),200(115)	200		**200(115),100/400(101→170)**	200	
	199	**50(98)**		199	**50(98)**
Inbound aggressive order (limit price, side and quantity)					
	200	**70(160)**			

The messages from this order flow are shown in the next table.

Message flow on WATS Market Data

Step	Message	Message Details	Note
1	OrderAdd	N/A	OrderAdd for the aggressive buy limit is suppressed (order 160).
2	OrderExecute	N/A	OrderExecute attendant to the aggressive order is suppressed (order 160).
3	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=101 TradeId=35 ExecutionPrice=200 ExecutionQuantity=70	OrderExecute related to the execution of the peak (order 101). The peak is exhausted and Quantity=0.
4	Trade	Message=Trade Price=200 Quantity=70 TradeId=35	Trade message for the partial execution of peak.
5	OrderAdd	Message=OrderAdd PublicOrderId=170 Side=Sell Price=200 Quantity=100	The peak is reloaded with display quantity of 100 and a new PublicOrderID (170) is assigned. The reloaded peak loses priority.

11.6.3. AGGRESSIVE ORDER REMOVES PEAK AND PART OF HIDDEN QUANTITY

A similar process unfolds when an aggressive order sweeps the peak and part of the hidden quantity. The peak of the iceberg is executed first. After the peak is exhausted the iceberg loses priority and moves to the end of the price level and the next order in the line of priority is processed. That order is fully filled with quantity of 50. Finally, the hidden part of the iceberg is processed as fully as possible. Since the hidden quantity is not exhausted, the iceberg's peak is reloaded and a new PublicOrderID is assigned. This example differs slightly from the previous ones in that the quantity left in the hidden part is less than the preset display quantity. The remaining volume of 80 is reloaded instead of the full peak of 100. The peak loses priority.

Initial Order Book State (Instrument A)			**Final Order Book State**		
Asks (sell)	**Price**	**Bids (buy)**	**Asks (sell)**	**Price**	**Bids (buy)**
50(102)	201		**50(102)**	201	
100/130(101),50(115)	200		**80/0(101→180)**	200	
	199	**50(98)**		199	**50(98)**
Inbound aggressive order (limit price, side and quantity)					
	200	**200(160)**			

The messages from this order flow are shown in the next table.

Message flow on WATS Market Data

Step	Message	Message Details	Note
1	OrderAdd	N/A	OrderAdd for the aggressive buy limit is suppressed (order 160).
2	OrderExecute	N/A	OrderExecute attendant to the aggressive order is suppressed (order 160).
3	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=101 TradeId=40 ExecutionPrice=200 ExecutionQuantity=100	OrderExecute related to the execution of the peak (order 101). The peak is exhausted and Quantity=0.
4	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=115 TradeId=41 ExecutionPrice=200 ExecutionQuantity=50	OrderExecute related to the order next in line (order 115). The order is exhausted and Quantity=0.
5	OrderExecute	N/A	OrderExecute attendant to the hidden part of the iceberg is suppressed (order 101).
6	Trade	Message=Trade Price=200 Quantity=100 TradeId=40	Trade message for the partial execution of peak.

7	Trade	Message=Trade Price=200 Quantity=50 TradeId=41	Trade message for the execution of the order with next priority (order 115).
8	Trade	Message=Trade Price=200 Quantity=50 TradeId=42	Trade message for the partial execution of the hidden quantity.
9	OrderAdd	Message=OrderAdd PublicOrderId=180 Side=Sell Price=200 Quantity=80	The peak is reloaded with the remainder of the hidden quantity. It's less than the nominal display quantity of 100. A new PublicOrderID (180) is assigned. The reloaded peak loses priority.

11.6.4. AGGRESSIVE ORDER REMOVES PEAK AND HIDDEN QUANTITY

When an aggressive order sweeps an entire iceberg the process is similar to when a part of the hidden quantity is consumed. The peak and the hidden quantity are executed in 2 separate steps.

Initial Order Book State (Instrument A)			Final Order Book State		
Asks (sell)	**Price**	**Bids (buy)**	**Asks (sell)**	**Price**	**Bids (buy)**
50(102)	201		**50(102)**	201	
100/100(101),50(115)	200			200	
	199	**50(98)**		199	**50(98)**
Inbound aggressive order (limit price, side and quantity)					
	200	**250(160)**			

The messages from this order flow are shown in the next table.

Message flow on WATS Market Data

Step	Message	Message Details	Note
1	OrderAdd	N/A	OrderAdd for the aggressive buy limit is suppressed (order 160).
2	OrderExecute	N/A	OrderExecute attendant to the aggressive order is suppressed (order 160).
3	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=101 TradeId=50 ExecutionPrice=200 ExecutionQuantity=100	OrderExecute related to the execution of the peak (order 101). The peak is exhausted and Quantity=0.
4	OrderExecute	Message=OrderExecute Quantity=0 PublicOrderId=115 TradeId=51 ExecutionPrice=200 ExecutionQuantity=50	OrderExecute related to the execution of order 115 with quantity 50. The order is fully filled.
5	OrderExecute	N/A	OrderExecute attendant to the hidden part of the iceberg is suppressed (order 101).
6	Trade	Message=Trade Price=200 Quantity=100 TradeId=50	Trade message for the partial execution of peak.
7	Trade	Message=Trade Price=200 Quantity=50 TradeId=51	Trade message for the execution of order 115.

8	Trade	Message=Trade Price=200 Quantity=100 TradeId=52	Trade message for the full execution of the hidden quantity.

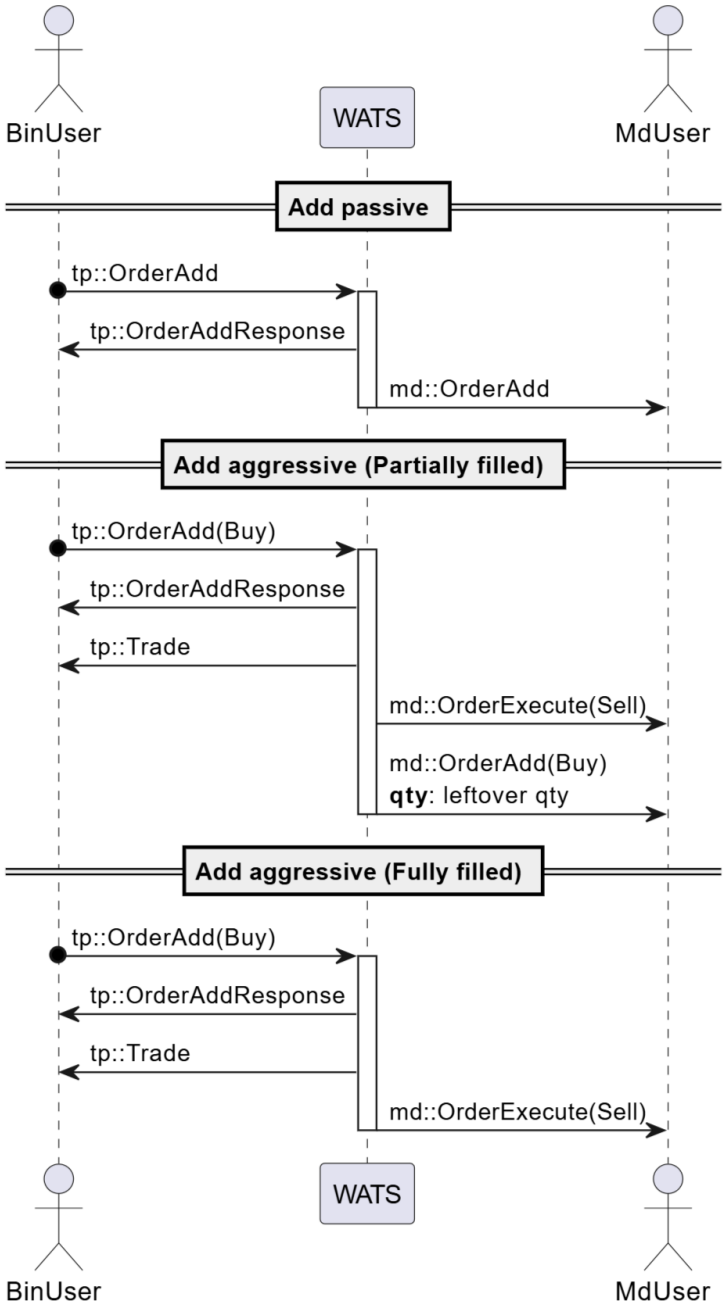
11.7. SEQUENCE DIAGRAMS

The sequence diagrams for messages on the Trading Port and Market Data are included below. They treat cases of new orders and order modifications.

The user submits OrderAdd or OrderModify messages on the native trading port. The sequence shows the sequence of messages on the Native and Market Data ports following aggressive order executions.

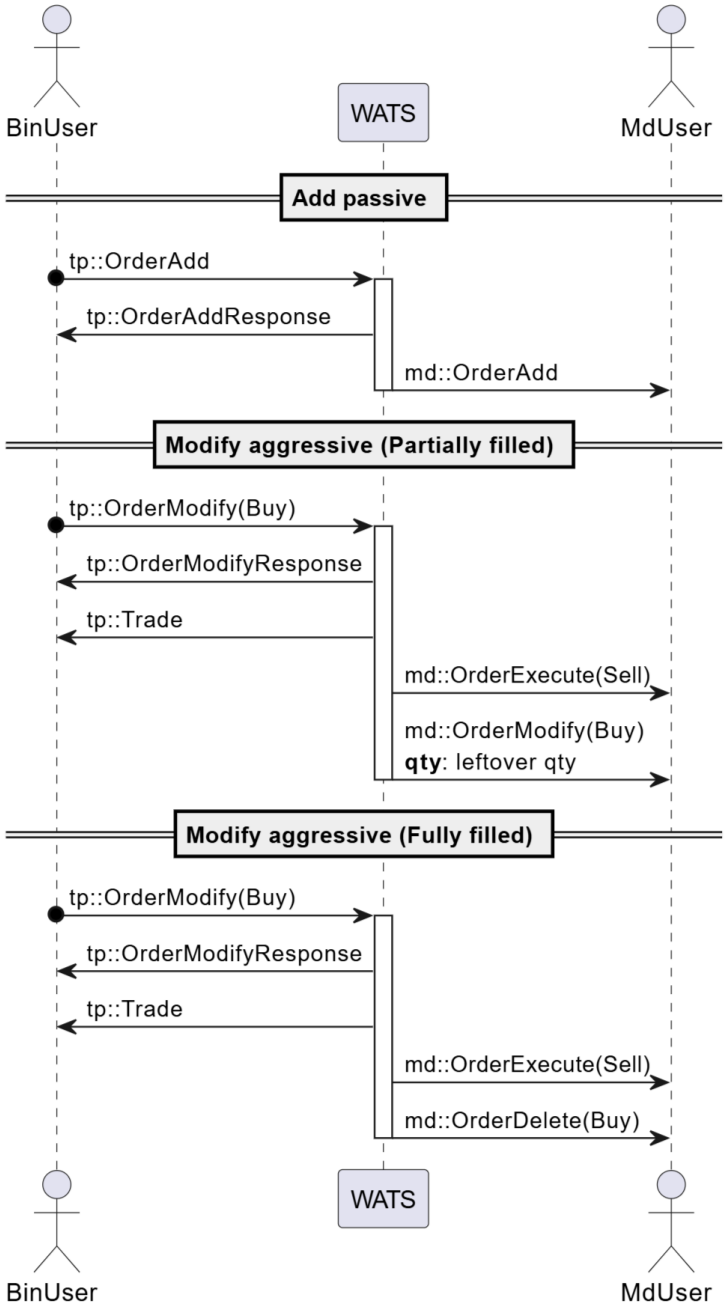
11.7.1. NEW ORDERS

Figure 9 - New Order



11.7.2. ORDER MODIFICATIONS

Figure 10- Order Modifications



12. General Considerations

12.1. GPW WATS Time

GPW WATS runs on UTC time and GPW WATS Market Data serves information signed by UTC time stamps. All messages and all fields which include a time stamp are to be assumed UTC. The timestamp and sourceTimestamp fields in the Header of Market Data messages represent UTC time.

12.2. Matching Engine Failure

Under normal operating conditions, the matching engine sends real time messages for each Market data stream from redundant sites A and B. Price-Time Priority Determination.

Each message sent on Market Data features a Header. The Header contains the SeqNum field which represents the sequence number assigned to an order by the Market Data Sequencer. The priority of an order should be determined using this value. For example, for orders at the same price level the order with the lower SeqNum has priority over an order with a higher SeqNum.

One should note that changes to order priority following order modification are communicated in the OrderModify message by the PriorityFlag value - an order can either retain or lose priority following modification.

12.3. TVTIC

Messages OrderExecute and Trade provide Trading Venue Transaction Identification Code which is a number generated by the WATS trading system for transactions reporting purposes in accordance with Article 12 of Commission Delegated Regulation (EU) 2017/580.

The TVTIC numbers are unique systemwide and assigned across all market segments and all financial instruments covering both book and off-book trades.

At the start of the trading day TVTIC is reset and starts from 1.

Note: for redistributed instruments, TVTICs in Trade messages are assigned by the outside trading system sending the message. The foreign TVTICs may therefore conflict with native WATS TVTICs.

The table shows Market Data messages relaying the TVTICs.

Message	Attribute containing the TVTIC
OrderExecute	tradeId
Trade	tradeId

12.4. PriceUpdate use cases

PriceUpdate is generated for redistributed instruments (TBSP) and WATS native instruments.

For redistributed instruments PriceUpdate contains fixing prices from TBSP. For WATS native instruments PriceUpdate carries updates to the reference price or midpoint for structured products (priceUpdateType= ReferencePrice, or MidPrice).

There are three circumstances where a PriceUpdate message may appear on Market Data:

- For redistributed instruments, PriceUpdate contains the prices redistributed TBSP instruments behaves as described in Chapter 8.
- For structured products, PriceUpdate contains the midpoint of the latest market maker quotes. Because base instruments may not be quoted on WATS and their prices may change overnight, the midpoint is published each morning to reflect the latest market maker quotes.
- In normal circumstances, the Instrument message conveys the reference price for an instrument. During the NoTradingEarlyMonitoring phase, the Market Operator may decide to update the reference price by sending a PriceUpdate for a selected instrument. This new reference price takes precedence over the reference price published in the Instrument reference data message.

12.5. Numbers and Decimal Places

In Market Data numbers are encoded differently depending on Type. The information is also available in the description fields of market_data.json for the relevant Types. The following table provides a guideline on how to interpret Types in Market Data. The related types follow the precision of the main types.

Type	Related Type	Kind	Number of decimal places	Decimal interpretation of i64 1
CollarValue		i64	8	0.00000001
Multiplier		i64	8	0.00000001
Price	PercentageChange IndexValue	i64	8	0.00000001
TickSize		i64	8	0.00000001
Value		i64	2	0.01

From a mathematical perspective, CollarValue, Multiplier, Price and TickSize should be divided by 10^8 and Value by 10^2. Please note that floating point operations on financial values are to be avoided.

The market_data.json file provides a description of type formats, as illustrated below:

Definition of Multiplier type in market_data.json

```
{
"kind": "Alias",
"name": "Multiplier",
"alias": "Number",
"description": "Multiplier as decimal(18,8) without decimal separator."
}
```

12.6. ENCRYPTION

Market Data is secured to prevent unauthorized access to data. Encryption is performed by the Encryption Engine component and involves encrypting sensitive data in online Market Data messages. The list of fields in messages subject to encryption is fixed. Encryption involves the use of a key which is generated per session and only valid for that session.

Encryption keys are disseminated by the Snapshot service before the start of each trading day by EncryptionKey messages. The encryptionKeyId is used to identify the appropriate encryption keys for different messages.

GPW WATS implements the ChaCha20 encryption algorithm in CTR (i.e., counter) mode. In this mode, the following chacha20 initialization should be performed:

- Encryption Key - the Online Market Data Header message refers to an encryption key (encryptionKeyId) which is distributed by EncryptionKey messages, as a part of Market Data snapshot messages,
- Nonce - The value for the nonce is all zeros (a vector of 12 bytes/96 bits filled with zeros). This may change in the future as new designs are adopted.
- The counter starts with the value 0 and is encoded as a 32 bit integer.

A provisioned stream cipher object is created for each captured key and is used to decrypt designated fields of messages. In most implementations there is no need to alter any of the initial parameters of the encryption object, such as counter, as they should be properly set by the implementation, which provides keystream for decryption as the stream of data progresses. Each Header message contains an encryptionOffset which determines the absolute offset in the encryption stream used to encrypt the message. The decryption process should only be performed if the isEncrypted field in the Header message is true.

The list of encrypted fields and messages is determined on a business basis.

12.7. STREAM MANAGEMENT

12.7.1. THROUGHPUT OF MARKET DATA

Market Data is throttled per multicast stream. The throttling levels are configured at system level.

Market Data throughput is limited to that stated in the Connectivity document. Currently each stream is limited to 40Mb/s.

Upon reaching the limit, the latency of the Market Data stream will increase until all the public data is sent out and normal behavior resumes.

Market Data throttling does not affect order entry gateways. Market participants will continue to be able to enter orders on the trading ports while Market Data is throttled. They will receive order acknowledgments and trade confirmations but will be unable to see the real-time state of the order book until Market Data catches up.

12.7.2. STREAMID AND INDIVIDUAL STREAMS

The MD header contains the streamID attribute which indicates the stream to which a particular MD message belongs.

StreamIDs enable GPW WATS to channel MD messages into different streams. Each stream is filtered at system level and contains non-nested/non-overlapping fragments of the market structure. The relevant market structure elements are identified by ElementIDs.

StreamIDs are static.The ReplayRequest message takes the StreamID as parameter to replay the requested market data content.

In the future streams and their content will be communicated to users through various channels. Please refer to the Connectivity document, GPW WATS 6.01 Connectivity, for more information on streams and their use.

12.7.3. ENCRYPTION OF STREAMS

Market Data implements encryption on the full depth and BBO streams.

Under this scheme, select Market Data messages are encrypted for clients who require personalized streams of public market information. Such clients will be able to choose from an array of service packages and receive them as encrypted streams.

12.7.4. DETAILS OF THE ENCRYPTION SCHEME

The reference data related to encryption is distributed by means of the Snapshot and each market participant receives their own keys through their own private Snapshot channel. The key messages distributed by Snapshot are:

MD message	Note
EncryptionKey	• Id: id of the encryption key. This id is referred to in the header (MD::Header::encryptionKeyId attribute) of the message featuring encrypted content • secretKey: the encryption key

Encryption is performed on a per instrument basis. This means that Market Data messages related to each individual instrument will be encrypted.

The header of each Market Data message contains attributes used in the encryption scheme. The header is tailored to each message and will vary accordingly. For encrypted messages the following header attributes are of interest:

MD Header attribute	Note
isEncrypted	Yes, if the message contains encrypted attributes. No if the message has no encrypted content.
encryptionOffset	Encryption offset for the algorithm.

MD Header attribute	Note
encryptionKeyId	The key to decrypt the message attributes. Each message header may feature a different encryption key specific to that particular message. This attribute refers to MD::EncryptionKey::id.

Encrypted attributes are those marked with a metadata": { "isEncrypted": true} tag in the JSON definition file.

12.8. Currency

In WATS, currencies are a dictionary of type Currency where currencies are specified according to the ISO-4217 standard. In line with ISO-4217, the code 'XXX' appears when a currency is not stipulated.

12.9. Order Book data after session close

After the trading session ends for an instrument (i.e. when the trading phase for an instrument changes from NoTradingLateMonitoring to NoTradingClosed) the order book updates (OrderAdd, OrderCancel, OrderModify, OrderExecute, PriceLevelSnapshot) cease to be disseminated.

From that time onward, Market Data recipients should disregard the state of the order book and consider it hidden.

The up-to-date order book (cleaned up from the orders that expired at the end of the previous day) will be published the next day in the morning, upon commencement of the first continuous trading market phase in the trading schedule.

12.10. HeartBeat

HeartBeat messages consist of the Header. They contain a sequence number seqNum. Heartbeat echoes back the sequence number of the last sent message. Heartbeat does not increment the sequence number.

12.11. Endianness

In GPW WATS, messages are formatted in little-endian byte order.

12.12. Market Data Suppression

Market Data does not broadcast all information related to trading activity and suppresses selected information. The data suppression scheme does not hide relevant information from Market participants but instead serves to streamline and enhance the performance of the Market Data service.

Order type	Note
Market	Market orders are suppressed and are not communicated by Market Data since they do not persist in the order book. Market Data broadcasts Trade and OrderExecute messages related to Market orders.
Limit	Limit orders, unexecuted or partially executed, are disclosed on Market Data.
Iceberg	Iceberg peaks are displayed since they function as limits.
Stop	The behavior of Stop-orders on Market Data varies. In general, Stop-orders are not published on Market Data until they activate and enter the Order Book. Stop-orders which activate into

Order type	Note
	Market orders follow rules for Market orders and are suppressed on Market Data. Stop-orders which activate into limit orders follow rules for Limits and are published on Market Data.

12.13. General Issues

12.13.1. MAXIMUM MESSAGE SIZE

The maximum message payload length that Market Data can handle is:

- 1500 - message header size (42 b) - IP and UDP header sizes (28 b) = 1430 bytes.

12.13.2. THE DEPTH OF PRICELEVELSNAPSHOT MESSAGE

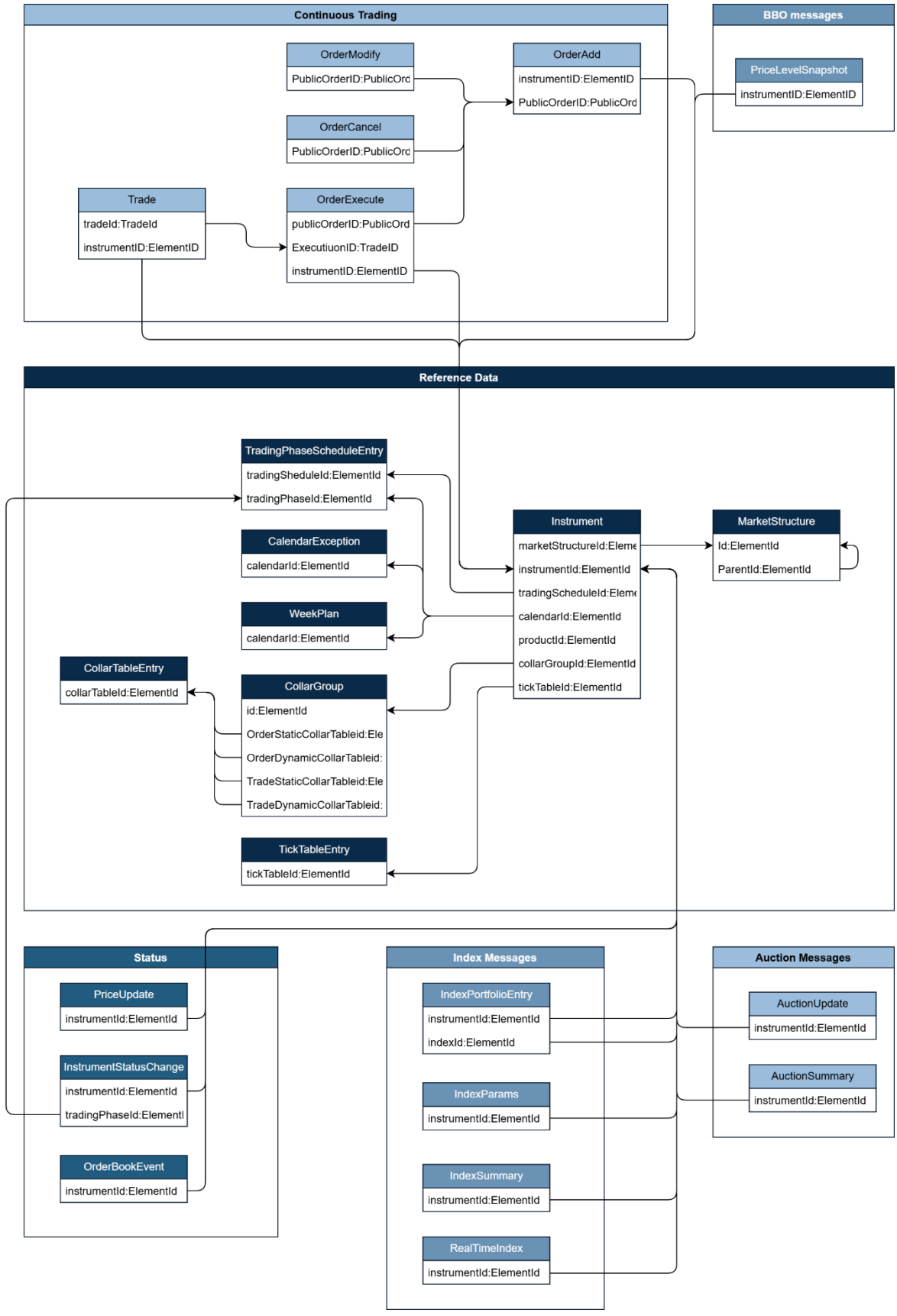
This is a configuration issue.

The MD message handles up to 5 levels, but fewer may be broadcast.

PriceLevelSnapshot::maxDepth shows how many price levels are broadcast at any time.

12.14. Market Data Interconnect Diagram

Figure 11 - Market Data Messages



12.15. Presence Maps in WATS

There is no presence map mechanism on WATS Market Data. WATS messages are of fixed length and can't be shortened.

On the other hand, Market Data implements presence flags in some messages. These flags may inform participants whether a message attribute contains values or no data. The presence flags are message specific and apply to specific message attributes. In addition, presence flags do not shorten the length of messages but only inform participants of the meaning of the data transmitted.

12.16. (MMT) Market Model Typology tags not used in WATS

The MD::Trade message contains attributes whose values are not used or implemented in WATS The attributes in question relate to the MMT (Market Model Typology). This section lists the unused attributes.

Note : the "not used" remark applies only to trades matched within WATS trading engine. Trades that are matched in other platforms and only redistributed by WATS market data service may utilize full range of values.

<table>
<tr><th>Attribute</th><th>Description</th></tr>
<tr><td>mmtMarketMechanism</td><td>MMT Market Mechanism
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>CentralLimitOrderBook</td><td>1</td><td>1 = Central Limit Order Book</td></tr>
<tr><td>QuoteDrivenMarket</td><td>2</td><td>Not used.</td></tr>
<tr><td>DarkOrderBook</td><td>3</td><td>Not used.</td></tr>
<tr><td>OffBook</td><td>4</td><td>4 = Off Book</td></tr>
<tr><td>PeriodicAuction</td><td>5</td><td>Not used.</td></tr>
<tr><td>RequestForQuotes</td><td>6</td><td>Not used.</td></tr>
<tr><td>Other</td><td>7</td><td>7 = Any other including Hybrid</td></tr>
</table></td></tr>
<tr><td>mmtTradingMode</td><td>MMT Trading Mode
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>UndefinedAuction</td><td>1</td><td>1 = Undefined Auction</td></tr>
<tr><td>ScheduledOpeningAuction</td><td>2</td><td>O = Scheduled Opening Auction</td></tr>
<tr><td>ScheduledClosingAuction</td><td>3</td><td>K = Scheduled Closing Auction</td></tr>
<tr><td>ScheduledIntradayAuction</td><td>4</td><td>Not used.</td></tr>
<tr><td>UnscheduledAuction</td><td>5</td><td>U = Unscheduled Auction</td></tr>
<tr><td>ContinuousTrading</td><td>6</td><td>2 = Continuous Trading</td></tr>
<tr><td>AtMarketCloseTrading</td><td>7</td><td>Not used.</td></tr>
<tr><td>OutOfMainSessionTrading</td><td>8</td><td>Not used.</td></tr>
<tr><td>TradeReportingOnExchange</td><td>9</td><td>Not used.</td></tr>
<tr><td>TradeReportingOffExchange</td><td>10</td><td>Not used.</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Attribute</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>TradeReportingSystematicInternaliser</td><td>11</td><td>Not used.</td></tr>
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</td></tr>
<tr><td>mmtTransactionCategory</td><td>
MMT Transaction Category
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>DarkTrade</td><td>1</td><td>Not used.</td></tr>
<tr><td>TradePriceImprovement</td><td>2</td><td>Not used.</td></tr>
<tr><td>PackageTrade</td><td>3</td><td>Not used.</td></tr>
<tr><td>ExchangeForPhysicalsTrade</td><td>4</td><td>Not used.</td></tr>
<tr><td>None</td><td>5</td><td>- = None apply (a standard trade for the Market Mechanism and Trading Mode)</td></tr>
</table>
</td></tr>
<tr><td>mmtNegotiationIndicator</td><td>
MMT Negotiation Indicator
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NegotiatedTrade</td><td>1</td><td>N = Negotiated Trade</td></tr>
<tr><td>NegotiatedTradeLiquid FinancialInstruments</td><td>2</td><td>Not used.</td></tr>
<tr><td>NegotiatedTradeIlliquid FinancialInstruments</td><td>3</td><td>2 = Negotiated Trade in Illiquid Financial Instruments</td></tr>
<tr><td>NegotiatedTradeOther</td><td>4</td><td>Not used.</td></tr>
<tr><td>NoNegotiatedTrade</td><td>5</td><td>Not used.</td></tr>
<tr><td>PreTradeTransparencyWaiver IlliquidInstrument</td><td>6</td><td>Not used.</td></tr>
<tr><td>PreTradeTransparencyWaiver StandardMarketSize</td><td>7</td><td>Not used.</td></tr>
<tr><td>PreTradeTransparencyWaivers IlqdAndSize</td><td>8</td><td>Not used.</td></tr>
</table>
</td></tr>
<tr><td>mmtAgencyCrossTradeIndicator</td><td>
MMT Agency Cross Trade Indicator
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>AgencyCrossTrade</td><td>1</td><td>Not used.</td></tr>
<tr><td>NoAgencyCrossTrade</td><td>2</td><td>Default value.</td></tr>
</table>
</td></tr>
<tr><td>mmtBenchmarkReferencePriceIndicator</td><td>
MMT Benchmark / Reference Price Indicator
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>BenchmarkTrade</td><td>1</td><td>Not used.</td></tr>
<tr><td>ReferencePriceTrade</td><td>2</td><td>Not used.</td></tr>
<tr><td>NoBenchmarkorReferencePriceTrade</td><td>3</td><td>Default value.</td></tr>
</table>
</td></tr>
<tr><td>mmtOrdinaryTradeIndicator</td><td>
MMT Ordinary Trade Indicator
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>PlainVanillaTrade</td><td>1</td><td>Default value.</td></tr>
<tr><td>NonPriceFormingTrade</td><td>2</td><td>Not used.</td></tr>
<tr><td>TradeNotContributingToPriceDiscovery</td><td>3</td><td>Not used.</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Attribute</th><th>Description</th></tr>
<tr><td></td><td>
<table>
<tr><td>PricePending</td><td>4</td><td>Not used.</td></tr>
</table>
</td></tr>
<tr><td>mmtPostTradeDeferralReason</td><td>
MMT Post-Trade Deferral Reason
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>ImmediatePublication</td><td>1</td><td>Default value.</td></tr>
<tr><td>NonImmediatePublication</td><td>2</td><td>Not used.</td></tr>
<tr><td>NonImmediatePublicationLargeInScale</td><td>3</td><td>Not used.</td></tr>
<tr><td>NonImmediatePublicationIlliquidInstrument</td><td>4</td><td>Not used.</td></tr>
<tr><td>NonImmediatePublicationSizeSpecific</td><td>5</td><td>Not used.</td></tr>
<tr><td>NonImmediatePublicationIlqdAndSize</td><td>6</td><td>Not used.</td></tr>
<tr><td>NonImmediatePublicationIlqdAndLrgs</td><td>7</td><td>Not used.</td></tr>
</table>
</td></tr>
<tr><td>mmtPostTradeDeferralType</td><td>
MMT Post-Trade Deferral Type
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>LimitedDetailsTrade</td><td>1</td><td>Not used.</td></tr>
<tr><td>DailyAggregatedTrade</td><td>2</td><td>Not used.</td></tr>
<tr><td>VolumeOmissionTrade</td><td>3</td><td>Not used.</td></tr>
<tr><td>FourWeeksAggregationTrade</td><td>4</td><td>Not used.</td></tr>
<tr><td>IndefiniteAggregationTrade</td><td>5</td><td>Not used.</td></tr>
<tr><td>VolumeOmissionTradeAggregated</td><td>6</td><td>Not used.</td></tr>
<tr><td>NotApplicable</td><td>7</td><td>Default value.</td></tr>
</table>
</td></tr>
</table>

13. Sending Rules

This table outlines the rules for the publication of messages by Market Data.

Category	Message type	Message	Sending Rule
Trading and execution	Auction	AuctionUpdate	Auctions only. An OrderAdd / OrderModify / OrderCancel entered the order book during an auction.
		AuctionSummary	Auctions only. Auction ended with an uncrossing, the final IMP/IMV are determined.
	Continuous Trading (Full Depth)	OrderAdd	A limit order entered the order book, was modified or cancelled. Valid for continuous trading. Suppressed during auctions.
		OrderCancel	
		OrderModify	
		OrderBookEvent	Instruction for participants to 1. clear the order book in anticipation of auction (OrderBookEventType=Clear) 2. expect the book rebuilding process to start upon completion of auction (OrderBookEventType= ResendStart) 3. prepare to continue to the next phase after the book rebuilding process has concluded (OrderBookEventType= ResendEnd)
		OrderExecute	Two orders matched.
	BBO	PriceLevelSnapshot	BBO price levels are sent according to a BBO publication interval. Published only in case of activity on order book. In case of no activity on book, the message is not published.
Status messages	Status	InstrumentStatusChange	1. Instrument enters a new trading phase 2. Stressed Market Conditions initiated for instrument 3. Instrument status has changed, for example suspensions 4. Hybrid market specific statuses: No Quotes and Knockout
		PriceUpdate	1. Redistributed instruments 2. Hybrid market midpoint 3. On CLOB, comes out during Early Monitoring to show a new reference price.
		OrderCollars / TradeCollars	New Order and / or Trade collars. 1. If dynamic collars are configured, updated collars come out after OrderExecutes 2. Static collars are updated at the start of each trading phase 3. After auction uncrossing
		Trade	Follows an OrderExecute message
		News	A news item communicated to the market.
		PositionReport	Redistributed data.

Category	Message type	Message	Sending Rule
			1. At start of trading day 2. End of day 3. Following a Trade in a derivative instrument 4. Predetermined intervals
Summary messages	Statistical	InstrumentSummary	1. End of trading session, after market close 2. Following a corporate auction
		ProductSummary	1. End of trading session, after market close
Reference Data	Market Structure	MarketStructure	Pre-session during reference data upload
		Instrument	
		TickTableEntry	
	Trading Calendar	TradingPhaseScheduleEntry	1. Pre-session during reference data upload 2. During Next Dy Reference Data after end of session
		CalendarException	
		WeekPlan	
	Order and Trade Collars	CollarTableEntry, CollarGroup	
	Pricing Messages	AccruedInterestTableEntry	
		IndexationTableEntry	
	Redistribution	ExternalUnderlying	During Next Dy Reference Data after end of session
Index	Index messages	IndexParams	Pre-session during reference data upload
		IndexPortfolioEntry	Pre-session during reference data upload
		RealTimeIndex	During the trading day whenever data is received
		IndexSummary	Session close
Session Management	Encryption	EncryptionKey	Only on Snapshot, in response to Login
	Technical	TradingSessionStatus	1. The beginning and end of the technical session 2. Beginning and end of reference data within a technical session 3. Start and end of next day reference data 4. Start and end of trading session for a MIC. StartOfTradingSessionMIC is sent whenever any instrument in a MIC transitioned from a NoTradingClosed phase to NoTradingEarlyMonitoringEarlyMontoring EndOfTradingSessionMIC is sent whenever all instruments in a MIC transitioned from the NoTradingLateMonitoring LateMonitoring phase to NoTradingClosed
		EndOfSnapshot	Snapshot has finished
		ReplayRequest	Not Applicable – this is a request message not a MD broadcast message

Category	Message type	Message	Sending Rule
		HeartBeat	1. whenever there is no traffic on MD for a preset period of time (currently 1s) 2. sent periodically thereafter at 1s intervals until inactivity ends
	Login	Login	Login to Snapshot
		LoginResponse	LoginResponse is sent in response to Login
		Logout	Logout sent by the client to Snapshot

14. Market Data Protocol

Name	Id	Length (bytes)	Description
AccruedInterestTableEntry	36	58	Message describing an accrued interest table entry.
AuctionSummary	270	63	Auction summary, including auction price and quantity.
AuctionUpdate	268	110	Updates IMP/IMV and BBO during an auction.
CalendarException	35	51	Upcoming calendar day exception.
CollarGroup	299	62	Message defining a collar group.
CollarTableEntry	108	96	Collar table definition.
EncryptionKey	24	78	Encryption key and ID.
EndOfSnapshot	670	46	Marks the end of a snapshot.
ExternalUnderlying	806	155	The message contains a list of underlyings not traded in WATS. It concerns underlyings for derivatives and structured products.
Heartbeat	1	42	A message type used to check connectivity.
IndexParams	503	187	The metadata message for the given index.
IndexPortfolioEntry	502	101	The message contains information about a product providing the index portfolio. The entire index portfolio is sent as successive IndexPortfolioEntry messages.
IndexSummary	501	119	The index summary message provides a summary of the data calculated in a stock index for a given day.
IndexationTableEntry	37	58	Message describing an indexation table entry.
Instrument	279	741	Definition of a financial instrument.
InstrumentStatusChange	26	53	Start of a new trading phase.
InstrumentSummary	671	137	Provides brief instrument summary.
Login	667	53	Login message to the Snapshot service.
LoginResponse	668	43	Response to a Login message.
Logout	669	42	A Logout message is sent by the client to initiate disconnection from the service.

Name	Id	Length (bytes)	Description
MarketStructure	275	105	Messages defining the Trading Venue's market structure.
News	517	932	News message.
OrderAdd	9	72	New (limit) order added to order book.
OrderBookEvent	410	47	Order book event for given instrument
OrderCancel	11	55	Order deleted.
OrderCollars	272	87	New order price collars.
OrderExecute	12	83	Execution report.
OrderModify	10	72	Order modified.
PositionReport	801	82	Report of open positions.
PriceLevelSnapshot	257	359	BBO price levels and volumes.
PriceUpdate	271	73	Message indicating a price or YTM update
ProductSummary	672	671	Session day product summary.
RealTimeIndex	500	130	Message providing real-time index values for instruments.
TestEvent	8192	243	Test event
TickTableEntry	32	62	Tick size definition.
TopPriceLevelUpdate	256	65	BBO Level 1.
Trade	38	220	Enriched trade report.
TradeCollars	273	71	New trade price collars.
TradingPhaseScheduleEntry	27	82	Trading phase definition.
TradingSessionStatus	23	55	The Trading Session Status provides information on the status of a market and on a trading day events.
WeekPlan	33	53	Trading week plan.

14.1.HEADER

Name	Data type	Kind	Length (bytes)	Description
length	MsgLength	Alias (u16)	2	Message length.
msgType	MsgType	Enum	2	Type of message (e.g. OrderExecute).
version	MsgVersion	Alias (u16)	2	Version of the Market Data protocol.
seqNum	SeqNum	Alias (u32)	4	Sequence number of the message added by the Market Data Sequencer.
timestamp	Timestamp	Alias (u64)	8	Time when message was sequenced on Market Data (Timestamp >= sourceTimestamp).
sourceTimestamp	Timestamp	Alias (u64)	8	Time of the business event that triggered or generated the message.
isEncrypted	bool	Primitive	1	True if message is encrypted.
encryptionOffset	u64	Primitive	8	Encryption byte offset.
encryptionKeyId	ElementId	Alias (u32)	4	Reference to an EncryptionKey message.
sessionId	SessionId	Alias (u16)	2	ID of the session.
streamId	StreamId	Alias (u8)	1	ID of the md stream.

14.2. ACCRUEDINTERESTTABLEENTRY

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
productId	ElementId	Alias (u32)	4	
accruedInterestTableEntryDate	Date	Alias (u32)	4	Accrued interest table entry date.
accruedInterestTableEntryValue	Number	Alias (i64)	8	Accrued interest table entry value.

14.3. AuctionSummary

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.
price	Price	Alias (i64)	8	Auction price (final auction price).
quantity	Quantity	Alias (u64)	8	Auction quantity (final auction volume).
auctionType	AuctionType	Enum	1	Type of auction.

Name	Value	Description
NotApplicable	1	NotApplicable.
AuctionOpening	2	Opening auction.
AuctionClosing	3	Closing auction.
AuctionIntraday	4	Intraday auction.
AuctionVolatilityStatic	5	Volatility auction after static collars breach.
AuctionVolatilityDynamic	6	Volatility auction after dynamic collars breach.
AuctionExtendedVolatilityStatic	7	Additional volatility auction after static collar breach. It can be activated independently or as an extension of the previous AuctionVolatilityStatic phase.
AuctionExtendedVolatilityDynamic	8	Additional volatility auction after dynamic collar breach. It can be activated independently or as an extension of the previous AuctionVolatilityDynamic phase.
UnsuspensionAuction	9	A phase triggered whenever an instrument's changes its status from RegulatorySuspension or MarketOperationSuspension before entering into continuous phase.

14.4. AuctionUpdate

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.
indicativeMatchingPrice	Price	Alias (i64)	8	Indicative Matching Price (IMP).
indicativeMatchingVolume	Quantity	Alias (u64)	8	Indicative Matching Volume (IMV).
totalSellQuantityAtImp	Quantity	Alias (u64)	8	Total sell quantity at IMP.
totalBuyQuantityAtImp	Quantity	Alias (u64)	8	Total buy quantity at IMP.
bestSellLevel	Price	Alias (i64)	8	Best sell price level (1st BBO level).
bestSellLevelQuantity	Quantity	Alias (u64)	8	Quantity at the best sell price level.
bestBuyLevel	Price	Alias (i64)	8	Best buy price level (1st BBO level).
bestBuyLevelQuantity	Quantity	Alias (u64)	8	Quantity at the best buy price level.

14.5. CalendarException

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
calendarId	ElementId	Alias (u32)	4	Calendar ID.
calendarExceptionDate	Date	Alias (u32)	4	Calendar exception date.
calendarExceptionType	CalendarExceptionType	Enum	1	Calendar exception type.

Name	Value	Description
Closed	1	The day is closed for trading.
Open	2	The day is open for trading.

14.6. CollarGroup

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
id	ElementId	Alias (u32)	4	Collar group ID.
orderStaticCollarTableId	ElementId	Alias (u32)	4	Collar table ID.
orderDynamicCollarTableId	ElementId	Alias (u32)	4	Collar table ID.
tradeStaticCollarTableId	ElementId	Alias (u32)	4	Collar table ID.
tradeDynamicCollarTableId	ElementId	Alias (u32)	4	Collar table ID.

14.7. CollarTableEntry

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
collarTableId	ElementId	Alias (u32)	4	Collar table ID.
collarMode	CollarMode	Enum	1	Collar mode (trade price collar or order price collar). **TradePriceCollar** (1): Trade price collar. **OrderPriceCollar** (2): Order price collar.
collarExpression	ExpressionType	Enum	1	Collar expression (percentage or absolute value in reference price). **Percentage** (1): Percentage. **AbsoluteValue** (2): Absolute value.
collarLowerBound	Bound	Alias (i64)	8	Collar lower bound.

Name	Data type	Kind	Length (bytes)	Description
collarValue	CollarValue	Alias (i64)	8	Collar value.
collarLowerBid	Bound	Alias (i64)	8	Lower bid collar.
collarUpperBid	Bound	Alias (i64)	8	Upper bid collar.
collarUpperAsk	Bound	Alias (i64)	8	Upper ask collar.
collarLowerAsk	Bound	Alias (i64)	8	Lower ask collar.

14.8. EncryptionKey

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
id	ElementId	Alias (u32)	4	Encryption key ID.
secretKey	EncryptionGroupKey	Array (u8)	32	Secret Key value.

14.9. EndOfSnapshot

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
lastSeqNum	SeqNum	Alias (u32)	4	Sequence number of the last Market Data message transmitted by the Snapshot.

14.10. ExternalUnderlying

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
externalUnderlyingId	ElementId	Alias (u32)	4	External underlying ID.
productIdentification	ProductIdentification	Array (AnsiChar)	30	Product identification.
productIdentificationType	ProductIdentificationType	Enum	1	Product identification type. **NotApplicable** (1): Not applicable **ISIN** (2): Financial instrument's ISIN.
productCode	Code	Array (AnsiChar)	20	Product code.
productName	Name	Array (AnsiChar)	50	Product name.
mic	MicCode	Array (AnsiChar)	4	Market Identifier Code.
nominalCurrency	Currency	Enum	2	Nominal currency.
tradingCurrency	Currency	Enum	2	Trading currency.

14.11. Heartbeat

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.

14.12. IndexParams

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	InstrumentID of the redistributed index.
indexBaseValue	IndexValue	Alias (i64)	8	Index base value.
indexBaseCapitalisation	Value	Alias (i64)	8	Index base capitalisation
indexBaseDate	Date	Alias (u32)	4	Index base date.
correctionCoefficient	Number	Alias (i64)	8	Index adjustment coefficient K(t)
numOfComponents	u16	Primitive	2	Total number of instruments in the portfolio.
indexPublicationSchedule	IndexPublicationSchedule	Enum	1	Mode of index dissemination. Name / Value / Description: Continuous / 1 / Index is quoted in determined time intervals. Fixing / 2 / Index is quoted on determined hours. Closing / 3 / Index is quoted on closing.
indexType	IndexType	Enum	1	Index product classification Name / Value / Description: **W** / 1 / WSE index from the main or alternative markets. **Z** / 2 / External index. **V** / 3 / Another WSE information product. **M** / 4 / WSE strategy dividend point index. **S** / 5 / WSE strategy index of short type.

Name	Data type	Kind	Length (bytes)	Description
				L 6 WSE strategy index of leverage type.
presenceFlags	IndexParamsPresenceFlags	Enum	1	Indicates the filling of fields: daysSinceLastPublication, numberOfDividends. **None** 0b00000000 **HasDaysSinceLastPublication** 0b00000001 **HasNumberOfDividends** 0b00000010
daysSinceLastPublication	u8	Primitive	1	Number of calendar days between current session and the previous session day.
numberOfDividends	u16	Primitive	2	Number of dividends included in the index calculation.
indexUnderlyings.count	u8	Primitive	1	How many elements does IndexUnderlyings contain.
indexUnderlyings.items	IndexUnderlying Array	Array (IndexUnderlying)	96	IndexUnderlying Array. IndexUnderlying:

Name	Data type	Kind	Length (bytes)	Description
productIdentification.productIdentification	ProductIdentification	Array (AnsiChar)	30	Product identification, for example its ISIN code.
productIdentification.productIdentificationType	ProductIdentificationType	Enum	1	Type of product identification. **NotApplicable** 1 Not applicable **ISIN** 2 Financial instrument's ISIN.
indexUnderlyingType	IndexUnderlyingType	Enum	1	Underlying type of index. **NotApplicable** 1 Not applicable **Index** 2 Index.

Name	Data type	Kind	Length (bytes)	Description
				ReferenceRate 3 Reference Rate.
publicationOrder	u16	Primitive	2	Index publication presentation order.
currency	Currency	Enum	2	Currency (e.g. USD).
dateValidity	Date	Alias (u32)	4	Date of validity of index portfolio.

14.13. IndexPortfolioEntry

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
id	ElementId	Alias (u32)	4	The identifier of the entry within the given index.
indexId	ElementId	Alias (u32)	4	ID of index.
presenceFlags	IndexPortfolioEntryPresenceFlags	Enum	1	Indicates the filling of fields: instrumentId. **None** 0b00000000 **HasInstrumentId** 0b00000001
instrumentId	ElementId	Alias (u32)	4	ID of the financial instrument (index participant).
publicProductIdentification.productIdentification	ProductIdentification	Array (AnsiChar)	30	Product identification, for example its ISIN code.
publicProductIdentification.productIdentificationType	ProductIdentificationType	Enum	1	Type of product identification. **NotApplicable** 1 Not applicable **ISIN** 2 Financial instrument's ISIN.

Name	Data type	Kind	Length (bytes)	Description
mic	MicCode	Array (AnsiChar)	4	Market structure's Market Identifier Code (MIC) as specified in ISO 10383.
currency	Currency	Enum	2	Currency (e.g. USD).
instrumentPacket	u64	Primitive	8	Instrument packet in index portfolio / of information product. Number of units (usually stocks/shares) of a given instrument in the index portfolio or in the information product.
suspended	bool	Primitive	1	Is instrument suspended.

14.14. IndexSummary

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	InstrumentID of the redistributed index.
openingIndexValue	IndexValue	Alias (i64)	8	Opening index value
sessionLow	IndexValue	Alias (i64)	8	Lowest value of the day.
sessionHigh	IndexValue	Alias (i64)	8	Highest value of the day.
closingIndexValue	IndexValue	Alias (i64)	8	Closing index value.
presenceFlags	IndexSummaryPresenceFlags	Enum	1	Indicates the filling of fields: sessionAvg. **None** — 0b00000000 **HasSessionAvg** — 0b00000001
sessionAvg	IndexValue	Alias (i64)	8	Average value of the day.

Name	Data type	Kind	Length (bytes)	Description
pctChangeIndexValPrevSession	PercentageChange	Alias (i64)	8	Percentage change of the index value in relation to the value of the closing index from the previous session.
pctChangeIndexValPrevYear	PercentageChange	Alias (i64)	8	Percentage change of the index value in relation to its value at the end of the previous year
closingCapitalisationPortfolio	Value	Alias (i64)	8	Capitalisation of the portfolio for the closing index expressed in the currency of the index, on basis of which the closing index is calculated.
indIndexOpeningPortfolio	PercentageChange	Alias (i64)	8	Indicator of the index opening portfolio W(t). Percentage share of instruments with at least one trade during the session.

14.15. IndexationTableEntry

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
productId	ElementId	Alias (u32)	4	Product ID.
indexationTableEntryDate	Date	Alias (u32)	4	Indexation table entry date.
indexationTableEntryValue	Number	Alias (i64)	8	Indexation table entry value.

14.16. Instrument

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.
productType	ProductType	Enum	1	Type of the product.

Name	Value	Description
Equity	1	Equity

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length (bytes)</th><th colspan="3">Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>FixedIncome</td><td>2</td><td>Fixed income</td></tr>
<tr><td></td><td></td><td></td><td></td><td>DerivativeFutures</td><td>3</td><td>Derivative Futures</td></tr>
<tr><td></td><td></td><td></td><td></td><td>DerivativeOptions</td><td>4</td><td>Derivative Options</td></tr>
<tr><td></td><td></td><td></td><td></td><td>Index</td><td>5</td><td>Index</td></tr>
<tr><td></td><td></td><td></td><td></td><td>Currency</td><td>6</td><td>Currency</td></tr>
<tr><td></td><td></td><td></td><td></td><td>StructuredProduct</td><td>7</td><td>Structured Product</td></tr>
<tr><td></td><td></td><td></td><td></td><td>Tracker</td><td>9</td><td></td></tr>
<tr><td>productSubtype</td><td>ProductSubtype</td><td>Enum</td><td>1</td><td colspan="3">Subtype of the product.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Share</td><td>1</td><td>Share</td></tr>
<tr><td>AllotmentRight</td><td>2</td><td>Allotment Right</td></tr>
<tr><td>SubscriptionRight</td><td>3</td><td>Subscription Right</td></tr>
<tr><td>TenderOffer</td><td>4</td><td>Tender Offer</td></tr>
<tr><td>IssueRight</td><td>5</td><td>Issue Right</td></tr>
<tr><td>SubscriptionWarrant</td><td>6</td><td>Subscription Warrant</td></tr>
<tr><td>BankSecurities</td><td>7</td><td>Bank Securities</td></tr>
<tr><td>Bond</td><td>8</td><td>Bond</td></tr>
<tr><td>TreasuryBond</td><td>9</td><td>Treasury Bond</td></tr>
<tr><td>AgencyBond</td><td>10</td><td>Bond issued by a government institution</td></tr>
<tr><td>MunicipalBond</td><td>11</td><td>Municipal Bond</td></tr>
<tr><td>CorporateBankBond</td><td>12</td><td>Corporate Bond - bank</td></tr>
<tr><td>CorporateFirmBond</td><td>13</td><td>Corporate Bond - firm</td></tr>
<tr><td>ConvertibleBond</td><td>14</td><td>Convertible Bond</td></tr>
<tr><td>MortgageBond</td><td>15</td><td>Mortgage Bond</td></tr>
</table></td></tr>
</table>

Name	Data type	Kind	Length (bytes)	Description		
				MortgageBackedBond	16	Mortgage Backed Bond
				PublicMortgageBond	17	Public Mortgage Bond
				Bill	18	Bill
				TreasuryBill	19	Treasury Bill
				CommercialBill	20	Commercial Bill
				ExchangeTradedFund	21	ETF - Exchange Traded Fund
				ExchangeTradedNote	22	ETN - Exchange Traded Note
				ExchangeTradedCommodity	23	ETC - Exchange Traded Commodity
				InvestmentCertificate	24	Investment Certificate
				IndexFutures	25	Index Futures
				StockFutures	26	Stock Futures
				CurrencyFutures	27	Currency Futures
				BondFutures	28	Bond Futures
				InterestRateFutures	29	Interest Rate Futures
				IndexOptions	30	Index Options
				StockOptions	31	Stock Options
				OptionWarrants	32	Option warrants
				FactorCertificates	33	Factor certificates
				TurboCertificates	34	Turbo certificates
				CapitalProtectionCertificates	35	Capital protection certificates
				BonusCertificates	36	Bonus certificates
				ReverseConvertibleCertificates	37	Reverse convertible certificates
				ExpressCertificates	38	Express certificates
				DiscountCertificates	39	Discount certificates

Name	Data type	Kind	Length (bytes)	Description		
				IndexTrackerCertificates	40	Index/tracker certificates
				OtherInvestmentCertificates	41	Other investment certificates (non leveraged)
				StructuredNotes	42	Structured notes
				ReferenceRate	43	Reference rate
				InterestRate	44	Interest rate
				ExchangeRate	45	Exchange rate
				PriceIndex	46	Price index
				TotalReturnIndex	47	Total return index
				SectorPriceIndex	48	Sector price index
				SectorTotalReturnIndex	49	Sector total return index
				DividendIndex	50	Dividend index
				StrategyPriceIndex	51	Strategy price index
				StrategyTotalReturnIndex	52	Strategy total return index
				CommodityIndex	53	Commodity index
				BondTotalReturnIndex	54	Bond total return index
				SupranationalBond	55	Supranational Bond
				WarrantWithKnockOut	56	Warrant with Knock Out
firstTradingDate	Date	Alias (u32)	4	First trading day of the financial instrument.		
lastTradingDate	Date	Alias (u32)	4	Last trading day of the financial instrument.		
productId	ElementId	Alias (u32)	4	Reference to financial product definition.		
currency	Currency	Enum	2	Trading currency (e.g. USD).		
icebergMinValue	Value	Alias (i64)	8	Minimum iceberg order value.		
lotSize	LotSize	Alias (u32)	4	Lot size for the instrument.		
priceExpressionType	PriceExpressionType	Enum	1	Price expression type for the financial instrument.		

Name	Data type	Kind	Length (bytes)	Description
				<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>**NotApplicable**</td><td>3</td><td>Not applicable.</td></tr><tr><td>**Price**</td><td>1</td><td>Price expressed as absolute value.</td></tr><tr><td>**Percentage**</td><td>2</td><td>Price expressed as percentage.</td></tr></table>
calendarId	ElementId	Alias (u32)	4	ID of trading calendar for the financial instrument.
marketStructureId	ElementId	Alias (u32)	4	ID of the financial instrument's market segment.
tickTableId	ElementId	Alias (u32)	4	ID of the tick table used for the financial instrument.
referenceInstrumentId	ElementId	Alias (u32)	4	Identifier of reference instrument.
collarGroupId	ElementId	Alias (u32)	4	Collar group ID.
tradingScheduleId	ElementId	Alias (u32)	4	Trading schedule ID.
description	InstrumentDescription	Array (AnsiChar)	64	Financial instrument code.
mic	MicCode	Array (AnsiChar)	4	Market structure's Market Identifier Code (MIC) as specified in ISO 10383.
nominalValue	Price	Alias (i64)	8	Nominal value of the financial instrument.
nominalValueType	NominalValueType	Enum	1	Type of the product's nominal value (no nominal, constant or unknown).<table><tr><th>Name</th><th>Value</th><th>Description</th></tr><tr><td>**NotApplicable**</td><td>1</td><td>Not applicable.</td></tr><tr><td>**Constant**</td><td>2</td><td>Indicates that the security has a constant nominal value.</td></tr></table>
multiplier	Multiplier	Alias (i64)	8	Product value multiplier.
strikePrice	Price	Alias (i64)	8	Product strike price.
strikePriceCurrency	Currency	Enum	2	Strike Price Currency
productIdentification	ProductIdentification	Array (AnsiChar)	30	Product identification, e.g. ISIN number.
productIdentificationType	ProductIdentificationType	Enum	1	Type of product identification.

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length (bytes)</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>1</td><td>Not applicable</td></tr>
<tr><td>ISIN</td><td>2</td><td>Financial instrument's ISIN.</td></tr>
</table>
</td></tr>
<tr><td>settlementCalendarId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>Settlement calendar ID.</td></tr>
<tr><td>bondCouponType</td><td>CouponType</td><td>Enum</td><td>1</td><td>Coupon type for a bond.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>1</td><td>NotApplicable.</td></tr>
<tr><td>Zero</td><td>2</td><td>Zero coupon.</td></tr>
<tr><td>Fixed</td><td>3</td><td>Fixed coupon.</td></tr>
<tr><td>Floating</td><td>4</td><td>Floating coupon.</td></tr>
<tr><td>Indexed</td><td>5</td><td>Indexed coupon.</td></tr>
</table>
</td></tr>
<tr><td>preTradeCheckMinPrice</td><td>Price</td><td>Alias (i64)</td><td>8</td><td>Minimum order price for pre-trade check.</td></tr>
<tr><td>preTradeCheckMaxPrice</td><td>Price</td><td>Alias (i64)</td><td>8</td><td>Maximum order price for pre-trade check.</td></tr>
<tr><td>preTradeCheckMinQuantity</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Minimum volume for pre-trade check.</td></tr>
<tr><td>preTradeCheckMaxQuantity</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Maximum volume for pre-trade check.</td></tr>
<tr><td>preTradeCheckMaxValue</td><td>Value</td><td>Alias (i64)</td><td>8</td><td>Maximum value for pre-trade check.</td></tr>
<tr><td>preTradeCheckMinValue</td><td>Value</td><td>Alias (i64)</td><td>8</td><td>Minimum value for pre-trade check.</td></tr>
<tr><td>liquidity</td><td>bool</td><td>Primitive</td><td>1</td><td>Liquidity flag, true - liquid, false - Illiquid.</td></tr>
<tr><td>marketModelType</td><td>MarketModelType</td><td>Enum</td><td>1</td><td>Market Model.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>1</td><td>Not applicable</td></tr>
<tr><td>CLOB</td><td>2</td><td>A Central Limit Order Book (CLOB)</td></tr>
<tr><td>BLOCK</td><td>3</td><td>BLOCK</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length (bytes)</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><td>HYBRID</td><td>4</td><td>HYBRID market model</td></tr>
<tr><td>CROSS</td><td>5</td><td>CROSS</td></tr>
<tr><td>IPO</td><td>6</td><td>Initial public offering</td></tr>
<tr><td>TenderOffer</td><td>7</td><td>Tender offer</td></tr>
<tr><td>Redistribution</td><td>8</td><td>Redistribution</td></tr>
</table>
</td></tr>
<tr><td>issuer</td><td>Issuer</td><td>Array (AnsiChar)</td><td>150</td><td>Name of the issuer.</td></tr>
<tr><td>issuerRegCountry</td><td>Country</td><td>Enum</td><td>2</td><td>Identifier of the country of the registration.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>AFG</td><td>4</td><td>Afghanistan</td></tr>
<tr><td>ALB</td><td>8</td><td>Albania</td></tr>
<tr><td>ATA</td><td>10</td><td>Antarctica</td></tr>
<tr><td>DZA</td><td>12</td><td>Algeria</td></tr>
<tr><td>ASM</td><td>16</td><td>American Samoa</td></tr>
<tr><td>AND</td><td>20</td><td>Andorra</td></tr>
<tr><td>AGO</td><td>24</td><td>Angola</td></tr>
<tr><td>ATG</td><td>28</td><td>Antigua and Barbuda</td></tr>
<tr><td>AZE</td><td>31</td><td>Azerbaijan</td></tr>
<tr><td>ARG</td><td>32</td><td>Argentina</td></tr>
<tr><td>AUS</td><td>36</td><td>Australia</td></tr>
<tr><td>AUT</td><td>40</td><td>Austria</td></tr>
<tr><td>BHS</td><td>44</td><td>Bahamas</td></tr>
<tr><td>BHR</td><td>48</td><td>Bahrain</td></tr>
<tr><td>BGD</td><td>50</td><td>Bangladesh</td></tr>
<tr><td>ARM</td><td>51</td><td>Armenia</td></tr>
</table>
</td></tr>
</table>

Name	Data type	Kind	Length (bytes)	Description		
				BRB	52	Barbados
				BEL	56	Belgium
				BMU	60	Bermuda
				BTN	64	Bhutan
				BOL	68	Bolivia
				BIH	70	Bosnia and Herzegovina
				BWA	72	Botswana
				BVT	74	Bouvet Island
				BRA	76	Brazil
				BLZ	84	Belize
				IOT	86	British Indian Ocean Territory
				SLB	90	Solomon Islands
				VGB	92	Virgin Islands (British)
				BRN	96	Brunei Darussalam
				BGR	100	Bulgaria
				MMR	104	Myanmar
				BDI	108	Burundi
				BLR	112	Belarus
				KHM	116	Cambodia
				CMR	120	Cameroon
				CAN	124	Canada
				CPV	132	Cabo Verde
				CYM	136	Cayman Islands
				CAF	140	Central African Republic

Name	Data type	Kind	Length (bytes)	Description		
				LKA	144	Sri Lanka
				TCD	148	Chad
				CHL	152	Chile
				CHN	156	China
				TWN	158	Taiwan
				CXR	162	Christmas Island
				CCK	166	Cocos (Keeling) Islands
				COL	170	Colombia
				COM	174	Comoros
				MYT	175	Mayotte
				COG	178	Congo
				COD	180	Democratic Republic of the Congo
				COK	184	Cook Islands
				CRI	188	Costa Rica
				HRV	191	Croatia
				CUB	192	Cuba
				CYP	196	Cyprus
				CZE	203	Czechia
				BEN	204	Benin
				DNK	208	Denmark
				DMA	212	Dominica
				DOM	214	Dominican Republic
				ECU	218	Ecuador
				SLV	222	El Salvador

Name	Data type	Kind	Length (bytes)	Description		
				GNQ	226	Equatorial Guinea
				ETH	231	Ethiopia
				ERI	232	Eritrea
				EST	233	Estonia
				FRO	234	Faroe Islands
				FLK	238	Falkland Islands (Malvinas)
				SGS	239	South Georgia and the South Sandwich Islands
				FJI	242	Fiji
				FIN	246	Finland
				ALA	248	AlandIslands
				FRA	250	France
				GUF	254	French Guiana
				PYF	258	French Polynesia
				ATF	260	French Southern Territories
				DJI	262	Djibouti
				GAB	266	Gabon
				GEO	268	Georgia
				GMB	270	Gambia
				PSE	275	Palestine
				DEU	276	Germany
				GHA	288	Ghana
				GIB	292	Gibraltar
				KIR	296	Kiribati
				GRC	300	Greece

Name	Data type	Kind	Length (bytes)	Description		
				GRL	304	Greenland
				GRD	308	Grenada
				GLP	312	Guadeloupe
				GUM	316	Guam
				GTM	320	Guatemala
				GIN	324	Guinea
				GUY	328	Guyana
				HTI	332	Haiti
				HMD	334	Heard Island and McDonald Islands
				VAT	336	Holy See
				HND	340	Honduras
				HKG	344	Hong Kong
				HUN	348	Hungary
				ISL	352	Iceland
				IND	356	India
				IDN	360	Indonesia
				IRN	364	Iran
				IRQ	368	Iraq
				IRL	372	Ireland
				ISR	376	Israel
				ITA	380	Italy
				CIV	384	Cote d'Ivoire
				JAM	388	Jamaica
				JPN	392	Japan

Name	Data type	Kind	Length (bytes)	Description		
				KAZ	398	Kazakhstan
				JOR	400	Jordan
				KEN	404	Kenya
				PRK	408	Democratic People's Republic of Korea
				KOR	410	Republic of Korea
				KWT	414	Kuwait
				KGZ	417	Kyrgyzstan
				LAO	418	Lao People's Democratic Republic
				LBN	422	Lebanon
				LSO	426	Lesotho
				LVA	428	Latvia
				LBR	430	Liberia
				LBY	434	Libya
				LIE	438	Liechtenstein
				LTU	440	Lithuania
				LUX	442	Luxembourg
				MAC	446	Macao
				MDG	450	Madagascar
				MWI	454	Malawi
				MYS	458	Malaysia
				MDV	462	Maldives
				MLI	466	Mali
				MLT	470	Malta
				MTQ	474	Martinique

Name	Data type	Kind	Length (bytes)	Description		
				MRT	478	Mauritania
				MUS	480	Mauritius
				MEX	484	Mexico
				MCO	492	Monaco
				MNG	496	Mongolia
				MDA	498	Republic of Moldova
				MNE	499	Montenegro
				MSR	500	Montserrat
				MAR	504	Morocco
				MOZ	508	Mozambique
				OMN	512	Oman
				NAM	516	Namibia
				NRU	520	Nauru
				NPL	524	Nepal
				NLD	528	Netherlands (the)
				CUW	531	Curacao
				ABW	533	Aruba
				SXM	534	Sint Maarten (Dutch part)
				BES	535	Bonaire, Sint Eustatius and Saba
				NCL	540	New Caledonia
				VUT	548	Vanuatu
				NZL	554	New Zealand
				NIC	558	Nicaragua
				NER	562	Niger

Name	Data type	Kind	Length (bytes)	Description		
				NGA	566	Nigeria
				NIU	570	Niue
				NFK	574	Norfolk Island
				NOR	578	Norway
				MNP	580	Northern Mariana Islands
				UMI	581	United States Minor Outlying Islands
				FSM	583	Micronesia, Federated States of
				MHL	584	Marshall Islands
				PLW	585	Palau
				PAK	586	Pakistan
				PAN	591	Panama
				PNG	598	Papua New Guinea
				PRY	600	Paraguay
				PER	604	Peru
				PHL	608	Philippines
				PCN	612	Pitcairn
				POL	616	Poland
				PRT	620	Portugal
				GNB	624	Guinea-Bissau
				TLS	626	Timor-Leste
				PRI	630	Puerto Rico
				QAT	634	Qatar
				REU	638	Réunion
				ROU	642	Romania

Name	Data type	Kind	Length (bytes)	Description		
				RUS	643	Russian Federation
				RWA	646	Rwanda
				BLM	652	Saint Barthélemy
				SHN	654	Saint Helena, Ascension and Tristan da Cunha
				KNA	659	Saint Kitts and Nevis
				AIA	660	Anguilla
				LCA	662	Saint Lucia
				MAF	663	Saint Martin (French part)
				SPM	666	Saint Pierre and Miquelon
				VCT	670	Saint Vincent and the Grenadines
				SMR	674	San Marino
				STP	678	Sao Tome and Principe
				SAU	682	Saudi Arabia
				SEN	686	Senegal
				SRB	688	Serbia
				SYC	690	Seychelles
				SLE	694	Sierra Leone
				SGP	702	Singapore
				SVK	703	Slovakia
				VNM	704	Viet Nam
				SVN	705	Slovenia
				SOM	706	Somalia
				ZAF	710	South Africa
				ZWE	716	Zimbabwe

Name	Data type	Kind	Length (bytes)	Description		
				ESP	724	Spain
				SSD	728	South Sudan
				SDN	729	Sudan
				ESH	732	Western Sahara
				SUR	740	Suriname
				SJM	744	Svalbard and Jan Mayen
				SWZ	748	Eswatini
				SWE	752	Sweden
				CHE	756	Switzerland
				SYR	760	Syrian Arab Republic
				TJK	762	Tajikistan
				THA	764	Thailand
				TGO	768	Togo
				TKL	772	Tokelau
				TON	776	Tonga
				TTO	780	Trinidad and Tobago
				ARE	784	United Arab Emirates
				TUN	788	Tunisia
				TUR	792	Türkiye
				TKM	795	Turkmenistan
				TCA	796	Turks and Caicos Islands
				TUV	798	Tuvalu
				UGA	800	Uganda
				UKR	804	Ukraine

Name	Data type	Kind	Length (bytes)	Description		
				MKD	807	North Macedonia
				EGY	818	Egypt
				GBR	826	United Kingdom of Great Britain and Northern Ireland (the)
				GGY	831	Guernsey
				JEY	832	Jersey
				IMN	833	Isle of Man
				TZA	834	Tanzania, United Republic of
				USA	840	United States of America (the)
				VIR	850	Virgin Islands (U.S.)
				BFA	854	Burkina Faso
				URY	858	Uruguay
				UZB	860	Uzbekistan
				VEN	862	Venezuela
				WLF	876	Wallis and Futuna
				WSM	882	Samoa
				YEM	887	Yemen
				ZMB	894	Zambia
underlyingInstrumentId	ElementId	Alias (u32)	4	Identifier of the underlying instrument.		
productCode	Code	Array (AnsiChar)	20	Unique code of the product.		
cfi	CfiCode	Array (AnsiChar)	6	CFI (Classification of Financial Instrument) code.		
fisn	FisnCode	Array (AnsiChar)	35	Product FISN (Financial Instrument Short Name) code.		
issueSize	u64	Primitive	8	Size of the issue (depends on the issue size type)		
nominalCurrency	Currency	Enum	2	Identifier of the nominal currency (ISO-4217).		

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length (bytes)</th><th>Description</th></tr>
<tr><td>usIndicator</td><td>UsIndicator</td><td>Enum</td><td>1</td><td>Identifier of the us indicator.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>1</td><td>No trading restrictions.</td></tr>
<tr><td>RegulationS</td><td>2</td><td>Trading restricted due to US Regulation S.</td></tr>
<tr><td>RegulationSPlusRule144A</td><td>3</td><td>Trading restricted due to US Regulation S with Rule 144A.</td></tr>
</table></td></tr>
<tr><td>expiryDate</td><td>Date</td><td>Alias (u32)</td><td>4</td><td>Expiration date.</td></tr>
<tr><td>versionNumber</td><td>u8</td><td>Primitive</td><td>1</td><td>Version of the derivative. Version is incremented when corporate event adjusting underlying closing price occurs.</td></tr>
<tr><td>settlementType</td><td>SettlementType</td><td>Enum</td><td>1</td><td>Identifier of settlement type.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>1</td><td>Not applicable.</td></tr>
<tr><td>CashSettlement</td><td>2</td><td>Derivatives are settled in cash.</td></tr>
<tr><td>PhysicalDelivery</td><td>3</td><td>Derivatives are settled as physical delivery of the underlying.</td></tr>
</table></td></tr>
<tr><td>optionType</td><td>OptionType</td><td>Enum</td><td>1</td><td>Identifier of option type.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>1</td><td>Not applicable.</td></tr>
<tr><td>Call</td><td>2</td><td>Call option.</td></tr>
<tr><td>Put</td><td>3</td><td>Put option.</td></tr>
</table></td></tr>
<tr><td>exerciseType</td><td>ExerciseType</td><td>Enum</td><td>1</td><td>Identifier of exercise style.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>AMER</td><td>1</td><td>AMER</td></tr>
<tr><td>EURO</td><td>2</td><td>EURO</td></tr>
<tr><td>NA</td><td>3</td><td>Not applicable</td></tr>
</table></td></tr>
<tr><td>productName</td><td>Name</td><td>Array (AnsiChar)</td><td>50</td><td>Name of the product.</td></tr>
</table>

Name	Data type	Kind	Length (bytes)	Description
referencePrice	Price	Alias (i64)	8	Reference price
status	InstrumentStatus	Enum	1	Instrument status. **Active** (1) — Financial instrument is active. **Inactive** (2) — Financial instrument is inactive and is not trading. **MarketOperationsSuspension** (3) — Manual suspension by Market Operations. **OutsideCollarsStatic** (4) — Outside static trade price collars. **OutsideCollarsDynamic** (5) — Outside dynamic trade price collars. **RegulatorySuspension** (6) — Regulatory suspension **HybridNoQuotes** (8) — Hybrid no valid quotes. **HybridKnockout** (9) — Hybrid knockout. **HybridKnockoutByIssuer** (10) — Hybrid knockout by issuer.
initialPhaseId	ElementId	Alias (u32)	4	Id of starting phase for the instrument
thresholdMax	Price	Alias (i64)	8	upper threshold for leveraged instruments
thresholdMin	Price	Alias (i64)	8	lower threshold for leveraged instruments
isLeverage	LeverageFlag	Enum	1	Identifier of leverage instruments **NotApplicable** (1) — Not applicable **Yes** (2) — Leveraged **No** (3) — Not leveraged
accruedInterestValue	Number	Alias (i64)	8	Accrued iterest value of the instrument
kid	Kid	Array (AnsiChar)	128	KID.
kidIssueDate	Date	Alias (u32)	4	KID issue date.

Name	Data type	Kind	Length (bytes)	Description
valueAtRisk	u8	Primitive	1	Value at risk.
externalUnderlyingId	ElementId	Alias (u32)	4	External underlying ID.

14.17. InstrumentStatusChange

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.
tradingPhaseId	ElementId	Alias (u32)	4	Trading phase ID.
tradingPhaseType	TradingPhaseType	Enum	1	Type of matching algorithm.

Name	Value	Description
ContinuousPriceTime	1	Continuous price and time.
ContinuousRefPriceTime	2	Continuous time at reference price.
ContinuousLastAuctionTime	3	Trade at last.
AuctionOpening	4	Opening auction.
BlockExactMatching	5	Block transaction, exact matching.
NoTradingClosed	6	Trading disabled, no matching.
Uncrossing	7	Uncrossing.
AuctionClosing	8	Closing auction.
AuctionIntraday	9	Intraday auction.
AuctionVolatilityStatic	10	Volatility auction after static collars breach.
AuctionVolatilityDynamic	11	Volatility auction after dynamic collars breach.

Name	Data type	Kind	Length (bytes)	Description
				AuctionExtendedVolatilityStatic 12 Additional volatility auction after static collar breach. It can be activated independently or as an extension of the previous AuctionVolatilityStatic phase.
				AuctionExtendedVolatilityDynamic 13 Additional volatility auction after dynamic collar breach. It can be activated independently or as an extension of the previous AuctionVolatilityDynamic phase.
				NoTradingEarlyMonitoring 14 Early Monitoring
				NoTradingLateMonitoring 15 Late Monitoring
				Hybrid 16 Hybrid normal trading
				HybridBuyOnly 17 Hybrid buy only phase
				HybridPreTrade 18 Hybrid pre trade phase
				UnsuspensionAuction 19 A phase triggered whenever an instrument's changes its status from RegulatorySuspension or MarketOperationSuspension before entering into continuous phase.
				Ipo 20 Initial public offering phase
				TenderOffer 21 Tender offer phase
				HybridPreTradeBuyOnly 22 Hybrid pre trade BuyOnly phase
				Distribution 23 Instrument distribution phase
status	InstrumentStatus	Enum	1	Financial instrument status.

Name	Value	Description
Active	1	Financial instrument is active.
Inactive	2	Financial instrument is inactive and is not trading.
MarketOperationsSuspension	3	Manual suspension by Market Operations.
OutsideCollarsStatic	4	Outside static trade price collars.
OutsideCollarsDynamic	5	Outside dynamic trade price collars.
RegulatorySuspension	6	Regulatory suspension
HybridNoQuotes	8	Hybrid no valid quotes.
HybridKnockout	9	Hybrid knockout.

Name	Data type	Kind	Length (bytes)	Description		
				HybridKnockoutByIssuer	10	Hybrid knockout by issuer.
stressedMarket	bool	Primitive	1	Stressed market conditions is called for when instrument experience high and short term intraday volatility.		

14.18. InstrumentSummary

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	Identifier of the instrument.
lastTradedPrice	Price	Alias (i64)	8	Last Traded Price (LTP).
closingPrice	Price	Alias (i64)	8	Closing Price (CP).
closingPriceType	ClosingPriceType	Enum	1	Closing Price Type. For OffBook means last transaction price. **InitialPrice** (1): Initial price of the instrument **LTP** (2): Last traded price **LastACP** (3): Last adjusted closing price **FairValue** (4): Fair value **DailySettlementPrice** (5): Daily Settlement Price **FinalSettlementPrice** (6): Final Settlement Price
adjustedClosingPrice	Price	Alias (i64)	8	Adjusted Closing Price (ACP).
adjustedClosingPriceReason	AdjustedClosingPriceReason	Enum	1	Reason for adjusting the Closing Price.

Name	Data type	Kind	Length (bytes)	Description
				Name / **Value** / **Description** **NotApplicable** / 1 / Not applicable. **Dividend** / 2 / Dividend **IssueRight** / 3 / Issue Right **Split** / 4 / Split **ReverseSplit** / 5 / Reverse Split **Bonus** / 6 / Bonus **SpinOff** / 7 / Spin-Off **TickSizeChange** / 8 / Tick Size Change **OrderPurge** / 9 / Order Purge **OtherReason** / 10 / Other reason
pctChange	PercentageChange	Alias (i64)	8	Percentage change from last session closing price.
vwap	Price	Alias (i64)	8	Volume-weighted average price for the market model in which there is an instrument.
noTrades	u64	Primitive	8	Total number of transactions on the current trading day.
totalVolume	Quantity	Alias (u64)	8	Total transaction volume.
totalValue	Value	Alias (i64)	8	Total transaction value.
openingPrice	Price	Alias (i64)	8	The price of the first trade on the current trading day.
maxPrice	Price	Alias (i64)	8	Highest price of the instrument on the current trading day.
minPrice	Price	Alias (i64)	8	Lowest price of the instrument on the current trading day.
isCorrection	bool	Primitive	1	Previous message correction flag.

14.19. Login

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
connectionId	ConnectionId	Alias (u16)	2	ID of the connection.
token	Token	Array (AnsiChar)	8	Client token.
filter	MessageFilter	Enum	1	Indicates which message types should be sent in stream.

Name	Value	Description
None	0b0000	Disable trades and ref_data.
Trades	0b0001	Enable trades.
RefData	0b0010	Enable ref_data.

14.20. LoginResponse

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
result	LoginResult	Enum	1	see the LoginResult description.

Name	Value	Description
Ok	1	Successful login.
InvalidToken	2	Authorization failure.
AlreadyLoggedIn	3	Already logged in.
LoginNotAllowed	4	Login is currently unavailable due to reasons such as service unavailability.
Other	5	Other errors.

14.21. Logout

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.

14.22. MarketStructure:

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
id	ElementId	Alias (u32)	4	ID of the market structure.
parentId	ElementId	Alias (u32)	4	ID of the market structure's parent market structure if the market structure is a sub-structure. Set to '0' (zero) if no parent market structure.
mic	MicCode	Array (AnsiChar)	4	Market structure's Market Identifier Code (MIC) as specified in ISO 10383.
name	Name	Array (AnsiChar)	50	Market structure name assigned by market operator.
marketModelType	MarketModelType	Enum	1	Market model type.

Name	Value	Description
NotApplicable	1	Not applicable
CLOB	2	A Central Limit Order Book (CLOB)
BLOCK	3	BLOCK
HYBRID	4	HYBRID market model
CROSS	5	CROSS
IPO	6	Initial public offering
TenderOffer	7	Tender offer
Redistribution	8	Redistribution

14.23. News

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header
marketStructureId	ElementId	Alias (u32)	4	ID of the instrument's market structure.
newsId	ElementId	Alias (u32)	4	News id - The news id serves as a unique reference for tracking purposes within the WATS system.
title	NewsTitle	Array (AnsiChar)	80	News title, unique per message.
entryNumber	u8	Primitive	1	News entry number.
total	u8	Primitive	1	Total number of news entries.
text	NewsText	Array (AnsiChar)	800	News text.

14.24. OrderAdd

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
publicOrderId	PublicOrderId	Alias (u64)	8	Order identifier (ID).
instrumentId	ElementId	Alias (u32) - encrypted	4	ID of financial instrument.
side	OrderSide	Enum	1	Order side (buy or sell). **Buy** (1) - Buy order. **Sell** (2) - Sell order.
price	Price	Alias (i64) - encrypted	8	Order (limit) price.
quantity	Quantity	Alias (u64) - encrypted	8	Order quantity.

Name	Data type	Kind	Length (bytes)	Description
mmQuote	bool	Primitive	1	Market maker quote flag.

14.25. OrderBookEvent

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of the instrument.
eventType	OrderBookEventType	Enum	1	Type/Kind of book event **Name** / **Value** / **Description** **Clear** / 1 / Clear order book **ResendStart** / 2 / Order book resend start (resend will begin shortly) **ResendEnd** / 3 / Order book resend end (resend is done)

14.26. OrderCancel

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.
publicOrderId	PublicOrderId	Alias (u64)	8	Order identifier (ID).
side	OrderSide	Enum	1	Order side.

Name	Data type	Kind	Length (bytes)	Description
				Name / **Value** / **Description** **Buy** / 1 / Buy order. **Sell** / 2 / Sell order.

14.27. OrderCollars

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.
collarType	CollarType	Enum	1	Collars type. **Name** / **Value** / **Description** **Static** / 1 / **Dynamic** / 2 /
price	Price	Alias (i64)	8	The reference price.
lowerBid	Bound	Alias (i64)	8	Value of the lower bid Price collar.
upperBid	Bound	Alias (i64)	8	Value of the upper bid Price collar.
lowerAsk	Bound	Alias (i64)	8	Value of the lower ask Price collar.
upperAsk	Bound	Alias (i64)	8	Value of the upper ask Price collar.

14.28. OrderExecute

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
quantity	Quantity	Alias (u64) - encrypted	8	Remaining un-executed order quantity.
instrumentId	ElementId	Alias (u32) - encrypted	4	ID of financial instrument.
publicOrderId	PublicOrderId	Alias (u64)	8	Order identifier (ID).
tradeId	TradeId	Alias (u32) - encrypted	4	ID of the underlying trade (equal to TradeID in Trade.)
executionPrice	Price	Alias (i64) - encrypted	8	Price at which the order was executed.
executionQuantity	Quantity	Alias (u64) - encrypted	8	The order's executed quantity.
side	OrderSide	Enum	1	Order side. **Buy** – 1 – Buy order. **Sell** – 2 – Sell order.

14.29. OrderModify

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32) - encrypted	4	ID of financial instrument.
publicOrderId	PublicOrderId	Alias (u64)	8	Order identifier (ID).
price	Price	Alias (i64) - encrypted	8	New order price.
quantity	Quantity	Alias (u64) - encrypted	8	New order quantity.
priorityFlag	PriorityFlag	Enum	1	Priority loss flag. **Lost** — 1 — Order modification caused loss of priority. **Retained** — 2 — Order modification did not cause loss of priority.
side	OrderSide	Enum	1	Order side. **Buy** — 1 — Buy order. **Sell** — 2 — Sell order.

14.30. PositionReport

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message Header.

Name	Data type	Kind	Length (bytes)	Description
publicProductIdentification.productIdentification	ProductIdentification	Array (AnsiChar)	30	Product identification, for example its ISIN code.
publicProductIdentification.productIdentificationType	ProductIdentificationType	Enum	1	Type of product identification. **NotApplicable** (1) – Not applicable **ISIN** (2) – Financial instrument's ISIN.
posType	PosType	Enum	1	Type of quantity returned. **SOD** (1) – Start-of-Day Qty. **ITD** (2) – Intraday Qty. **FIN** (3) – End-of-Day Qty.
openPositions	u32	Primitive	4	Number of open positions.
tradeId	TradeId	Alias (u32)	4	Trade ID.

14.31. PRICELEVELSNAPSHOT

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.
maxDepth	u8	Primitive	1	The number of BBO levels contained in the message.
mmBuyQuoteLevel	u8	Primitive	1	Id of an level on which mm quote rests (0 means no mm).
mmSellQuoteLevel	u8	Primitive	1	Id of an level on which mm quote rests (0 means no mm).
buy	PriceLevels	Array (PriceLevel)	155	Price levels for buy side. PriceLevel:

Name	Data type	Kind	Length (bytes)	Description
				Name / **Data type** / **Kind** / **Length (bytes)** / **Description** **isEncrypted** / bool / Primitive / 1 / A boolean value specifying whether the level is encrypted. **encryptionOffset** / u64 / Primitive / 8 / Encryption byte offset. **encryptionKeyId** / ElementId / Alias (u32) / 4 / Encryption Key ID - reference to a EncryptionKey message. **price** / Price / Alias (i64) - encrypted / 8 / Price level in currency units. **quantity** / Quantity / Alias (u64) - encrypted / 8 / Quantity of securities available at a given price level. **orderCount** / OrderCount / Alias (u16) - encrypted / 2 / Number of open orders at a given price level.
sell	PriceLevels	Array (PriceLevel)	155	Price levels for sell side. PriceLevel: **Name** / **Data type** / **Kind** / **Length (bytes)** / **Description** **isEncrypted** / bool / Primitive / 1 / A boolean value specifying whether the level is encrypted. **encryptionOffset** / u64 / Primitive / 8 / Encryption byte offset. **encryptionKeyId** / ElementId / Alias (u32) / 4 / Encryption Key ID - reference to a EncryptionKey message. **price** / Price / Alias (i64) - encrypted / 8 / Price level in currency units. **quantity** / Quantity / Alias (u64) - encrypted / 8 / Quantity of securities available at a given price level. **orderCount** / OrderCount / Alias (u16) - encrypted / 2 / Number of open orders at a given price level.

14.32. PRICEUPDATE

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message Header.
instrumentId	ElementId	Alias (u32)	4	Instrument to which the update refers.
priceUpdateType	PriceUpdateType	Enum	1	Indicates price update message variant.

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length (bytes)</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>ReferencePrice</td><td>1</td><td>Reference price placed in the value field</td></tr>
<tr><td>ReferenceIndexPrice</td><td>2</td><td>TBSP.Price price placed in the value field</td></tr>
<tr><td>ReferenceFixingIndexPrice</td><td>3</td><td>TBSP.FixPrice price placed in the value field</td></tr>
<tr><td>MidPrice</td><td>4</td><td>MidPrice price placed in the value field</td></tr>
<tr><td>FixingPrice</td><td>5</td><td>The fixing prices placed appropriately in the value field as the average price, valueBid as the bid price, and valueAsk as the ask price.</td></tr>
<tr><td>FixingYtm</td><td>6</td><td>The fixing YTM (i.e. yield to maturity) appropriately placed in the value field as the average YTM, valueBid as the bid YTM, and valueAsk as the ask YTM.</td></tr>
</table>
</td></tr>
<tr><td>priceYtmPresence</td><td>PriceYtmPresence</td><td>Enum</td><td>2</td><td>Indicates the filling of fields: value, valueBid, valueAsk.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>None</td><td>0b000000000</td><td>No values provided.</td></tr>
<tr><td>value</td><td>0b000000001</td><td>A price was provided.</td></tr>
<tr><td>valueBid</td><td>0b000000010</td><td>A bid price was provided.</td></tr>
<tr><td>valueAsk</td><td>0b000000100</td><td>An ask price was provided.</td></tr>
<tr><td>YtmNoCalc</td><td>0b000001000</td><td>YTM was not provided due to the absence of its calculation.</td></tr>
<tr><td>YtmBidNoCalc</td><td>0b000010000</td><td>Bid YTM was not provided due to the absence of its calculation.</td></tr>
<tr><td>YtmAskNoCalc</td><td>0b000100000</td><td>Ask YTM was not provided due to the absence of its calculation.</td></tr>
</table>
</td></tr>
<tr><td>value</td><td>Price</td><td>Alias (i64)</td><td>8</td><td>Indicates the updated price or YTM.</td></tr>
<tr><td>valueBid</td><td>Price</td><td>Alias (i64)</td><td>8</td><td>Indicates the updated bid price or bid YTM.</td></tr>
<tr><td>valueAsk</td><td>Price</td><td>Alias (i64)</td><td>8</td><td>Indicates the updated ask price or ask YTM.</td></tr>
</table>

14.33. ProductSummary

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
clobInstrument.instrumentId	ElementId	Alias (u32) - encrypted	4	Identifier of the instrument.
clobInstrument.marketModelType	MarketModelType	Enum	1	Market model type. **NotApplicable** (1) – Not applicable **CLOB** (2) – A Central Limit Order Book (CLOB) **BLOCK** (3) – BLOCK **HYBRID** (4) – HYBRID market model **CROSS** (5) – CROSS **IPO** (6) – Initial public offering **TenderOffer** (7) – Tender offer **Redistribution** (8) – Redistribution
clobInstrument.lastTradedPrice	Price	Alias (i64) - encrypted	8	Last Traded Price (LTP).
clobInstrument.closingPrice	Price	Alias (i64) - encrypted	8	Closing Price (CP).
clobInstrument.closingPriceType	ClosingPriceType	Enum	1	Closing Price Type. **InitialPrice** (1) – Initial price of the instrument **LTP** (2) – Last traded price **LastACP** (3) – Last adjusted closing price

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length (bytes)</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><td>FairValue</td><td>4</td><td>Fair value</td></tr>
<tr><td>DailySettlementPrice</td><td>5</td><td>Daily Settlement Price</td></tr>
<tr><td>FinalSettlementPrice</td><td>6</td><td>Final Settlement Price</td></tr>
</table>
</td></tr>
<tr><td>clobInstrument.adjustedClosingPrice</td><td>Price</td><td>Alias (i64) - encrypted</td><td>8</td><td>Adjusted Closing Price (ACP).</td></tr>
<tr><td>clobInstrument.adjustedClosingPriceReason</td><td>AdjustedClosingPriceReason</td><td>Enum</td><td>1</td><td>Reason for adjusting the Closing Price.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>1</td><td>Not applicable.</td></tr>
<tr><td>Dividend</td><td>2</td><td>Dividend</td></tr>
<tr><td>IssueRight</td><td>3</td><td>Issue Right</td></tr>
<tr><td>Split</td><td>4</td><td>Split</td></tr>
<tr><td>ReverseSplit</td><td>5</td><td>Reverse Split</td></tr>
<tr><td>Bonus</td><td>6</td><td>Bonus</td></tr>
<tr><td>SpinOff</td><td>7</td><td>Spin-Off</td></tr>
<tr><td>TickSizeChange</td><td>8</td><td>Tick Size Change</td></tr>
<tr><td>OrderPurge</td><td>9</td><td>Order Purge</td></tr>
<tr><td>OtherReason</td><td>10</td><td>Other reason</td></tr>
</table>
</td></tr>
<tr><td>clobInstrument.pctChange</td><td>PercentageChange</td><td>Alias (i64) - encrypted</td><td>8</td><td>Percentage change from last session closing price.</td></tr>
<tr><td>clobInstrument.vwap</td><td>Price</td><td>Alias (i64) - encrypted</td><td>8</td><td>Volume-weighted average price for the market model in which there is an instrument.</td></tr>
<tr><td>clobInstrument.noTrades</td><td>u64</td><td>Primitive</td><td>8</td><td>Total number of transactions on the current trading day.</td></tr>
<tr><td>clobInstrument.totalVolume</td><td>Quantity</td><td>Alias (u64) - encrypted</td><td>8</td><td>Total transaction volume.</td></tr>
</table>

Name	Data type	Kind	Length (bytes)	Description
clobInstrument.totalValue	Value	Alias (i64) - encrypted	8	Total transaction value.
clobInstrument.openingPrice	Price	Alias (i64) - encrypted	8	The price of the first trade on the current trading day.
clobInstrument.maxPrice	Price	Alias (i64) - encrypted	8	Highest price of the instrument on the current trading day.
clobInstrument.minPrice	Price	Alias (i64) - encrypted	8	Lowest price of the instrument on the current trading day.
clobInstrument.settlementPrice	Price	Alias (i64) - encrypted	8	Settlement price.
clobInstrument.settlementValue	Price	Alias (i64) - encrypted	8	Settlement value.
clobInstrument.endTradingDate	Date	Alias (u32)	4	End of trading date.
clobInstrument.lastTradeDate	Date	Alias (u32)	4	The date of execution of the last trade for the given instrument.
crossInstrument.instrumentId	ElementId	Alias (u32) - encrypted	4	Identifier of the instrument.
crossInstrument.marketModelType	MarketModelType	Enum	1	Market model type. **NotApplicable** (1): Not applicable **CLOB** (2): A Central Limit Order Book (CLOB) **BLOCK** (3): BLOCK **HYBRID** (4): HYBRID market model **CROSS** (5): CROSS **IPO** (6): Initial public offering

Name	Data type	Kind	Length (bytes)	Description
				TenderOffer 7 Tender offer **Redistribution** 8 Redistribution
crossInstrument.lastTradedPrice	Price	Alias (i64) - encrypted	8	Last Traded Price (LTP).
crossInstrument.closingPrice	Price	Alias (i64) - encrypted	8	Closing Price (CP).
crossInstrument.closingPriceType	ClosingPriceType	Enum	1	Closing Price Type. **InitialPrice** 1 Initial price of the instrument **LTP** 2 Last traded price **LastACP** 3 Last adjusted closing price **FairValue** 4 Fair value **DailySettlementPrice** 5 Daily Settlement Price **FinalSettlementPrice** 6 Final Settlement Price
crossInstrument.adjustedClosingPrice	Price	Alias (i64) - encrypted	8	Adjusted Closing Price (ACP).
crossInstrument.adjustedClosingPriceReason	AdjustedClosingPriceReason	Enum	1	Reason for adjusting the Closing Price. **NotApplicable** 1 Not applicable. **Dividend** 2 Dividend **IssueRight** 3 Issue Right **Split** 4 Split **ReverseSplit** 5 Reverse Split **Bonus** 6 Bonus

Name	Data type	Kind	Length (bytes)	Description		
				SpinOff	7	Spin-Off
				TickSizeChange	8	Tick Size Change
				OrderPurge	9	Order Purge
				OtherReason	10	Other reason
crossInstrument.pctChange	PercentageChange	Alias (i64) - encrypted	8	Percentage change from last session closing price.		
crossInstrument.vwap	Price	Alias (i64) - encrypted	8	Volume-weighted average price for the market model in which there is an instrument.		
crossInstrument.noTrades	u64	Primitive	8	Total number of transactions on the current trading day.		
crossInstrument.totalVolume	Quantity	Alias (u64) - encrypted	8	Total transaction volume.		
crossInstrument.totalValue	Value	Alias (i64) - encrypted	8	Total transaction value.		
crossInstrument.openingPrice	Price	Alias (i64) - encrypted	8	The price of the first trade on the current trading day.		
crossInstrument.maxPrice	Price	Alias (i64) - encrypted	8	Highest price of the instrument on the current trading day.		
crossInstrument.minPrice	Price	Alias (i64) - encrypted	8	Lowest price of the instrument on the current trading day.		
crossInstrument.settlementPrice	Price	Alias (i64) - encrypted	8	Settlement price.		
crossInstrument.settlementValue	Price	Alias (i64) - encrypted	8	Settlement value.		
crossInstrument.endTradingDate	Date	Alias (u32)	4	End of trading date.		

Name	Data type	Kind	Length (bytes)	Description
crossInstrument.lastTradeDate	Date	Alias (u32)	4	The date of execution of the last trade for the given instrument.
blockInstrument.instrumentId	ElementId	Alias (u32) - encrypted	4	Identifier of the instrument.
blockInstrument.marketModelType	MarketModelType	Enum	1	Market model type. **Name** / **Value** / **Description** **NotApplicable** / 1 / Not applicable **CLOB** / 2 / A Central Limit Order Book (CLOB) **BLOCK** / 3 / BLOCK **HYBRID** / 4 / HYBRID market model **CROSS** / 5 / CROSS **IPO** / 6 / Initial public offering **TenderOffer** / 7 / Tender offer **Redistribution** / 8 / Redistribution
blockInstrument.lastTradedPrice	Price	Alias (i64) - encrypted	8	Last Traded Price (LTP).
blockInstrument.closingPrice	Price	Alias (i64) - encrypted	8	Closing Price (CP).
blockInstrument.closingPriceType	ClosingPriceType	Enum	1	Closing Price Type. **Name** / **Value** / **Description** **InitialPrice** / 1 / Initial price of the instrument **LTP** / 2 / Last traded price **LastACP** / 3 / Last adjusted closing price **FairValue** / 4 / Fair value **DailySettlementPrice** / 5 / Daily Settlement Price

Name	Data type	Kind	Length (bytes)	Description
				FinalSettlementPrice | 6 | Final Settlement Price
blockInstrument.adjustedClosingPrice	Price	Alias (i64) - encrypted	8	Adjusted Closing Price (ACP).
blockInstrument.adjustedClosingPriceReason	AdjustedClosingPriceReason	Enum	1	Reason for adjusting the Closing Price. **NotApplicable** | 1 | Not applicable. **Dividend** | 2 | Dividend **IssueRight** | 3 | Issue Right **Split** | 4 | Split **ReverseSplit** | 5 | Reverse Split **Bonus** | 6 | Bonus **SpinOff** | 7 | Spin-Off **TickSizeChange** | 8 | Tick Size Change **OrderPurge** | 9 | Order Purge **OtherReason** | 10 | Other reason
blockInstrument.pctChange	PercentageChange	Alias (i64) - encrypted	8	Percentage change from last session closing price.
blockInstrument.vwap	Price	Alias (i64) - encrypted	8	Volume-weighted average price for the market model in which there is an instrument.
blockInstrument.noTrades	u64	Primitive	8	Total number of transactions on the current trading day.
blockInstrument.totalVolume	Quantity	Alias (u64) - encrypted	8	Total transaction volume.
blockInstrument.totalValue	Value	Alias (i64) - encrypted	8	Total transaction value.

Name	Data type	Kind	Length (bytes)	Description
blockInstrument.openingPrice	Price	Alias (i64) - encrypted	8	The price of the first trade on the current trading day.
blockInstrument.maxPrice	Price	Alias (i64) - encrypted	8	Highest price of the instrument on the current trading day.
blockInstrument.minPrice	Price	Alias (i64) - encrypted	8	Lowest price of the instrument on the current trading day.
blockInstrument.settlementPrice	Price	Alias (i64) - encrypted	8	Settlement price.
blockInstrument.settlementValue	Price	Alias (i64) - encrypted	8	Settlement value.
blockInstrument.endTradingDate	Date	Alias (u32)	4	End of trading date.
blockInstrument.lastTradeDate	Date	Alias (u32)	4	The date of execution of the last trade for the given instrument.
hybridInstrument.instrumentId	ElementId	Alias (u32) - encrypted	4	Identifier of the instrument.
hybridInstrument.marketModelType	MarketModelType	Enum	1	Market model type. **NotApplicable** (1): Not applicable **CLOB** (2): A Central Limit Order Book (CLOB) **BLOCK** (3): BLOCK **HYBRID** (4): HYBRID market model **CROSS** (5): CROSS **IPO** (6): Initial public offering **TenderOffer** (7): Tender offer **Redistribution** (8): Redistribution

Name	Data type	Kind	Length (bytes)	Description
hybridInstrument.lastTradedPrice	Price	Alias (i64) - encrypted	8	Last Traded Price (LTP).
hybridInstrument.closingPrice	Price	Alias (i64) - encrypted	8	Closing Price (CP).
hybridInstrument.closingPriceType	ClosingPriceType	Enum	1	Closing Price Type. **Name** / **Value** / **Description** **InitialPrice** / 1 / Initial price of the instrument **LTP** / 2 / Last traded price **LastACP** / 3 / Last adjusted closing price **FairValue** / 4 / Fair value **DailySettlementPrice** / 5 / Daily Settlement Price **FinalSettlementPrice** / 6 / Final Settlement Price
hybridInstrument.adjustedClosingPrice	Price	Alias (i64) - encrypted	8	Adjusted Closing Price (ACP).
hybridInstrument.adjustedClosingPriceReason	AdjustedClosingPriceReason	Enum	1	Reason for adjusting the Closing Price. **Name** / **Value** / **Description** **NotApplicable** / 1 / Not applicable. **Dividend** / 2 / Dividend **IssueRight** / 3 / Issue Right **Split** / 4 / Split **ReverseSplit** / 5 / Reverse Split **Bonus** / 6 / Bonus **SpinOff** / 7 / Spin-Off **TickSizeChange** / 8 / Tick Size Change

Name	Data type	Kind	Length (bytes)	Description		
				OrderPurge	9	Order Purge
				OtherReason	10	Other reason
hybridInstrument.pctChange	PercentageChange	Alias (i64) - encrypted	8	Percentage change from last session closing price.		
hybridInstrument.vwap	Price	Alias (i64) - encrypted	8	Volume-weighted average price for the market model in which there is an instrument.		
hybridInstrument.noTrades	u64	Primitive	8	Total number of transactions on the current trading day.		
hybridInstrument.totalVolume	Quantity	Alias (u64) - encrypted	8	Total transaction volume.		
hybridInstrument.totalValue	Value	Alias (i64) - encrypted	8	Total transaction value.		
hybridInstrument.openingPrice	Price	Alias (i64) - encrypted	8	The price of the first trade on the current trading day.		
hybridInstrument.maxPrice	Price	Alias (i64) - encrypted	8	Highest price of the instrument on the current trading day.		
hybridInstrument.minPrice	Price	Alias (i64) - encrypted	8	Lowest price of the instrument on the current trading day.		
hybridInstrument.settlementPrice	Price	Alias (i64) - encrypted	8	Settlement price.		
hybridInstrument.settlementValue	Price	Alias (i64) - encrypted	8	Settlement value.		
hybridInstrument.endTradingDate	Date	Alias (u32)	4	End of trading date.		
hybridInstrument.lastTradeDate	Date	Alias (u32)	4	The date of execution of the last trade for the given instrument.		
productIdentificationType	ProductIdentificationType	Enum	1	Type of product identification.		

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length (bytes)</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>1</td><td>Not applicable</td></tr>
<tr><td>ISIN</td><td>2</td><td>Financial instrument's ISIN.</td></tr>
</table>
</td></tr>
<tr><td>productIdentification</td><td>ProductIdentification</td><td>Array (AnsiChar) - encrypted</td><td>30</td><td>Product identification, e.g. ISIN number.</td></tr>
<tr><td>productId</td><td>ElementId</td><td>Alias (u32) - encrypted</td><td>4</td><td>ID of the product.</td></tr>
<tr><td>mic</td><td>MicCode</td><td>Array (AnsiChar)</td><td>4</td><td>Market structure's Market Identifier Code (MIC) as specified in ISO 10383.</td></tr>
<tr><td>accumulatedInterest</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Accumulated interest on the bonds or for mortgage-backed bonds on the day of settling the transaction.</td></tr>
<tr><td>interestRate</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Interest rate is determined on the basis of WIBOR/WIBID/WIRON rates for each expiration date of options and futures.</td></tr>
<tr><td>currency</td><td>Currency</td><td>Enum</td><td>2</td><td>Price currency (e.g. USD).</td></tr>
<tr><td>sessionDate</td><td>Date</td><td>Alias (u32)</td><td>4</td><td>Stock Exchange session date.</td></tr>
<tr><td>tradingValueCurrency</td><td>Value</td><td>Alias (i64) - encrypted</td><td>8</td><td>The field comprises the trading value expressed in trading currency.</td></tr>
<tr><td>status</td><td>InstrumentStatus</td><td>Enum</td><td>1</td><td>Financial instrument status.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Active</td><td>1</td><td>Financial instrument is active.</td></tr>
<tr><td>Inactive</td><td>2</td><td>Financial instrument is inactive and is not trading.</td></tr>
<tr><td>MarketOperationsSuspension</td><td>3</td><td>Manual suspension by Market Operations.</td></tr>
<tr><td>OutsideCollarsStatic</td><td>4</td><td>Outside static trade price collars.</td></tr>
<tr><td>OutsideCollarsDynamic</td><td>5</td><td>Outside dynamic trade price collars.</td></tr>
<tr><td>RegulatorySuspension</td><td>6</td><td>Regulatory suspension</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length (bytes)</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><td>HybridNoQuotes</td><td>8</td><td>Hybrid no valid quotes.</td></tr>
<tr><td>HybridKnockout</td><td>9</td><td>Hybrid knockout.</td></tr>
<tr><td>HybridKnockoutByIssuer</td><td>10</td><td>Hybrid knockout by issuer.</td></tr>
</table>
</td></tr>
<tr><td>sector</td><td>u16</td><td>Primitive</td><td>2</td><td>Defines the sector of the economy that the company belongs to. Possible values for shares, the field assumes were described in the WATS Market Data documentation.</td></tr>
<tr><td>market</td><td>u8</td><td>Primitive</td><td>1</td><td>Defines the market the instrument belongs to. The field assumes were described in the WATS Market Data documentation.</td></tr>
<tr><td>markerPriceChange</td><td>ChangeIndicator</td><td>Enum - encrypted</td><td>1</td><td>The field contains the direction of the percentage change in the instrument closing price from the current session in relation to the reference price.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Increase</td><td>1</td><td>Increase.</td></tr>
<tr><td>Decrease</td><td>2</td><td>Decrease.</td></tr>
<tr><td>NoChange</td><td>3</td><td>No change.</td></tr>
<tr><td>NotApplicable</td><td>4</td><td>Not Applicable.</td></tr>
</table>
</td></tr>
<tr><td>lowerLiquidity</td><td>bool</td><td>Primitive</td><td>1</td><td>The positive field value (true) informs if the company was qualified to Lower Liquidity Space. A negative value has the opposite meaning.</td></tr>
<tr><td>multiplier</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Instrument multiplier.</td></tr>
<tr><td>dividendRate</td><td>Number</td><td>Alias (i64)</td><td>8</td><td>Dividend rate for the company is given on the basis of dividends paid by companies being the base instrument for futures and options.</td></tr>
<tr><td>delta</td><td>Number</td><td>Alias (i64) - encrypted</td><td>8</td><td>Value of Delta indicator from the current session.</td></tr>
<tr><td>gamma</td><td>Number</td><td>Alias (i64) - encrypted</td><td>8</td><td>Value of Gamma indicator from the current session.</td></tr>
<tr><td>rho</td><td>Number</td><td>Alias (i64) - encrypted</td><td>8</td><td>Value of Rho indicator from the current session.</td></tr>
</table>

Name	Data type	Kind	Length (bytes)	Description
theta	Number	Alias (i64) - encrypted	8	Value of Theta indicator from the current session.
vega	Number	Alias (i64) - encrypted	8	Value of Vega indicator from the current session.
volatility	Number	Alias (i64) - encrypted	8	Volatility is calculated according to algorithm worked out by the WSE.
optionsStrikePrice	Price	Alias (i64)	8	Option product strike price.
dividend	bool	Primitive	1	The positive field value (true) informs that the instruments are traded the first time after determining the right to dividend. A negative value has the opposite meaning.
subscriptionRight	bool	Primitive	1	The positive field value (true) informs that the instruments are traded the first time after determining the subscription right. A negative value has the opposite meaning.
interimDividendRight	bool	Primitive	1	The positive field value (true) informs that the instruments are traded the first time after determining the interim dividend right. A negative value has the opposite meaning.
split	bool	Primitive	1	The positive field value (true) informs that the instruments are traded the first time after the change of the nominal value of shares. A negative value has the opposite meaning.
quotationSystem	QuotationSystem	Enum	1	The field informing about the system of listing the instrument. **NA** — 1 — Not Applicable. **SinglePriceSingleFixing** — 2 — Single price quotation system with a single fixing. **ContinuousTrading** — 3 — Continuous trading. **SinglePriceTwoFixings** — 4 — Single price quotation system with two fixings.
liquiditySupportPge	bool	Primitive	1	The field informs if the company entered to Liquidity Support Programme. True - participation in the liquidity Programme, false - no participation in the liquidity Programme.

14.34. RealTimeIndex

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	InstrumentID of the redistributed index.
indexLevelCode	IndexLevelCode	Enum	1	Level of the index. **Name** / **Value** / **Description** **PreviousClosingValue** / 1 / Closing index value from previous session (reference index). **TheoreticalValueBeforeOpening** / 2 / Theoretical opening index before opening the session. **ValueBeforeOpening** / 3 / Theoretical value of the index before its opening. **OpeningValue** / 4 / Index opening value. **CurrentValue** / 5 / Current index value. **PreClosingValue** / 6 / Preliminary closing value. **ClosingValue** / 7 / Closing index value. **ValueWithoutOpening** / 8 / Index value without opening.
numOfActiveInstruments	u16	Primitive	2	Number of instruments – participants of the index, for which the quotation was set during the session.
indIndexOpeningPortfolio	PercentageChange	Alias (i64)	8	Indicator of the index opening portfolio W(t). Percentage share of instruments with at least one trade during the session.
indexValue	IndexValue	Alias (i64)	8	The value of the last level for the index according to the definition provided in the indexLevelCode.
pctChangeIndexValPrevSession	PercentageChange	Alias (i64)	8	Percentage change of the index value in relation to the value of the closing index from the previous session.
sessionLow	IndexValue	Alias (i64)	8	Lowest value of the day.
sessionHigh	IndexValue	Alias (i64)	8	Highest value of the day.

Name	Data type	Kind	Length (bytes)	Description
presenceFlags	RealTimeIndexPresenceFlags	Enum	1	Indicates the filling of fields: tradingValue, indexValueBid, indexValueAsk, midSpreadIndex, differenceCentralSpread.

Name	Value	Description
None	0b00000000	
HasTradingValue	0b00000001	
HasIndexValueBid	0b00000010	
HasIndexValueAsk	0b00000100	
HasMidSpreadIndex	0b00001000	
HasDifferenceCentralSpread	0b00010000	

tradingValue	Value	Alias (i64)	8	Value of trading in the instruments from the given index.
indexValueBid	IndexValue	Alias (i64)	8	Value of the buy index based on the best sell offers of instruments.
indexValueAsk	IndexValue	Alias (i64)	8	Value of the sell index based on the best sell offers of instruments.
midSpreadIndex	IndexValue	Alias (i64)	8	Mean of indexValueBid and indexValueAsk.
differenceCentralSpread	IndexValue	Alias (i64)	8	The point difference between midSpreadIndex and indexValue.

14.35. TickTableEntry

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
tickSize	TickSize	Alias (i64)	8	Size of the tick.
lowerBound	Bound	Alias (i64)	8	Tick size lower bound.
tickTableId	ElementId	Alias (u32)	4	Tick table ID.

14.36. TopPriceLevelUpdate

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32) - encrypted	4	ID of financial instrument.
price	Price	Alias (i64) - encrypted	8	BBO level 1 price level in currency units.
quantity	Quantity	Alias (u64) - encrypted	8	BBO level 1 quantity at price level.
side	OrderSide	Enum	1	Side (buy or sell). **Name** / **Value** / **Description** **Buy** / 1 / Buy order. **Sell** / 2 / Sell order.
orderCount	OrderCount	Alias (u16) - encrypted	2	Number of open orders at BBO level 1.

14.37. Trade

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.
settlementValue	Value	Alias (i64)	8	Settlement value.
settlementDate	Date	Alias (u32)	4	Settlement date.
tradingTimestamp	Timestamp	Alias (u64)	8	Date and time when the transaction was executed.

Name	Data type	Kind	Length (bytes)	Description
publicProductIdentification.productIdentification	ProductIdentification	Array (AnsiChar)	30	Product identification, for example its ISIN code.
publicProductIdentification.productIdentificationType	ProductIdentificationType	Enum	1	Type of product identification. **NotApplicable** (1) – Not applicable **ISIN** (2) – Financial instrument's ISIN.
price	Price	Alias (i64)	8	Trade price.
priceCurrency	Currency	Enum	2	Currency in which the price is expressed (applicable if the price is expressed as monetary value).
priceNotation	PriceExpressionType	Enum	1	Indication as to whether the price is expressed in monetary value, in percentage or in yield. **NotApplicable** (3) – Not applicable. **Price** (1) – Price expressed as absolute value. **Percentage** (2) – Price expressed as percentage.
quantity	Quantity	Alias (u64)	8	Trade quantity.
nominalValue	Value	Alias (i64)	8	Nominal value.
nominalCurrency	Currency	Enum	2	Currency in which the notional is denominated.
mic	MicCode	Array (AnsiChar)	4	Identification of the venue where the transaction was executed.
publicationTimestamp	Timestamp	Alias (u64)	8	Date and time when the transaction was published by a trading venue.
tradeId	TradeId	Alias (u32)	4	ID of the trade.
tradeToBeCleared	bool	Primitive	1	Code to identify whether the transaction will be cleared.

Name	Data type	Kind	Length (bytes)	Description
pctChange	PercentageChange	Alias (i64)	8	Percentage change.
vwap	Price	Alias (i64)	8	Volume-weighted average price.
noTrades	u64	Primitive	8	Total number of transactions.
totalVolume	Quantity	Alias (u64)	8	Total transaction volume.
totalValue	Value	Alias (i64)	8	Total transaction value.
openingPrice	Price	Alias (i64)	8	The price of the first trade on the current trading day.
maxPrice	Price	Alias (i64)	8	Highest price of the instrument on the current trading day.
minPrice	Price	Alias (i64)	8	Lowest price of the instrument on the current trading day.
mmtMarketMechanism	MmtMarketMechanism	Enum	1	MMT Market Mechanism
mmtTradingMode	MmtTradingMode	Enum	1	MMT Trading Mode

MMT Market Mechanism

Name	Value	Description
CentralLimitOrderBook	1	1 = Central Limit Order Book
QuoteDrivenMarket	2	2 = Quote Driven Market NOT USED
DarkOrderBook	3	3 = Dark Order Book NOT USED
OffBook	4	4 = Off Book
PeriodicAuction	5	5 = Periodic Auction NOT USED
RequestForQuotes	6	6 = Request For Quotes NOT USED
Other	7	7 = Any other including Hybrid

MMT Trading Mode

Name	Value	Description
UndefinedAuction	1	1 = Undefined Auction NOT USED
ScheduledOpeningAuction	2	O = Scheduled Opening Auction

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length (bytes)</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><td>ScheduledClosingAuction</td><td>3</td><td>K = Scheduled Closing Auction</td></tr>
<tr><td>ScheduledIntradayAuction</td><td>4</td><td>I = Scheduled Intraday Auction NOT USED</td></tr>
<tr><td>UnscheduledAuction</td><td>5</td><td>U = Unscheduled Auction</td></tr>
<tr><td>ContinuousTrading</td><td>6</td><td>2 = Continuous Trading</td></tr>
<tr><td>AtMarketCloseTrading</td><td>7</td><td>3 = At Market Close Trading NOT USED</td></tr>
<tr><td>OutOfMainSessionTrading</td><td>8</td><td>4 = Out Of Main Session Trading NOT USED</td></tr>
<tr><td>TradeReportingOnExchange</td><td>9</td><td>5 = Trade Reporting (On Exchange)</td></tr>
<tr><td>TradeReportingOffExchange</td><td>10</td><td>6 = Trade Reporting (Off Exchange) NOT USED</td></tr>
<tr><td>TradeReportingSystematicInternaliser</td><td>11</td><td>7 = Trade Reporting (Systematic Internaliser) NOT USED</td></tr>
</table>
</td></tr>
<tr><td>mmtTransactionCategory</td><td>MmtTransactionCategory</td><td>Enum</td><td>1</td><td>MMT Transaction Category
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>DarkTrade</td><td>1</td><td>D = Dark Trade NOT USED</td></tr>
<tr><td>TradePriceImprovement</td><td>2</td><td>R = Trade that has received price improvement NOT USED</td></tr>
<tr><td>PackageTrade</td><td>3</td><td>Z = Package Trade (excluding Exchange For Physicals) NOT USED</td></tr>
<tr><td>ExchangeForPhysicalsTrade</td><td>4</td><td>Y = Exchange For Physicals Trade NOT USED</td></tr>
<tr><td>None</td><td>5</td><td>- = None apply (a standard trade for the Market Mechanism and Trading Mode)</td></tr>
</table>
</td></tr>
<tr><td>mmtNegotiationIndicator</td><td>MmtNegotiationIndicator</td><td>Enum</td><td>1</td><td>MMT Negotiation Indicator</td></tr>
</table>

Name	Data type	Kind	Length (bytes)	Description

Name	Value	Description
NegotiatedTrade	1	N = Negotiated Trade
NegotiatedTradeLiquidFinancialInstruments	2	1 = Negotiated Trade in Liquid Financial Instruments
NegotiatedTradeIlliquidFinancialInstruments	3	2 = Negotiated Trade in Illiquid Financial Instruments
NegotiatedTradeOther	4	3 = Negotiated Trade subject to conditions other than the current market price NOT USED
NoNegotiatedTrade	5	- = No Negotiated Trade NOT USED
PreTradeTransparencyWaiverIlliquidInstrument	6	4 = Pre-Trade Transparency Waiver for Illiquid Instrument on an SI (for RTS 1 only) NOT USED

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length (bytes)</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><td>PreTradeTransparencyWaiverStandardMarketSize</td><td>7</td><td>5 = Pre-Trade Transparency Waiver for above Standard Market Size on an SI (for RTS 1 only) NOT USED</td></tr>
<tr><td>PreTradeTransparencyWaiversIlqdAndSize</td><td>8</td><td>6 = Pre-Trade Transparency Waivers of ILQD and SIZE (for RTS 1 only) NOT USED</td></tr>
</table>
</td></tr>
<tr><td>mmtAgencyCrossTradeIndicator</td><td>MmtAgencyCrossTradeIndicator</td><td>Enum</td><td>1</td><td>MMT Agency Cross Trade Indicator
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>AgencyCrossTrade</td><td>1</td><td>X = Agency Cross Trade NOT USED</td></tr>
<tr><td>NoAgencyCrossTrade</td><td>2</td><td>- = No Agency Cross Trade</td></tr>
</table>
</td></tr>
<tr><td>mmtModificationIndicator</td><td>MmtModificationIndicator</td><td>Enum</td><td>1</td><td>MMT Modification Indicator
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>TradeCancellation</td><td>1</td><td>C = Trade Cancellation</td></tr>
<tr><td>TradeAmendment (for future use)</td><td>2</td><td>A = Trade Amendment</td></tr>
<tr><td>NewTrade</td><td>3</td><td>- = New Trade</td></tr>
</table>
</td></tr>
<tr><td>mmtBenchmarkReferencePriceIndicator</td><td>MmtBenchmarkReferencePriceIndicator</td><td>Enum</td><td>1</td><td>MMT Benchmark / Reference Price Indicator
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>BenchmarkTrade</td><td>1</td><td>B = Benchmark Trade NOT USED</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length (bytes)</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><td>ReferencePriceTrade</td><td>2</td><td>S = Reference Price Trade NOT USED</td></tr>
<tr><td>NoBenchmarkorReferencePriceTrade</td><td>3</td><td>- = No Benchmark or Reference Price Trade</td></tr>
</table>
</td></tr>
<tr><td>mmtSpecialDividendIndicator</td><td>MmtSpecialDividendIndicator</td><td>Enum</td><td>1</td><td>MMT Special Dividend Indicator
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>SpecialDividendTrade</td><td>1</td><td>E = Special Dividend Trade</td></tr>
<tr><td>NoSpecialDividendTrade</td><td>2</td><td>- = No Special Dividend Trade</td></tr>
</table>
</td></tr>
<tr><td>mmtOffBookAutomatedIndicator</td><td>MmtOffBookAutomatedIndicator</td><td>Enum</td><td>1</td><td>MMT Off Book Automated Indicator
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>UnspecifiedOrNotApply</td><td>1</td><td>'- = Unspecified or does not apply</td></tr>
<tr><td>OffBookNonAutomated</td><td>2</td><td>M = Off Book Non-Automated</td></tr>
<tr><td>OffBookAutomated</td><td>3</td><td>Q = Off Book Automated</td></tr>
</table>
</td></tr>
<tr><td>mmtOrdinaryTradeIndicator</td><td>MmtOrdinaryTradeIndicator</td><td>Enum</td><td>1</td><td>MMT Ordinary Trade Indicator
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>PlainVanillaTrade</td><td>1</td><td>P = Plain-Vanilla Trade</td></tr>
<tr><td>NonPriceFormingTrade</td><td>2</td><td>T = Non-Price Forming Trade (formerly defined as a Technical Trade) NOT USED</td></tr>
<tr><td>TradeNotContributingToPriceDiscovery</td><td>3</td><td>J = Trade not contributing to the price discovery process (formerly defined as a Technical Trade) NOT USED</td></tr>
<tr><td>PricePending</td><td>4</td><td>N = Price is currently not available but pending NOT USED</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length (bytes)</th><th>Description</th></tr>
<tr><td>mmtAlgorithmicIndicator</td><td>MmtAlgorithmicIndicator</td><td>Enum</td><td>1</td><td>MMT Algorithmic Indicator
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>AlgorithmicTrade</td><td>1</td><td>H = Algorithmic Trade</td></tr>
<tr><td>NoAlgorithmicTrade</td><td>2</td><td>- = No Algorithmic Trade</td></tr>
</table></td></tr>
<tr><td>mmtPostTradeDeferralReason</td><td>MmtPostTradeDeferralReason</td><td>Enum</td><td>1</td><td>MMT Post-Trade Deferral Reason
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>ImmediatePublication</td><td>1</td><td>- = Immediate Publication</td></tr>
<tr><td>NonImmediatePublication</td><td>2</td><td>1 = Non-Immediate Publication NOT USED</td></tr>
<tr><td>NonImmediatePublicationLargeInScale</td><td>3</td><td>2 = Non-Immediate Publication : Deferral for 'Large in Scale' NOT USED</td></tr>
<tr><td>NonImmediatePublicationIlliquidInstrument</td><td>4</td><td>3 = Non-Immediate Publication : Deferral for 'Illiquid Instrument' (for RTS 2 only) NOT USED</td></tr>
<tr><td>NonImmediatePublicationSizeSpecific</td><td>5</td><td>4 = Non-Immediate Publication : Deferral for 'Size Specific' (for RTS 2 only) NOT USED</td></tr>
<tr><td>NonImmediatePublicationIlqdAndSize</td><td>6</td><td>5 = Non-Immediate Publication : Deferrals of ILQD and SIZE (for RTS 2 use only) NOT USED</td></tr>
<tr><td>NonImmediatePublicationIlqdAndLrgs</td><td>7</td><td>6 = Non-Immediate Publication : Deferrals of ILQD</td></tr>
</table></td></tr>
</table>

Name	Data type	Kind	Length (bytes)	Description
				and LRGS (for RTS 2 use only) NOT USED
mmtPostTradeDeferralType	MmtPostTradeDeferralType	Enum	1	MMT Post-Trade Deferral Type

Name	Value	Description
LimitedDetailsTrade	1	1 = Limited Details Trade NOT USED
DailyAggregatedTrade	2	2 = Daily Aggregated Trade NOT USED
VolumeOmissionTrade	3	3 = Volume Omission Trade NOT USED
FourWeeksAggregationTrade	4	4 = Four Weeks Aggregation Trade NOT USED
IndefiniteAggregationTrade	5	5 = Indefinite Aggregation Trade NOT USED
VolumeOmissionTradeAggregated	6	6 = Volume Omission Trade, eligible for subsequent enrichment in aggregated form NOT USED
NotApplicable	7	- = Not applicable / No relevant deferral or enrichment type NOT USED

14.38. TradeCollars

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
instrumentId	ElementId	Alias (u32)	4	ID of financial instrument.

Name	Data type	Kind	Length (bytes)	Description
collarType	CollarType	Enum	1	Collar type. Name: Static, Value: 1 Name: Dynamic, Value: 2
price	Price	Alias (i64)	8	The reference price.
lower	Bound	Alias (i64)	8	Value of the lower Price collar.
upper	Bound	Alias (i64)	8	Value of the upper Price collar.

14.39. TradingPhaseScheduleEntry

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
tradingScheduleId	ElementId	Alias (u32)	4	Trading schedule ID.
tradingPhaseId	ElementId	Alias (u32)	4	Trading phase ID.
tradingPhaseSettlementCycle	u8	Primitive	1	Settlement cycle for transactions made during a continuous phase.
tradingPhaseMaxBlockSettlementCycle	u8	Primitive	1	Max settlement cycle for block transactions.
staticCollarVolatilityAuctionId	ElementId	Alias (u32)	4	Static collar volatility auction ID.
dynamicCollarVolatilityAuctionId	ElementId	Alias (u32)	4	Dynamic collar volatility auction ID.
tradingPhaseStartTime	Timestamp	Alias (u64)	8	Trading phase start time.

Name	Data type	Kind	Length (bytes)	Description
tradingPhaseType	TradingPhaseType	Enum	1	Type of matching algorithm.

Name	Value	Description
ContinuousPriceTime	1	Continuous price and time.
ContinuousRefPriceTime	2	Continuous time at reference price.
ContinuousLastAuctionTime	3	Trade at last.
AuctionOpening	4	Opening auction.
BlockExactMatching	5	Block transaction, exact matching.
NoTradingClosed	6	Trading disabled, no matching.
Uncrossing	7	Uncrossing.
AuctionClosing	8	Closing auction.
AuctionIntraday	9	Intraday auction.
AuctionVolatilityStatic	10	Volatility auction after static collars breach.
AuctionVolatilityDynamic	11	Volatility auction after dynamic collars breach.
AuctionExtendedVolatilityStatic	12	Additional volatility auction after static collar breach. It can be activated independently or as an extension of the previous AuctionVolatilityStatic phase.
AuctionExtendedVolatilityDynamic	13	Additional volatility auction after dynamic collar breach. It can be activated independently or as an extension of the previous AuctionVolatilityDynamic phase.
NoTradingEarlyMonitoring	14	Early Monitoring
NoTradingLateMonitoring	15	Late Monitoring
Hybrid	16	Hybrid normal trading
HybridBuyOnly	17	Hybrid buy only phase
HybridPreTrade	18	Hybrid pre trade phase
UnsuspensionAuction	19	A phase triggered whenever an instrument's changes its status from RegulatorySuspension or MarketOperationSuspension before entering into continuous phase.

Name	Data type	Kind	Length (bytes)	Description		
				Ipo	20	Initial public offering phase
				TenderOffer	21	Tender offer phase
				HybridPreTradeBuyOnly	22	Hybrid pre trade BuyOnly phase
				Distribution	23	Instrument distribution phase
uncrossing	bool	Primitive	1	True if the phase includes an uncrossing.		
extStaticCollarVolatilityAuctionId	ElementId	Alias (u32)	4	ID of Static Collar Volatility Auction.		
extDynamicCollarVolatilityAuctionId	ElementId	Alias (u32)	4	ID of Dynamic Collar Volatility Auction.		
clatId	ElementId	Alias (u32)	4	ID of last auction phase.		

14.40. TradingSessionStatus

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
marketId	MicCode	Array (AnsiChar)	4	Market structure's Market Identifier Code (MIC) as specified in ISO 10383.
marketStructureId	ElementId	Alias (u32)	4	ID of the financial instrument's market segment.
tradingSessionEvent	TradingSessionEvent	Enum	1	Identifies an event related to the trading status of a trading session.

Name	Value	Description
NA	1	Not applicable.
StartOfTechnicalSession	2	Start of technical session.
EndOfTechnicalSession	3	End of technical session.
InitialReferenceDataStart	4	Initial reference data start.

Name	Data type	Kind	Length (bytes)	Description		
				InitialReferenceDataEnd	5	Initial reference data end.
				PreviousDayRestateStart	6	Previous day restate start.
				PreviousDayRestateEnd	7	Previous day restate end.
				NextSessionReferenceDataStart	8	Next session reference data start.
				NextSessionReferenceDataEnd	9	Next session reference data end.
				StartOfTradingSessionMIC	12	Start of trading session MIC.
				EndOfTradingSessionMIC	13	End of trading session MIC.
date	Date	Alias (u32)	4	Date of the business session when provided and 0 for StartOfTechnicalSession and EndOfTechnicalSession events.		

14.41. WeekPlan

Name	Data type	Kind	Length (bytes)	Description
header	Header	Struct	42	Message header.
calendarId	ElementId	Alias (u32)	4	Calendar ID.
weekdays	Weekdays	Array (bool)	7	Boolean array of weekdays (Monday to Sunday) open or closed for trading; true – weekday open for trading, false – weekday closed for trading.

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